

# defumat

what it computes, and how to ask for it

Plane-wave DFT in Python and JAX, validated against Quantum ESPRESSO

September 15, 2026

## Abstract

User documentation for the feature set: every capability, the equation behind it, a snippet that runs it, what it was validated against, and *what it refuses*. Two conventions run through the whole document.

**Everything is validated against Quantum ESPRESSO on the same input**, and the agreement is quoted per feature rather than claimed in general. Where a number is missing it is because the comparison does not exist, and that is said rather than hidden.

**Anything not implemented is refused by name**, with an error saying what is missing — never silently approximated. The “Refuses” notes are as much of the documentation as the features: they are the promise that a run which starts is a run whose physics is there.

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**One equation is written down and the rest are derived from it.** Almost everything below is a derivative of the total energy, taken by the compiler rather than by hand:

$$E[\{\psi\}, \tau, \varepsilon] = T_s + E_H + E_{xc} + E_{\text{loc}} + E_{\text{nl}} + E_{\text{Ewald}}$$

| quantity                                 | what it is                                                      | how it is obtained                        |
|------------------------------------------|-----------------------------------------------------------------|-------------------------------------------|
| $v_{xc}(\mathbf{r})$                     | $\delta E_{xc} / \delta n$                                      | jax.grad of the energy density            |
| $\mathbf{F}_I$                           | $-\partial E / \partial \boldsymbol{\tau}_I$                    | jax.grad at frozen $\psi$                 |
| $\sigma_{ij}$                            | $-\Omega^{-1} \partial E / \partial \varepsilon_{ij}$           | jax.grad along a strain                   |
| $C_{I\alpha, J\beta}$                    | $\partial^2 E / \partial \tau_{I\alpha} \partial \tau_{J\beta}$ | jvp of the force                          |
| $Z_{I, \alpha\beta}^*$                   | $\partial F_{I\alpha} / \partial \mathcal{E}_\beta$             | jvp of the force along the field response |
| $\partial \epsilon_{ij} / \partial \tau$ | Raman                                                           | jvp of the second-order energy            |

The “frozen  $\psi$ ” is what makes this exact rather than approximate: at self-consistency the energy is stationary with respect to the wavefunctions, so the terms from  $\partial \psi / \partial \lambda$  vanish. That is the Hellmann-Feynman argument, and taking it one order up is the  $2n+1$  theorem, which is why a *third* derivative needs only first-order wavefunctions.

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# 1 Input and invocation

defumat reads **pw.x input files**. The same file that runs in Quantum ESPRESSO runs here, which is what makes every comparison in this document possible.

```
from defumat import Calculator

calc = Calculator.from_file("scf.in", pseudo_dir="pseudo", conv_thr=1.0e-8)
result = calc.get_scf()

print(result.total_energy, "Ry in", result.iterations, "iterations")
```

conv\_thr means what it means in a pw.x input: it is compared against QE's dr2, the Hartree energy of the density residual.

It need not be given at all. from\_file and from\_text **adopt the input's own &electrons namelist** — conv\_thr, mixing\_beta, mixing\_mode, mixing\_fixed\_ns and electron\_maxstep — so an input file that states how to converge itself converges the same way here as under pw.x. A keyword argument given to the constructor still wins over the file, which is how the snippet above overrides it. One variable of that namelist is read by pw.x and deliberately *not* adopted: diagonalization, because this package offers one eigensolver, so honouring diagonalization = 'cg' would turn a run that works into an error, and mapping it silently onto Davidson would be the kind of substitution refused everywhere else here. Name a solver at construction instead.

**The adopted namelist reaches a relaxation too**, and did not until 2026-09-11: electron\_maxstep, diago\_full\_acc and mixing\_fixed\_ns were read off the file and then dropped on the way into get\_relax, so every SCF inside a relaxation ran at the default hundred iterations whatever the input asked for. On a DFT+U relaxation the dropped mixing\_fixed\_ns decides which minimum of the +U functional the run lands in, so the relaxed *geometry* could differ with nothing saying the request had been ignored.

**An input variable that changes the physics and is not implemented is refused by name**, never read and ignored — a run that quietly drops one is wrong by far more than the agreement with pw.x this code is checked to. That covers a charged cell (tot\_charge), the saw-tooth field and the dipole correction (tefield, dipfield), the Berry-phase finite field (lelfield), the isolated-system corrections (assume\_isolated), two chemical potentials (twochem, in either namelist), per-orbital atomic occupations (one\_atom\_occupations), the constant-cutoff kinetic functional (qcutz), a hand-pinned FFT grid (nr1-nr3s), and the four **symmetry** switches no\_t\_rev, force\_symmorphic, use\_all\_frac and nosym\_evc.

Those last four were read by *nothing* until 2026-09-11. Each says that some part of the group is not a symmetry of the state being asked for, so ignoring one left the k-mesh reduced with operations the input had excluded and sym\_rho averaging over them — a total energy that will not match a benchmark generated from the same file, with nothing saying why. Anything QE defaults to is silent, so an ordinary input never reaches the refusal.

## One object, bound methods

A Calculator is a system together with its pseudopotentials, and every calculation in this document is a method on it. The pseudopotentials are read from the names the ATOMIC\_SPECIES

card gave, resolved against `pseudo_dir` — which `from_file` defaults to the input file's own directory — so a script names an input and then names physics:

```
calc = Calculator.from_file("scf.in", pseudo_dir="pseudo")

bands = calc.get_bands(kpoints=path)      # runs the SCF first if none is cached
dos = calc.get_dos(grid=(8, 8, 8))
forces = calc.get_forces()
epsilon = calc.get_dielectric_tensor()
phonons = calc.get_phonons()

bands.plot()                             # and each result draws itself
```

Three properties are worth knowing before relying on it.

**The ground state is cached, and computed on demand.** A method that needs one and finds no cache runs an SCF, printing a line to `stderr` saying so; `get_scf()` is the explicit path and is the same code. The cache is a *single slot* holding the result together with the options that produced it — calling `get_scf` again with different options reruns and replaces it — because what it holds is the wavefunctions, and a cache keyed by option sets would quietly hold several sets of them. `calc.scf_result` reads the slot without triggering anything, and a state that did not converge is refused rather than differentiated.

**Nothing mutates.** `with_positions`, `with_cell`, `with_spin` and `with_moments` return a *new* calculator with an empty cache; the converged state crosses as a starting guess where that is sound, which for `with_spin` is the spin-regime continuation described under *Spin, magnetism and spin-orbit* and converges in one iteration rather than twenty-five. A cached result under a moved atom would otherwise answer for the geometry it was converged at.

**The refusals pass through untouched.** Every restriction in this document — PAW Born charges, a metal's Raman tensor, a spiral with spin-orbit coupling — raises through the bound method exactly as through the function beneath it.

The functional entry points the rest of this document names — `run_scf`, `run_bands`, `dielectric_tensor` and the others — are unchanged and remain the way to drive a calculation whose state is being managed by hand:

```
from defumat.io.pwin import read_pw_input
from defumat.pseudo import read_upf
from defumat.scf.driver import Calculation, run_scf
from defumat.system import build_system

system = build_system(read_pw_input("scf.in"))
pseudos = tuple(read_upf(f"pseudo/{s.pseudo_file}") for s in system.structure.species)
result = run_scf(system, pseudos, conv_thr=1.0e-8)
```

What such a call has to carry with the density is the rest of the converged state — `becsum` for a PAW dataset, `ns` under a Hubbard  $U$ , `tau` under a meta-GGA. None of the three can be rebuilt from the density, and the entry points refuse without them rather than computing something else. The calculator passes them.

## Every method, in one place

The rest of this document introduces each quantity where its physics belongs, and names the functional entry point beneath it. This is the index the other way round — what to *call*, for a reader who knows the physics and wants the line. It is complete: a method absent from it is a documentation bug, and a test says so (tests/unit/test\_facade\_documentation.py).

---

### *The ground state*

---

|              |                                                        |
|--------------|--------------------------------------------------------|
| get_scf      | the self-consistent field loop, or the cached result   |
| get_elk_seed | Elk's converged density on this run's grid, as a guess |

### *Band structure and densities of states*

---

|           |                                                                 |
|-----------|-----------------------------------------------------------------|
| get_bands | diagonalise on a k-path at the converged density                |
| get_nscf  | diagonalise on a k-grid at the converged density, and occupy it |
| get_dos   | the density of states, on a denser grid if asked                |
| get_pdos  | the projected density of states, with Löwdin charges            |

### *Derivatives of the energy*

---

|            |                                                        |
|------------|--------------------------------------------------------|
| get_forces | the forces on the atoms                                |
| get_stress | the stress tensor                                      |
| get_relax  | relax the geometry, and the whole optimisation with it |

### *Response, vibrations and spectra*

---

|                          |                                                                               |
|--------------------------|-------------------------------------------------------------------------------|
| get_band_velocities      | $d\varepsilon/d\mathbf{k}$ for every band                                     |
| get_effective_mass       | $(1/m^*)_{ab}$ at one k-point, in $1/m_e$                                     |
| get_angular_momenta      | $\langle L \rangle$ , $\langle S \rangle$ , $\langle J \rangle$ on every atom |
| get_dielectric_tensor    | $\epsilon^\infty$ , with no empty states                                      |
| get_born_charges         | the Born effective charges $Z^*$                                              |
| get_phonons              | the dynamical matrix at $\Gamma$ , and its modes                              |
| get_phonons_at_q         | the dynamical matrix at one wavevector                                        |
| get_raman_tensors        | $d\epsilon/d\tau$ , per atom                                                  |
| get_vibrational_spectrum | per-mode Raman and infrared activities                                        |
| get_strain_response      | the first-order response to a homogeneous strain                              |
| get_elastic_constants    | $C_{ijkl}$                                                                    |
| get_piezoelectric_tensor | $e_{(k)ij}$ , clamped-ion, in C/m <sup>2</sup>                                |
| get_electrostriction     | $d\chi/d(\text{strain})$ and its four tensors                                 |
| get_absorption           | an optical spectrum, with excitons                                            |
| get_optical_conductivity | $\sigma_{ab}(\omega)$ , the Kerr angle, the anomalous Hall                    |
| get_structure_factors    | $F(\mathbf{H})$ , X-ray and magnetic                                          |
| get_stm                  | a Tersoff-Hamann scanning-tunnelling image                                    |
| get_sts                  | the same, over an energy axis: a tunnelling spectrum                          |
| get_vertical_transport   | tunnelling <i>through</i> a 2D material                                       |
| get_momentum_transport   | which k-points that current comes out of                                      |
| get_nesting              | $N(\mathbf{q})$ , the Fermi-surface nesting function                          |
| get_spin_susceptibility  | $\chi^{+-}(\mathbf{q}, \omega)$                                               |
| get_magnon_dispersion    | $\omega(\mathbf{q})$ along a path                                             |
| get_shift_current        | the bulk photovoltaic effect, in A/V <sup>2</sup>                             |
| get_shg                  | $\chi^{(2)}(-2\omega; \omega, \omega)$ , in pm/V                              |

### *Topology, polarization and magnetism*

---

|                            |                                                   |
|----------------------------|---------------------------------------------------|
| get_berry_curvature        | $\Omega(\mathbf{k})$ on a plane mesh              |
| get_chern                  | the Chern number of one plane                     |
| get_z2                     | the 2D $Z_2$ invariant, by Wilson loops or parity |
| get_z2_3d                  | the four $Z_2$ indices of a 3D crystal            |
| get_polarization           | the Berry-phase polarization                      |
| get_orbital_magnetization  | $M_{\text{orb}}$ , by the modern theory           |
| get_magnetoelectric_tensor | $\alpha_{ij} = dP_i/dB_j$                         |

|                     |                                                        |
|---------------------|--------------------------------------------------------|
| get_anisotropy      | the magnetocrystalline anisotropy                      |
| get_torque          | the magnetic torque, and $K$ from one angle            |
| get_first_order_soc | the spin-orbit expectation at coupling-free states     |
| get_force_theorem   | one direction of the anisotropy: its band energy alone |

### *Spin spirals*

|                       |                                                     |
|-----------------------|-----------------------------------------------------|
| get_spiral_scan       | one SCF per spiral wavevector                       |
| get_spiral_relaxation | the spiral wavevector relaxed down $dE/d\mathbf{q}$ |

### *Modulations over many cells*

|                         |                                                      |
|-------------------------|------------------------------------------------------|
| get_ultracell           | a density or potential modulated over $N$ unit cells |
| get_ultracell_stm       | a Tersoff-Hamann image of that modulation            |
| get_ultracell_sts       | a tunnelling spectrum across that modulation         |
| get_ultracell_transport | tunnelling through it, tip to substrate              |

A command line covers the common workflows:

```
python3 -m defumat.cli inspect <qe-output> # summarise what the parser reads
python3 -m defumat.cli dos <input> # density of states
python3 -m defumat.cli pdos <input> # projected DOS
python3 -m defumat.cli relax <input> # structural relaxation
python3 -m defumat.cli stress <input> # stress tensor
python3 -m defumat.cli spiral <input> # spin-spiral scan
```

Standard `pw.x` variables — `ecutwfc`, `ecutrho`, `ibrav`, `nbnd`, `occupations`, `degauss` and the rest — mean what they mean there and are not re-documented here. A handful of knobs are worth knowing because they are specific to this code or to a feature below:

|                                                   |                                                           |
|---------------------------------------------------|-----------------------------------------------------------|
| <code>mbj_c</code>                                | fixes Tran-Blaha's $c$ instead of taking the cell average |
| <code>london_s6</code> , <code>london_rcut</code> | Grimme D2's scaling and real-space cutoff                 |
| <code>cell_dofree</code>                          | which cell degrees of freedom a vc-relax may move         |
| <code>starting_ns_eigenvalue</code>               | seeds the DFT+U occupation matrix                         |
| <code>spiral_q</code>                             | the spin-spiral wavevector, in lattice coordinates        |
| <code>LOCAL_MAGNETIC_FIELDS</code>                | a card with no <code>pw.x</code> counterpart              |

**Units are Rydberg atomic units throughout** (Ry, bohr), matching QE. The only layer that speaks anything else is `io/`: Hubbard  $U$  in the HUBBARD card is in eV and is converted at the boundary, as `pw.x` does it.

## What a run will cost, before it is run

Calculation builds the G-vectors, the plane waves and the projectors on the device, so an input too large for the machine dies in setup without saying what was too large. `Calculator.estimate()` answers that from the cutoffs and the datasets alone, allocating nothing:

```
print(Calculator.from_file("scf.in").estimate().report())
```

or `python3 -m defumat.cli size scf.in`. **It sizes the run the input describes**, which means it reads that input's own `diago_david_ndim` and resolves `k_batch` the way the SCF would — an estimate that quietly assumed a library default would report a run that does not happen,

and nobody relying on one is in a position to notice. The arguments and the CLI flags are *overrides* of the file, and the report's header states which assumptions it used. It reports  $n_{gm}$ ,  $n_{gms}$ ,  $n_{pwx}$ ,  $n_{bnd}$ ,  $n_{kb}$  and both FFT grids — *exactly*, from the same predicates the setup selects with, which is asserted against a real calculation rather than trusted — and a floor on the bytes for the wavefunctions, the projectors, the eigensolver's subspace and the fields. The G-vector enumeration runs in slabs, so an input destined for a GPU can be sized on a laptop.

Two lines of that report are worth reading before believing a hand estimate. The Davidson subspace is  $2 n_{vecx} n_{dim}$  for  $\psi$  and  $H\psi$  at QE's  $n_{vecx} = 4 n_{bnd}$ , plus the Ritz block, and it is usually the largest single array — on a 157-atom slab at  $E_{cut} = 60$  Ry it is 139 GB against 23 GB of wavefunctions. And the eigensolver is *not* doubled by spin, because the channels are solved one after another, where the wavefunctions are; the report separates the two.

The one lever that changes that floor without changing the physics is `diago_david_ndim` — QE's own name for it, read from `&electrons`, and settable per call or at construction:

```
calc = Calculator.from_file("scf.in", david=2) # or diago_david_ndim = 2
```

It halves  $\psi$  and  $H\psi$  exactly, for one more SCF iteration on silicon and the same total energy to 7.3e-11 Ry — the trade QE's documentation describes. Note that `INPUT_PW.txt` gives its default as 2 and is stale: `input_parameters.f90` sets 4 and `input.f90` assigns it straight through, so the number `pw.x` runs is **4**, and that is the default here.

## A converged state on disk

`result.save(path)` writes the state and `SCFResult.load(path, system=...)` reads it back for `run_scf(starting_from=...)` so a job that hits its wall clock does not cost the run. The system is supplied rather than stored — a resume already has the input file — and is checked against a fingerprint. `run_relax(checkpoint_dir=...)` does the same every ionic step *including the optimizer's own history*, which is the half that is easy to omit and most of the value: BFGS earns its convergence rate from the inverse Hessian and the trust radius, so a resume from positions alone takes its next step as if it were the first. Stopping a six-step relaxation at step two and resuming costs 2 + 4 steps, not 2 + 6.

## Gamma-only storage: half the plane waves

At  $k = 0$  a Kohn-Sham state can be chosen real, so  $c(-G) = c^*(G)$  and one plane wave of each  $(G, -G)$  pair carries all of it. `K_POINTS gamma` stores that half, which halves  $n_{pwx}$  and with it every array a band lives in — the wavefunctions, `vkb`, the Davidson subspace. On a 157-atom slab at  $E_{cut} = 60$  Ry that is **189 GB against 96**. It is a memory feature, not a speed one: the two-bands-per-FFT packing of `vloc_psi_gamma` is deliberately not implemented.

**Only the wavefunction sphere halves.** The dense  $G$  set stays whole, where `pw.x` halves both. The memory is entirely in the plane-wave-sized arrays, so halving the dense set as well saves a fifth of a per cent and puts a conjugate fill inside every consumer of a real field. The consequence to know is that  $n_{gm}$  here is not the one `pw.x` prints: the relation is  $n_{gm} = 2n_{gm}^{QE} - 1$ , and the test suite asserts it.

Three things carry the trick: the field is rebuilt from both halves, every sum over plane waves becomes  $2 \operatorname{Re}(\sum) - (G=0)$  — that vector being its own conjugate partner rather than half a pair — and  $\operatorname{Im} c(0)$  must stay zero. **The last is the silent one:** a complex  $c(0)$  makes the rebuilt field complex and the run converges to a plausible wrong answer.

**The validation needs no other code.** An explicit  $k = 0$  on the whole sphere is the same calculation in the other storage, so the two agree to round-off: totals to **1e-14** Ry, eigenvalues to **3e-15** and forces to **3e-16** Ry/bohr, on LDA, PBE, LSDA and PBE+LSDA alike.

**Refuses.** K\_POINTS gamma is *substituted* by an explicit  $k = 0$  on the full sphere — the same physics at twice the storage, with a warning — for an ultrasoft or PAW dataset (addusdens and newd need their own fact = 2), for a run that uses symmetry (a rotation carries a stored  $G$  onto an unstored  $-G'$ , so the permutation would need a conjugation; nosym = .true. is the way out), for a spinor or spiral run, and for a **Hubbard  $U$** . The last is the only one of them that would go wrong *during* the run rather than after it:  $n_\sigma$  is built from a plain  $\langle\phi|\psi\rangle$  over the stored list, so the Hubbard potential and energy would follow occupations about a quarter of their true size and the SCF would converge silently to a different ground state. estimate() makes the same decision the driver makes, so it sizes the run either way.

**Four things after the run refuse rather than substitute**, because by then the states exist and the honest answer is to say the sum is not written: the projected density of states, the sum-over-states  $\chi_0$  (so an optical spectrum of a molecule in a box, which is the archetypal gamma case), an STM image and the vertical tunnelling transmission. Each is a plane-wave sum that would have to be  $2 \operatorname{Re}(\sum) - (G=0)$  and is not; the escape is the same explicit  $k = 0$ , which is an *exact* substitution rather than an approximate one.

**The dynamical matrix works under gamma storage**, including a partial one — the combination a molecule on a fixed slab needs. The Sternheimer solve is gamma-aware throughout (the projector, the CG's products, the preconditioner, the response density), and a displacement's source term is one jvp through at\_positions, so it inherits the corrected  $h_{\text{psi}}$  with no branch of its own. Against the whole sphere: frequencies to **9.6e-10**  $\text{cm}^{-1}$  and the matrix to **1.5e-13** Ry/bohr<sup>2</sup>, with the acoustic residue identical; a partial matrix to **5.9e-11**. And  $\chi_0$  against a central difference of the density — which shares no machinery with either storage — to **7.0e-6**.

A **field** response is still refused, and the distinction is the reason: a field enters through the commutator  $[H, \mathbf{r}]$ , which builds plane-wave sums of its own. The dielectric tensor, the Raman tensor, the strain response and the Born charges all stand on it. It does not fail — with the guard lifted silicon's dielectric constant is 501.7/213.1/253.1 against an isotropic 190.8 — which is why it is a refusal. A checkpoint refuses rather than drops a *driven* magnetic field (a fixed-spin-moment scheme's, which no input file determines) and a *Hubbard setup* (without it ns is an array about nothing); an applied field is saved normally. stress, solver and history are deliberately not stored, so what comes back is a state and not a report.

## 2 Ground state

The Kohn-Sham problem, generalised because ultrasoft and PAW make the overlap non-trivial:

$$\begin{aligned} \hat{H} |\psi_{n\mathbf{k}}\rangle &= \epsilon_{n\mathbf{k}} \hat{S} |\psi_{n\mathbf{k}}\rangle, & \hat{S} &= 1 + \sum_{I,ij} q_{ij} |\beta_i^I\rangle \langle \beta_j^I| \\ \hat{H} &= -\nabla^2 + v_{\text{loc}}(\mathbf{r}) + v_{\text{H}}[n] + v_{xc}[n] & &+ \sum_{I,ij} D_{ij}^I |\beta_i^I\rangle \langle \beta_j^I| \end{aligned}$$

with  $n(\mathbf{r}) = \sum_{n\mathbf{k}} w_{\mathbf{k}} f_{n\mathbf{k}} (|\psi_{n\mathbf{k}}|^2 + \sum_{ij} Q_{ij}^I \langle \psi | \beta_i \rangle \langle \beta_j | \psi \rangle)$ , the augmentation term being what ultrasoft adds.  $D_{ij}^I$  is rebuilt from the potential every iteration, which is why the SCF is

a fixed point in  $(n, D)$  and not in  $n$  alone. Convergence is measured as QE measures it — the Hartree energy of the residual:

$$dr^2 = \iint \frac{\delta n(\mathbf{r}) \delta n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' < \text{conv\_thr}$$

```
result = run_scf(system, pseudos, conv_thr=1.0e-10, mixing_mode="anderson")
print(result.total_energy, result.energy_terms) # QE's own term-by-term split
print(result.fermi_energy, result.homo, result.lumo)
```

**Metals** work with every smearing QE has (gaussian, marzari-vanderbilt, methfessel-paxton, fermi-dirac) and with the tetrahedron family — both linear and Bloechl-corrected — usable as an occupation scheme *inside* the SCF, not only for a density of states.

**Continuation across a change of spin regime.** Give `run_scf` a previous result as `starting_from` — `System.with_spin` underneath — and it promotes a converged state into a target's variables: non-magnetic into collinear, collinear into noncollinear, spin-orbit switched on. It reaches the *same* solution as a fresh run ( $\leq 4e-8$  Ry on six cases) in as little as one iteration. The magnetization is **seeded** from the target's `starting_magnetization` when the source has none, because nothing in the SCF breaks spin symmetry on its own and an unseeded promotion would converge back to the unpolarized solution and report success.

A HUBBARD card comes along: the occupation matrix promotes with the density, a collinear pair going into the two *diagonal* spin blocks of the spinor `ns` with the off-diagonal ones zero. That closes the staged route into a hard magnet — converge the collinear antiferromagnet, promote it, let the moments cant — and, because a spinor-to-spinor promotion is the same code path, the **checkpoint resume** of a noncollinear DFT+U run. Coming back the other way keeps the diagonal, which is exact only while the shell is polarised along  $z$ , so a transverse block above  $10^{-8}$  is refused rather than dropped: the diagonal of a canted occupation matrix is a different state, not a coarser one.

**Convergence.** `mixing_mode` selects `anderson` (default), `linear`, `broyden`, Kerker/Thomas-Fermi preconditioning (TF), `local-TF`, OR `adaptive`; `scf_solver` offers a Newton-Krylov alternative that is slower but reaches solutions the mixer cannot hold.

TF screens the residual with one Thomas-Fermi length for the whole cell. `local-TF` makes that length a function of  $\rho(\mathbf{r})$ , which is what a **slab** needs: the metal wants strong screening and the vacuum wants none, and a single compromise value over-screens one and under-screens the other — charge sloshing between the surfaces. The screened residual is the solution of

$$4\pi e^2 v(\mathbf{G}) + |\mathbf{G}|^2 (\alpha v)(\mathbf{G}) = |\mathbf{G}|^2 (\alpha \delta n)(\mathbf{G}), \quad \alpha(\mathbf{r}) = 3 \left( \frac{2\pi}{3} \right)^{5/3} r_s(\mathbf{r}),$$

with  $(\alpha f)$  multiplied in *real* space, so the operator is not diagonal in  $\mathbf{G}$  and is inverted iteratively at each step. On a cobalt film it converges at `mixing_beta` = 0.7 where plain Anderson at the same  $\beta$  diverges and Kerker at 0.3 is still far out after 250 iterations.

`adaptive` is Elk's `mixadapt` and answers the opposite failure. Every other mode here damps an iteration that is *overshooting*; this one lengthens one that is crawling. It carries a separate step length for each component  $j$  of the mixed vector and moves it by the sign of that component's residual,

$$\beta_j \leftarrow \min(\beta_j + \beta_0, \beta_{\max}) \quad \text{if } r_j \text{ kept its sign,} \quad \beta_j \leftarrow \frac{1}{2}(\beta_j + \beta_0) \quad \text{if it changed,}$$

and then steps as  $\beta_j \nu_j + (1 - \beta_j) \mu_j$ . A component pushed the same way for many iterations is one the SCF is crawling along, so its step grows until it moves; one that overshoots and

comes back has its step cut. That makes it a stall detector with no threshold in it, which is why it is worth reaching for on a **magnet**: turning every moment in a cell together costs no energy without spin-orbit coupling, so the residual has no component along that rotation and the flat direction has to be crossed rather than descended, and twisting the moments slowly costs only the spin-wave energy  $Dq^2$ , which vanishes as the cell grows. The whole packed vector adapts independently, so `becsum` and the Hubbard `ns` get their own step lengths rather than the density's.

**mixing\_beta does not mean the same thing under adaptive**, and this is the one thing to know before using it. It is Elk's `beta0`: the increment, the starting value and the floor the halving relaxes towards, all three, not a step length. Elk's default is 0.05 and it is small on purpose; 0.7 saturates at  $\beta_{\max}$  within two iterations and leaves plain linear mixing at 0.85 under an adaptive name, which is what the warning above that value is for. Each mode supplies its own default when `mixing_beta` is unset, 0.7 for the QE family and 0.05 here, so an input that names neither gets the right one; an input that *does* set `mixing_beta` has that number adopted as  $\beta_0$ , and the same figure therefore means two different things depending on the mode.

**Elk's 0.05 is a safe default rather than the best value**, and the measurement says so: on bcc iron with its moment in the plane (`tests/data/qe/fe-noncollin-pbe-stress.in`, ultrasoft, PBE, noncollinear)  $\beta_0 = 0.2$  converges in **16** iterations where  $\beta_0 = 0.05$  takes 24, because the soft directions want the step to grow faster than Elk's default lets it. Anderson on the same cell takes 43 at the input's own `mixing_beta` and 24 at the best one for it, and `pw.x` takes 19; all of them reach the same energy within  $3 \times 10^{-10}$  Ry and the same moment *length* within  $3 \times 10^{-5} \mu_B$ , with the transverse components that symmetry makes zero coming out below  $2.4 \times 10^{-4} \mu_B$ . So raising  $\beta_0$  is the knob to try on a magnet that will not move, and it is the opposite of the reflex a slow SCF usually gets. The saving is smaller than the iteration count suggests, because the diagonalisation threshold is scheduled from the density error and a mixer that drops it faster is handed a tighter eigenproblem: the Davidson work falls by  $1.6\times$  where the iteration count falls by  $2.7\times$ .

On an easy cell it loses, which is what a robustness mixer does: two-atom silicon (`benchmarks/si-1k.in`) at `conv_thr = 1e-10` takes 17 iterations against Anderson's 7, to the same total energy to ten digits.

**A mixer can change which solution the run finds**, and this is the thing to know before trusting an adaptive result on a magnet. On a four-cell hydrogen ultracell under a field turning  $90^\circ$  per cell — the case where rotating every moment together costs nothing — that cell has **three** self-consistent solutions, and every one of them reports `converged = True` with an accuracy three orders below `conv_thr`. Anderson at  $\beta = 0.1$  and 0.7 finds one of them, agreeing to eight digits of the band energy; Anderson at  $\beta = 0.3$  finds a second; adaptive at both its values finds the third, the one whose moments follow the applied field's ladder. **Which is the ground state is not something the run reports**: the moments differ by tens of degrees per cell, and nothing in a converged result orders them.

**The step length does this too, which is the part not to file under “adaptive”**. Anderson's own  $\beta$  moves between solutions here, so this is not a property of one mixer and running a second mixer is not a test that the basin was kept. What a growing step buys is more of the same freedom, not a new kind of it.

So on a cell with a soft or degenerate magnetic direction, vary **both** the mixer and  $\beta$ , and compare the **moments** rather than the iteration count: a converged run is not evidence that the basin was kept, and “fewer iterations” is satisfied by a run that found something else. `SCFResult.site_moments` and `UltracellResult.cell_moments()` are what

to compare.

`mixing_mode = 'adaptive'` also refuses a **preconditioner** rather than ignoring one. Its step is pointwise in real space with its own factor per point, and Kerker or `local-TF` is a single scalar acting in **G**-space; the two do not compose, so a run asking for both raises. Choose the one that matches the direction the cell is slow in: Kerker for long-wavelength charge, this for a residual that is small and will not turn around.

It is registered under its own name and is **not** aliased to any `pw.x` mode. QE has no adaptive mixer, so a `pw.x` input reaching one of QE's names here still gets what QE would do.

`max_iterations` defaults to 100 — **a hard SCF may need more**, and hitting the cap is reported as not converged rather than as an answer: `run_scf` issues a `RuntimeWarning` naming the accuracy it reached, and `calculator` raises. Both matter, because a script driving the functional entry point — a spiral scan, a  $\lambda$  ramp, a moment-versus-field sweep — is exactly what would otherwise read `total_energy` off an unconverged density without looking at converged.

`mixing_ndim` sets how many previous iterations the extrapolation spans, default 8 as in `pw.x`. Raising it is the standard first move on a cell that will not converge, and *on the magnetic benchmark here it does not help*: `benchmarks/fe-mag-1k.in` at `conv_thr = 1e-8` takes 27 iterations at 4, **25 at 8**, 33 at 12 and 39 at 20, so the default is already the best value and the deeper histories are worse. The nonmagnetic twin of the same cell (`fe-unstable-nonmagnetic.in`, same  $\beta$ ) has the same shape — 26, **21**, 30, 26 — which says the sensitivity is not about magnetism.

**dr2 is reported as its two halves** as well as its total, in `scf.history` and on the console. `rho_ddot` weights the charge residual by  $1/G^2$  and the magnetization residual by a constant — a factor of 13.6 apart at  $G_{\min}$  on `fe-mag-1k` — so a single accuracy below `conv_thr` bounds the moment much more weakly than it bounds the charge, and only the split can say whether it is the charge or the magnetization that is still moving.

### 3 Bands, densities of states, projections

A band structure is one diagonalisation per **k** on a fixed density — `run_bands`, or `run_nscf` for a non-self-consistent run on a grid rather than a path. A density of states is

$$g(E) = \sum_{n\mathbf{k}} w_{n\mathbf{k}} \delta(E - \epsilon_{n\mathbf{k}})$$

with  $\delta$  replaced by a smearing function or evaluated by tetrahedron interpolation. The projected DOS weights each term by  $|\langle \phi_\mu | \hat{S} | \psi_{n\mathbf{k}} \rangle|^2$ , and the spilling parameter is what is *not* captured,  $\sum_{n\mathbf{k}} w_{n\mathbf{k}} f_{n\mathbf{k}} (1 - \sum_\mu |\langle \phi_\mu | \hat{S} | \psi \rangle|^2)$ .

```
from defumat.workflows.bands import run_bands
from defumat.workflows.dos import run_dos
from defumat.workflows.pdos import run_pdos

bands = run_bands(system, pseudos, scf.density, kpoints=path)  # a k-path

dos, _ = run_dos(system, pseudos, scf.density, grid=(12, 12, 12),
                 scheme="tetrahedra_opt")  # -> (DensityOfStates, NSCFResult)

pdos, _ = run_pdos(system, pseudos, scf)  # -> (ProjectedDOS, NSCFResult)
```

```
print(pdos.channels[0])          # resolved by atom, l and m
print(pdos.charges.charges, pdos.charges.spilling)
```

The DOS schemes live behind a name registry (DOS\_SCHEMES), and the projected DOS feeds *the same* registry as a per-band weight, so a PDOS and a DOS are the same integration. Projections are  $\langle \phi | \hat{S} | \psi \rangle$  on Löwdin-orthogonalised pseudo-atomic orbitals. Validated against a purpose-built projwfc.x on seven cases: **6.9e-4** on a projection, **4.7e-5** on a Löwdin charge, and the spilling parameter to all four decimals it prints.

## Resolved by $j$ and $m_j$ for a spin-orbit run

With noncolin and lspinorb the orbitals projected onto are the **spin-angle functions**  $|l j m_j\rangle$  rather than  $|l m\rangle$  times a spin, so a channel is labelled by  $j$  and  $m_j$  and the projection separates the two branches spin-orbit coupling splits a shell into. For platinum that is  $5d_{3/2}$  against  $5d_{5/2}$ : one number, resolved, where an  $l$ -resolved projection reports their sum and cannot see the splitting at all. The functions are

$$|l j m_j\rangle = \sum_{\sigma=\uparrow,\downarrow} C(l, \frac{1}{2}, j; m_j - \sigma, \sigma) Y_{l, m_j - \sigma} \chi_{\sigma},$$

Clebsch-Gordan coefficients against the harmonics, and the number of columns is  $\sum_{\text{shells}} (2j+1)$  — which is *not* in general twice the scalar count, because it is read off the dataset's own  $j$  channels.

```
from defumat import Calculator

calc = Calculator.from_file("pt-soc-paw-nosym.in", # noncolin + lspinorb
                           pseudo_dir="../pseudo")

pdos = calc.get_pdos()

print(pdos.channels[0])          # -> #1 Pt1 s_j0.5 m_j=-0.5
print(pdos.charges.format())     # -> s = 0.5032, p = 0.9944, d = 8.4903
print(pdos.charges.charges_lm)  # -> None: a spin-angle function has no m
```

Validated on fully-relativistic **ultrasoft and PAW** platinum datasets against projwfc.x runs generated for them, four ways. The *columns* — how many, and each one's  $(l, j, m_j)$  — match QE's own label table exactly. The two datasets differ there in a way worth seeing: the PAW file carries both  $j$  partners of every  $l$ , so its 18 columns are exactly twice the  $j$ -averaged 9, while the ultrasoft one has 12 against a scalar 11 — the count is  $\sum (2j+1)$  read off the file and not a doubling. The *projections* agree to **6.3e-6**, compared as sums over each degenerate multiplet: a spinor band structure is Kramers-degenerate everywhere, and an individual  $|\langle \phi | \hat{S} | \psi_n \rangle|^2$  is not invariant under the mixing inside a multiplet that either code's eigensolver is free to choose, where the sum is. The **Löwdin charges and the spilling parameter agree to every digit projwfc.x prints**. And the *curves* — the total and each shell's, on projwfc.x's own energy grid and broadening — agree to **0.16 per cent of the peak** everywhere below the lowest empty band, above which the topmost states are the ones neither eigensolver converges. Timing, one core each from a converged ground state: get\_pdos costs **1.65 s** against projwfc.x's 0.62 s, and on both sides that is mostly rebuilding from the pseudopotential files rather than projecting — the projection and its integration are 0.137 s of it here.

Norm-conserving is untested rather than refused — no reference case is committed for it.

## Two spin channels for a noncollinear run without spin-orbit coupling

With `noncolin` but `lspinorb = .false.` there is no  $j$  to resolve by: nothing couples  $s_z$  to the orbital motion, so  $s_z$  is still a good quantum number and the orbitals are  $|l m\rangle\chi_\sigma$  — an up and a down copy of every harmonic, `2 natomwfc` columns where a collinear run has `natomwfc`. What comes back is nonetheless *not* twice as many channels: the columns are routed into an **up and a down density of states**, which is `partialdos_nc`'s `nspin0 = 2` and is the shape an LSDA projection has. This is the regime of any magnetic run with scalar-relativistic datasets, and it is what `h10-chain-noncolin.in` and `alas-magnetoelectric-nosoc.in` are.

```
from defumat import Calculator

calc = Calculator.from_file("h-atom-noncolin.in", # noncolin, no lspinorb
                           pseudo_dir="..pseudo")
pdos = calc.get_pdos(symmetrize=False)

print(pdos.nspin)           # -> 2: an up channel and a down one
print(pdos.channels[0])     # -> #1 H1 s: no s_z, the spin is an axis now
print(pdos.charges.polarization) # -> [0.99857905]: the moment the s shell carries
```

The decomposition is along the **global**  $z$  and not along the local moment, exactly as QE's is: an in-plane texture reports two equal channels, which is a statement about the frame the orbitals are built in rather than a defect.

**No projwfc.x reference for this regime is committed here**, so what validates it is an identity instead, and a sharper one: without spin-orbit coupling a state polarized along  $+z$  block-diagonalises the noncollinear Hamiltonian into the two collinear ones, so the projection of such a run must reproduce an LSDA run of the same cell channel for channel — and the LSDA route is itself validated against `projwfc.x`. On a hydrogen atom in a 12 bohr box the two runs agree to **3e-12 Ry** in total energy, their majority curves to **2e-5 of the peak** and their minority ones to **0.4 per cent**, and their Löwdin charges to **5e-3** on a polarization of **0.9986** electrons — which is the number a code that binned both spins into one channel, or split the weight evenly, would get wrong while passing every sum rule. The two bounds differ for an arithmetic reason rather than a chosen tolerance: the noncollinear branch reaches  $\rho_\downarrow$  as  $(n - |m|)/2$ , a cancellation of two numbers of order 0.1 where the collinear branch carries  $\rho_\downarrow$  itself, and that is worth  $1.1 \times 10^{-4}$  Ry on the *empty* minority eigenvalue where every occupied one agrees to  $10^{-6}$ .

Unlike a spin-orbit run this case *does* have an  $m$  to report, so `charges_lm` is filled where a  $j$ -resolved projection leaves it `None`. `print_lowdin` allocates it only IF (`nspin != 4`) and so prints none for either noncollinear branch; the reason it gives — a spin-angle function has no  $m$  — is true of the spin-orbit branch alone, and the table here is the richer one on purpose.

**Refuses.** A **symmetrised** spinor projection, in either noncollinear branch — run with `nosym` and the whole **k**-grid, where both codes use every point and neither averages, or pass `symmetrize=False`. *The two branches are missing different things, and only one of them is missing a matrix.* Without spin-orbit coupling the columns are  $|lm\rangle \otimes |\sigma\rangle$  and the operator is the tensor product  $D^l \otimes S$ , both factors of which now exist (the harmonic rotations and the  $SU(2)$  ones); what is left there is the coefficient table, which is real over  $2l + 1$  columns and would have to be complex over  $2(2l + 1)$ . With `lspinorb` the columns are  $|j m_j\rangle$  and the operator is  $D^j$  for  $j = \frac{1}{2} \dots \frac{7}{2}$ , which is a different matrix

and is not that tensor product. A **fully-relativistic dataset** with `lspinorb = .false.:` QE dispatches the orbitals on `has_so` and their labels on `lspinorb`, so it builds  $j$ -resolved columns and calls them up/down ones. An  $m$ -resolved charge is *absent* rather than refused for a **spin-orbit** run: `charges_lm` is `None` there, since a spin-angle function has an  $m_j$  and no  $m$ , and there is no  $p_x$  weight to report.

## 4 Pseudopotentials and functionals

**Kinds:** norm-conserving, **ultrasoft** and **PAW** — the two-grid split, the augmentation charge, the overlap operator, self-consistent  $D_{ij}$ , and PAW’s one-centre terms including its radial Poisson solve and spherical quadrature. UPF v2 is read.

**Functionals:** LDA (PZ), **PBE**, **revPBE**, **PBEsol**, on the plane-wave grid and on the PAW spheres. The functional comes from the pseudopotential headers unless `input_dft` overrides it. Only the *energy* is written down;  $v_{xc}$ , and a GGA’s  $v_1$  and  $v_2$ , come from `jax.grad` of it.

**Meta-GGA, potential-only branch:** `input_dft = 'tb09'` (Tran-Blaha) and `'bj06'` (Becke-Johnson). These invert the rule above — they *are* potentials, there is no energy — so the consequences are enforced rather than documented: `run_scf` warns that its total is not the value of any functional it minimised, and forces, stress, phonons and response all refuse. Silicon’s gap goes from LDA’s 0.49 eV to **1.13 eV** against an experimental 1.17.

**Van der Waals:** Grimme **D2** (`vdw_corr = 'grimme-d2'`). Bilayer graphene matches `pw.x` to **3.1e-9 Ry** and binds at 3.23 Å where PBE alone has no minimum.

**Refuses.** UPF v1; an unimplemented functional (rather than silently substituting LDA); energy-carrying meta-GGAs (TPSS, SCAN, M06L), whose potential has a  $\partial E/\partial \tau$  piece acting on the wavefunction that is not written; plain *ultrasoft* under a meta-GGA, which has no partial waves to reconstruct  $\tau$  from inside the sphere where PAW does; EXX; and **D3**, **Tkatchenko-Scheffler**, **MBD** and **XDM** — where `pw.x` warns and silently runs with no correction at all.

## 5 Spin, magnetism and spin-orbit

| regime       | how to ask                       | note                                                         |
|--------------|----------------------------------|--------------------------------------------------------------|
| unpolarised  | default                          |                                                              |
| collinear    | <code>nspin = 2</code>           | one Fermi level, or two under <code>tot_magnetization</code> |
| noncollinear | <code>noncolin = .true.</code>   | magnetization is a vector                                    |
| spin-orbit   | + <code>lspinorb = .true.</code> | $j$ -resolved projectors; NC, US and PAW                     |

**Three spin numbers are kept apart**, and `System` exposes all three: `nspin` says which regime is in force, `npol` is the number of spinor components of a *wavefunction*, and `nspin_mag` the number of components of a *density*. They coincide at 1 and 2 and come apart at 4, where `npol = 2` but `nspin_mag` is **one** for a nonmagnetic spin-orbit run — which is why such a run costs about what a doubled unpolarized one costs.

**Fixed occupations under `nspin = 2`** fill each channel to its own count rather than searching for a level: `tot_magnetization` gives `nelup` and `nelow`, and `iweights_only` fills `NINT( $n_{\uparrow}$ )` bands in one channel and `NINT( $n_{\downarrow}$ )` in the other. That is the *only* shape the combination has — `pw.x` refuses fixed-occupation LSDA without a `tot_magnetization`, and requires an integer

one — and this package makes the same two refusals at input, before anything is allocated. Each channel's highest occupied level is reported as `fermi_energy_up / fermi_energy_down` beside the overall `homo` and `lumo`, which is what `iweights` returns. Validated on an oxygen atom at `tot_magnetization = 2`, whose four-up / two-down filling is the Hund's-rule ground state: **4.9e-9 Ry** against `pw.x`.

**Magnetic fields and constraints:** a uniform field over the cell, a field inside one atom's sphere (through a `LOCAL_MAGNETIC_FIELDS` card that `pw.x` has no counterpart for), Elk's `reducebf`, and all four of QE's `constrained_magnetization` schemes. **The field's energy is not part of the reported total** — QE prints `etcon` and never adds it, and Elk excludes its external field's energy by the same convention.

*A seed field has no natural size, so the symmetry filter has no scale in it.* An infinitesimal field is the standard way to start a texture: small enough not to bias the energy, large enough to pick the state. What the magnetic filter asks of a field is a *pattern* — does the operation map it onto itself, up to time reversal — so each field is compared after being divided by its own largest component, and it is ignored only below  $10^{-12}$  Ry, which is the same floor that decides whether the run is magnetic at all. One rule: a field big enough to make a run magnetic is big enough to cut its group. (It used to be two, seven orders of magnitude apart, and a field in between turned the run magnetic and then contributed nothing to the filter — which is the one thing worse than no field, because the full crystal group then averages away the texture the field was applied to create.)

*reducebf decays the field after the convergence test*, so a run can stop with the field still substantially on. The state reported is then the ground state of a functional that includes a Zeeman term, and **the error in the total is second order in the residual field**, not the field energy — the state is stationary. On a hydrogen atom with a 0.1 Ry local field, `reducebf = 0.5` stops with  $6 \times 10^{-6}$  Ry still applied and a total right to  $5 \times 10^{-14}$  Ry, while `reducebf = 0.99` stops after six iterations with  $9.5 \times 10^{-2}$  Ry applied and a total  $4.7 \times 10^{-5}$  Ry out. A run that stops above  $10^{-4}$  Ry now says so, naming `field_scale`, the residual and `field_energy`; a field held at full strength (`reducebf = 1`) is a deliberate calculation and is left alone. `reducebf` is also held to Elk's own `[0.5, 1]` (`readinput.f90:1264`), which this used to accept any value outside: above 1 the symmetry-breaking field *grows* every iteration, and below 0.5 it is gone before the density has responded to it.

**Refused.** `constrained_magnetization = 'fsm'` together with `lspinorb = .true.` The fixed-spin-moment search measures  $\chi = \delta m_a / \delta B_a$  one cartesian component at a time and inverts it the same way, which models  $\partial m_a / \partial B_b$  as *diagonal* — three independent one-dimensional searches. Spin-orbit coupling ties the moment to the lattice and is exactly what makes that untrue: pushing along  $x$  moves the moment along  $z$  as well. The scheme would still converge sometimes and find the wrong field the rest of the time, with nothing in the output to say which. Use a penalty scheme (`'total', 'atomic'`) to hold a moment under spin-orbit coupling.

**Also refused:** `noncolin = .true.` with `angle1/angle2` set and no `starting_magnetization`. The angles are the *direction* of a moment whose length is `starting_magnetization`, so with no length there is no moment: `domag` goes false, `nspin_mag` collapses to 1, and what runs is an unpolarized calculation with spinor wavefunctions — converged, successful, and not the calculation the angles asked for. `pw.x` runs that input silently. A spinor run with nothing magnetic in it stays accepted, because that is an ordinary calculation; the refusal is about a *stated* direction with nothing to point.

**The polar-angle constraint is guarded on the axis.** `'total direction'` used to return a NaN potential in all three components whenever the moment lay along  $z$  — which is where a run seeded from `starting_magnetization` with no angles starts. `arccos'` diverges

at the poles while the transverse moment's own derivative vanishes there, so the chain is  $0 \times \infty$ ; and at a *round-off* transverse moment  $m_z/|m|$  rounds to bit-exactly 1 and the clamp halves the tangent instead. The angle is written  $\text{atan2}(|m_\perp|, m_z)$  with both arguments masked at the value the derivative is taken at, which reproduces QE's `fact1` to  $1\text{e-}12$  off the axis and QE's zeroed transverse factors on it, plus its literal  $10^{-14}$  escape field along  $x$  — without which the moment sits at an unstable stationary point of its own penalty and never turns (`add_bfield.f90:184-192`).

**Restarting an SCF from the middle** (`checkpoint_dir`, `checkpoint_every`, `max_seconds`). A job killed at its wall clock used to lose every iteration it had run: a state reached disk only after `run_scf` returned. It now writes on a cadence and *resumes from the same directory*, so a resubmitted batch script is the same command and continues.

```
from defumat import Calculator
calculator = Calculator.from_file("scf.in", checkpoint_dir="ckpt")
scf = calculator.get_scf()           # run again after a kill: it continues
```

A restart is three things, and the third is the one with no obvious home: the state (density, wavefunctions, `becsum`, `ns`, `tau`); the *mixer's history*, without which Anderson restarts empty and pays back the iterations the checkpoint saved; and the *loop state* — the iteration number, `dr2` and the diagonalisation threshold `ethr`. That last one matters because QE's threshold schedule is indexed on the iteration number, so a resume re-entering at iteration 1 with a fresh threshold converges on a different schedule than the one it left. With all three carried the restart is exact: on silicon at `conv_thr = 1e-12`, stopping and resuming costs the *same* total iteration count as an uninterrupted run at three different mixing parameters (17/17, 13/13, 11/11), with energies agreeing to  $2 \times 10^{-15}$  Ry. Both Quantum ESPRESSO and Elk restart this way, and carry the same three parts.

`max_seconds` lets the loop stop *itself* before the scheduler does, writing a checkpoint on the way out, so a wall-clock kill lands after a checkpoint rather than between two. It is checked between iterations, so set it at least one iteration short of the real limit.

A *field is carried, and only one kind of field is refused*. An applied `LOCAL_MAGNETIC_FIELDS` card, and a penalty constraint of any flavour, leave the field exactly as the input built it, so the resume rebuilds it from `scf.in` with the rest of the calculator. Elk's `reducebf` fades a *scale* rather than the field, and that scale is saved and restored — a run whose field had decayed over forty iterations comes back where it left, not at full field.

**Refused.** Checkpointing is switched off, once and with a message, for a run whose field is *driven* rather than applied: the fixed-spin-moment schemes (`constrained_magnetization = 'fsm', 'atomic fsm', 'atomic fsm direction'`), where the field is the controller's own state and is replaced after every iteration. No input file says what it reached, so a resume would start again from the input field and converge somewhere else without saying so. A *Hubbard setup* is refused too. `checkpoint_every` must be at least 1.

**Know the cost.** The payload is the wavefunctions, which is tens of gigabytes on a large cell, so this is a scratch-directory operation and the cadence is a knob rather than every iteration. It has not been timed on a large cell; the default of 10 is a guess until it is.

**A starting magnetic texture** (`STARTING_MOMENTS` card): one starting moment per *atom*, cartesian, in Bohr magnetons, one line per atom in the order of `ATOMIC_POSITIONS`. *Bohr magnetons* is meant literally: a row of `0.0 0.0 2.0` seeds a  $2\mu_B$  moment on that atom whatever its valence charge, which is the same number the `atomic` constraint aims at and the same one the per-site report gives back. A row asking for more moment than the atom

has valence electrons is clamped to it, with a warning — a superposition of atomic charges cannot express more, and the alternative is a negative spin channel. `starting_magnetization` in the namelist follows Quantum ESPRESSO: below 1 it is a *fraction* of the valence charge, and at or above 1 it is Bohr magnetons and every species is divided by its own valence charge (input.f90:1448). QE's `starting_magnetization`, `angle1` and `angle2` are per *species*, so a helix, a cycloid or a skyrmion cannot be stated in a `pw.x` input at all.

*It decides three things*, and the first is why it matters more than a starting guess usually does. It decides the **symmetry group**; it decides whether the run is **magnetic at all** (a noncollinear input whose texture is given only through this card is a magnetic run, with a four-component density — this used to be read off the per-species `starting_magnetization` alone, so such an input converged, reported a total energy and never had a magnetization); and it seeds the **starting density**, one moment per atom rather than one per species. The magnetic group is filtered by the per-atom moments  $\mathbf{m}_i$  — an operation survives only if it maps that axial vector field onto itself, possibly followed by time reversal — and it is built from the input, once, never from the converged state. Built from per-species values it is a *ferromagnet's* group, which keeps operations a texture does not have, and `sym_rho` then averages the texture away a few iterations in while the charge converges and `dr2` reports nothing.

A per-atom *applied* field is filtered with the moments, together, under one choice of time reversal, exactly as Elk's `findsym.f90` tests `bfcmt0`. So a `LOCAL_MAGNETIC_FIELDS` card that asks for a texture now cuts the group by itself and needs no card of its own.

#### STARTING MOMENTS

```
0.00000000  1.50000000  0.00000000
-0.75000000  1.29903811  0.00000000
-1.29903811  0.75000000  0.00000000
```

**A texture from Python, without writing a file** (`Calculator.with_moments`, `System.with_moments`).

The card above is the only way to *state* a texture in an input, and a sweep over magnetic configurations — a 120-degree Néel state against a collinear one, a cone at five angles — wants to state several from one script. `with_moments` takes an `(nat, 3)` array of Bohr magnetons in `ATOMIC_POSITIONS` order, or `None` to drop the card, and returns a new calculator whose k-points have been rebuilt for the new moments' magnetic group.

```
import numpy as np
from defumat import Calculator
```

```
ferro = Calculator.from_file("h4-chain-ferro.in") # tests/data/qe
angles = np.deg2rad([0.0, 90.0, 180.0, 270.0])
cycloid = 0.6 * np.stack([np.sin(angles), np.zeros(4), np.cos(angles)], axis=1)
```

```
print(ferro.system.symmetry_group().nsym, ferro.system.kpoints.nk) # 16 9
textured = ferro.with_moments(cycloid)
print(textured.system.symmetry_group().nsym, textured.system.kpoints.nk) # 4 12
```

*The obvious workaround is silently wrong*, which is why this is a method rather than an attribute. `dataclasses.replace(system, starting_moments=...)` swaps the field and leaves kpoints reduced with the group the *old* moments had, while `symmetry_group()` is a property recomputed from the new ones — 9 k-points reduced with 16 operations, symmetrised with 4, measured on the cell above. On a system built nonmagnetic it also flips `domag`, so the k-set carries a `time_reversal` a magnetic run must not have. Nothing compares the two.

**Holding a texture** (`constrained_magnetization = 'atomic texture'`) constrains the *full* unit vec-

tor of every atom's moment,

$$E_{\text{con}} = \lambda \sum_i \left( 1 - \frac{\mathbf{m}_i \cdot \hat{\mathbf{n}}_i}{|\mathbf{m}_i|} \right),$$

with  $\hat{\mathbf{n}}_i$  the normalised `STARTING_MOMENTS` rows. It is not one of QE's schemes and has no `i_cons` number. QE's nearest, 'atomic direction' (`i_cons` = 2), constrains  $m_z/|m|$  — the polar angle *alone* — so it is exactly satisfied by every texture lying in a plane that contains  $z$ , a helix and a collinear state alike, at zero penalty: it cannot hold one, and reports success while failing to.

**Or hold it with a field instead of a penalty.** A penalty is a term added to the energy, so at convergence it is still pushing — the residual angle *is* that push, and it cannot be made zero by choosing  $\lambda$  better. Elk's fixed-spin-moment scheme does the other thing: it drives an external field until the moment sits where it was asked to, and the converged state is then a stationary point of the *unconstrained* functional under that field. 'atomic fsm' and 'atomic fsm direction' are that scheme resolved by atom (Elk's `fsmttype` = 2 and -2),

$$\mathbf{B}_i \leftarrow \mathbf{B}_i - \tau (\mathbf{m}_i - \mathbf{m}_{\text{fix},i}),$$

with the targets read from `STARTING_MOMENTS` in Bohr magnetons. The direction-only variant finishes by projecting  $\mathbf{B}_i$  perpendicular to  $\mathbf{m}_{\text{fix},i}$  (Elk's `r3vo`), so the field can *turn* a moment and can never lengthen or shorten one — and note what that buys: it has no  $1/|\mathbf{m}|$  anywhere in it, where the direction-only *penalty* above has one in its gradient and runs away.

**It is transcribed and it is not yet shown to win.** The argument above is sound and the implementation is checked against Elk's own routines, but on the two-hydrogen cell measured below it does not converge at all: at half Elk's default gain the site residual rings over 2000 iterations with an envelope that *grows*, from 0.15 to between 0.3 and  $0.66 \mu_B$ , while the inner calculation's own accuracy swings between  $10^{-6}$  and  $6 \times 10^{-2}$ . That is an unstable loop rather than a slow one. The cause is the cell rather than the scheme: left alone this pair carries  $0.000235 \mu_B$ , so its  $m(B)$  is nearly a step and no fixed-gain controller is stable against a nearly vertical response — and the cell was chosen for a *penalty*, which does not care.

The secant update fails in a different and more interesting way. It is perfectly well behaved, reaching  $6.5 \times 10^{-6}$  in the density, and it converges to *the wrong state*: the moment lengths come out right to eight per cent and the angles wrong by 145 degrees per site, with the residual sitting on a plateau for 900 iterations. That is its model rather than its stability. It measures  $d\mathbf{m}/d\mathbf{B}$  one atom and one cartesian component at a time, and what sets a texture's angles is the exchange *between* atoms, which is exactly the off-diagonal part a diagonal model discards. A diagonal controller can size a moment and cannot orient a pair of them.

So use a robust magnet, and read the vector penalty's table below as the answer until one has been measured.

**lambda means two different things three orders of magnitude apart, and this is the trap of the family.** For a *penalty* it is a stiffness, and 1 to 10 is the useful range. For a *feedback* scheme it is Elk's `taufsm`, a gain that multiplies a moment error in Bohr magnetons to give a field in Rydberg, and Elk's own default is **0.01**. Leave it unset and you get that; write a penalty-sized value and the run warns by name. Measured: at  $\tau = 0.2$  the vector scheme runs 200 iterations without converging and drives the moments to  $0.82 \mu_B$  against a target of 0.26.

**Which penalty holds a texture, measured.** *Not this one.* 'atomic texture' constrains a direction and not a length, so its potential carries a  $1/|\mathbf{m}|$  and *grows* as a site's moment

shrinks — positive feedback, and on the two-sublattice cell below no  $\lambda$  converged. 'atomic' constrains the moment as a vector, so its potential is  $2\lambda(\mathbf{m} - \mathbf{m}_{\text{target}})$  and bounded, and it is what to reach for. Two hydrogen atoms of one species asked to sit 120 degrees apart (tests/data/qe/h2-texture-120.in), conv\_thr = 1e-8, targets at the converged sphere moment of  $0.26 \mu_B$ :

| scheme         | $\lambda$ | converged  | iterations | angle          | error/site   |
|----------------|-----------|------------|------------|----------------|--------------|
| (none)         | —         | <b>yes</b> | 10         | 180.00°        | 30.00°       |
| atomic         | 0.05      | yes        | 10         | 170.95°        | 25.47°       |
| atomic         | 0.2       | yes        | 7          | 153.52°        | 16.76°       |
| atomic         | 1.0       | yes        | 9          | 130.30°        | 5.15°        |
| atomic         | 3.0       | yes        | 17         | 123.65°        | 1.83°        |
| atomic         | 10.0      | yes        | 38         | <b>121.13°</b> | <b>0.55°</b> |
| atomic         | 30.0      | <b>no</b>  | 200        | 11.91°         | 64.41°       |
| atomic texture | 0.1       | <b>no</b>  | 200        | 118.39°        | 4.30°        |
| atomic texture | 0.02      | <b>no</b>  | 400        | 140.03°        | 10.02°       |
| atomic texture | $\geq 2$  | <b>no</b>  | 200        | collapsed      | —            |

Read the first row and the sixth together: left alone the 120-degree state relaxes to the collinear antiferromagnet in ten iterations and *reports success*, and at the largest  $\lambda$  the SCF tolerates the vector penalty holds it to **0.55 degrees per site** in 38. So a magnetic structure that is not the ground state can be converged here, which is what a user mapping out  $E(\theta)$  needs and what the cone and spiral collapses recorded elsewhere in this document have no constraint applied to them. The recipe is 'atomic', a STARTING\_MOMENTS card scaled to the moment you *expect* rather than the moment you seed with, and  $\lambda$  raised until the angle stops moving or the SCF stops converging — which on this cell are within a factor of three of each other. 'atomic texture' warns and names this table.

*How far each atom is from its target* is on SCFResult.site\_residuals, one number per atom — degrees for the two direction schemes, Bohr magnetons for 'atomic'. constraint\_energy is a single scalar over every site, so under 'atomic texture' one flipped site out of fifteen reads as a small number indistinguishable from partial convergence everywhere; the per-site array is the only form of the comparison that can tell those apart.

**Refused.** 'atomic texture' needs noncolin = .true. (a collinear moment has no direction beyond its sign) and a STARTING\_MOMENTS card with no zero row (a zero vector carries no direction). STARTING\_MOMENTS itself needs noncolin for any  $x$  or  $y$  component, the same rule B\_field and LOCAL\_MAGNETIC\_FIELDS are held to, and one row per atom.

**Checked, not remembered:** the card above parses through build\_system, reaches system.starting\_moments unchanged, becomes m\_loc verbatim, and constraint\_targets returns it normalised — three unit rows. The starting density built from it carries opposite moment lobes on two antiparallel sites and integrates to zero over the cell.

**It reaches all three consumers**, which it did not until recently: the charge density, the Hubbard occupation matrix, and a PAW or ultrasoft dataset's atomic becsun. The last two used to be split by the per-species starting\_magnetization and so started every site of a species pointing the same way, while the charge carried the texture — so iteration 1 contradicted itself on exactly the systems the card exists for. For PAW that is not merely a poorer guess: the one-centre terms are a *function* of becsun, so the two sites got different Hamiltonians than the density asked for. And nothing in the SCF turns a moment, so a spinor DFT+U run — which is forced to nosym, leaving the starting matrix as the only steering there is — could converge with the shells polarised along the wrong axes and report success.

**Warned about, with a number:** a textured starting *density* handed to `run_scf` by the caller. The symmetry group is decided from the input and never from the density, so a texture that reaches the run only that way is invisible to the filter and *will* be symmetrised away. `run_scf` measures  $\|\rho - \text{sym}(\rho)\|/\|\rho\|$  on the seed before iteration 1 and warns above  $10^{-3}$ , reporting the charge and the magnetization apart. It is not a small effect and it is not a rounding: a four-atom 90-degree cycloid handed to the same cell's *ferromagnetic* group has its magnetization residual come out at **1.0** — averaging four directions 90 degrees apart over a group that permutes the four sites gives exactly zero, and every site moment goes from  $0.27 \mu_B$  to  $10^{-19}$  — while its charge residual is  $5.7 \times 10^{-16}$ , so the charge notices nothing. From a `STARTING_MOMENTS` card the same figure is  $6 \times 10^{-16}$ . Give the texture as a card, or set `nosym = .true.`

**The moment on each atom** (`SCFResult.site_moments, .site_charges`). The two magnetic numbers a run has always printed are integrals over the *cell*, and both are zero for every compensated state — for an antiferromagnet, for a spiral, for a cycloid, and equally for the non-magnetic state any of those can collapse into. What separates them is the charge and the moment integrated in a sphere around each atom, which is what Quantum ESPRESSO's `report_mag` prints and what Elk computes as `mommt`:

$$q_a = \int w_a(\mathbf{r}) n(\mathbf{r}) d^3r, \quad \mathbf{m}_a = \int w_a(\mathbf{r}) \mathbf{m}(\mathbf{r}) d^3r,$$

with  $w_a$  one inside a radius  $r_m$  around atom  $a$ , falling linearly to zero at  $1.2 r_m$ , and every grid point belonging to at most one atom.

**from** `defumat` **import** `Calculator`

```
calculator = Calculator.from_file("h2-mirror-afm.in")
scf = calculator.get_scf()
```

```
print(scf.magnetization)          # -2.3e-06 — the cell total, zero
print(scf.absolute_magnetization) # 0.806933
for a, (q, m) in enumerate(zip(scf.site_charges, scf.site_moments)):
    print(f"atom_{a}: charge_{q:.4f} moment_{m[0]:+.4f} mu_B")
# atom 0: charge 0.6240 moment +0.2909 mu_B
# atom 1: charge 0.6240 moment -0.2909 mu_B
```

*They are recorded at every iteration*, in `scf.history`, and not only at convergence. That is the point rather than a convenience: a texture that is going to be lost goes early and the run then converges cleanly, so the converged value is exactly the one that cannot show it. The per-iteration console line carries `|m|_site = min..max`, the shortest pair that tells a live compensated state from a dead one, and the full block is printed at the end with  $\theta$  and  $\phi$  per atom — angles Quantum ESPRESSO does not print and that nobody should be reconstructing by hand from three components.

**A collinear run has a magnetic symmetry group too.** Until recently it did not: the filter was built only for `noncolin`, so a compensated magnet stated on *one* species kept the operation carrying one sublattice onto the other, and the density symmetriser then averaged  $n_\uparrow$  over both sites — which is the statement that the staggered moment is zero. This is not an exotic case: an *altermagnet* is by definition compensated order whose sublattices are related by a rotation, so the swap is a point-group operation of the chemical structure. On two hydrogen atoms related by a mirror with  $\pm 0.6$  from the card, the moments were gone at iteration 1 and the run converged to a perfectly good total energy **6.6 meV** above the antiferromagnet, reporting an absolute magnetization of  $7 \times 10^{-6}$ .

With the filter the group is cut from 16 operations to the 8 that preserve the moment pattern, and the run agrees with the `nosym = .true.` spelling — the only one that used to work — to  $10^{-8}$  Ry and  $10^{-6} \mu_B$  in the same iteration count.

The collinear filter compares a *number* per site, not a vector. Without spin-orbit coupling a spatial operation does not turn the spin, so the moment is permuted and never rotated; filtering it as an axial vector instead would cut a  $C_{2x}$  or a mirror on a  $z$ -ferromagnet and leave a group that is too *small*.

**Refused.** An operation that reverses *every* moment is a symmetry of the crystal followed by time reversal. Those are **dropped** rather than used, which is Quantum ESPRESSO’s own `colin_mag = 1` behaviour: using one means swapping the two spin channels while symmetrising, and the density symmetriser here does not. The result is a subgroup of the true magnetic group — correct, and paid for in k-points rather than in physics.

**Where the site moments are not reported:** the response stack. All three relaxation drivers carry them now — `RelaxStep`, `VCRelaxStep` and `SpiralRelaxStep` each hold `site_charges` and `site_moments` at their own step, which is the record a *relaxation* needs for the same reason the SCF needs one per iteration: everything else a step reports is zero-sum over the sites, so a compensated magnet that lost its moments between two geometries reads exactly like one that kept them.

**The spheres are a choice, not a definition.** A plane-wave basis has no muffin tins, so “the moment on this atom” is whatever region you integrate over;  $r_m$  defaults to a little under half the nearest-neighbour distance so the tapered spheres are disjoint. The site charges therefore sum to *less* than the cell’s charge — the interstitial belongs to nobody — and the site moments sum to less than the cell moment. Compare them against another code only when both used the same radius.

**DFT+U:** both rotationally-invariant functionals, on atomic, ortho-atomic and norm-atomic projectors, read from the HUBBARD card. Forces come from differentiating through projectors that move with the atoms.

*The card selects which functional runs*, exactly as `pw.x` does it — the namelist’s `lda_plus_u_kind` is obsolete there and warned about. A card carrying only  $U$ ,  $J_0$ , ALPHA and BETA is Dudarev’s simplified functional; any  $J$ ,  $B$ ,  $E_2$  or  $E_3$  selects Liechtenstein’s full one, whose interaction is the whole Coulomb matrix

$$\text{vee}(m_1 m_2 m_3 m_4) = \sum_k F^k \frac{4\pi}{2k+1} \sum_{q=-k}^k a_{kq}(m_1 m_3) a_{kq}(m_2 m_4)$$

rather than one number, with the Slater integrals  $F^k$  fixed by  $U$  and  $J$ . Against `pw.x` on antiferromagnetic FeO: **4.2e-9 Ry** at  $J = 1$  eV. *Forces work for both*, and for the full functional `pw.x` does not have them at all (`force_hubb.f90` stops on “forces in the DFT+U+J scheme are not implemented”); a central difference of the SCF energy agrees to  $1.3\text{e-}4$  Ry/bohr. **Seed that difference from the undisplaced run** — with a full interaction matrix the DFT+U landscape has more than one solution, and cold-started displacements land on a different one, which does not look like an error, it looks like a force.

```
from defumat import Calculator
scf = Calculator.from_file("feo.in").get_scf()
print(scf.energy_terms["hubbard"])
```

```
HUBBARD {atomic}
U Fe1-3d 4.3
J Fe1-3d 1.0      # selects the full functional
```

**Around-mean-field double counting** (`hubbard_double_counting = 'amf'`) is the alternative to the fully-localised limit that both QE's functionals assume, and `pw.x` has neither the option nor the code. FLL treats the occupations as integers, which is right for a localised magnetic insulator and wrong for a metal; AMF subtracts the shell's mean occupation before the interaction, so a uniformly filled shell is corrected by *exactly* nothing — 0 to  $10^{-14}$  at any filling, where FLL is not. On FeO its Hubbard energy is  $-0.005$  Ry against FLL's  $+0.232$ . It needs the full functional, there being no interaction matrix in the simplified one to apply the shift to.

**Slater integrals from the orbital** (`hubbard_slater = 'yukawa'`) computes the interaction instead of parameterising it:  $F^k$  is the double radial integral of the manifold's own all-electron partial wave against a screened Coulomb kernel,

$$F^k = \iint \mathrm{d}r \mathrm{d}r' [r\phi(r)]^2 [r'\phi(r')]^2 (2k+1) \lambda i_k(\lambda r_{<}) \tilde{k}_k(\lambda r_{>}),$$

which becomes the bare  $r_{<}^k/r_{>}^{k+1}$  as  $\lambda \rightarrow 0$ . Give  $\lambda$  on the card (`LAMBDA Ni-3d 2.1`) or give a  $U$  and let the screening length be solved for — either way  $F^0$ ,  $F^2$ ,  $F^4$  and  $J$  all come out of the orbital. Nickel's unscreened 3d gives  $F^4/F^2 = 0.626$  against the 0.625 that QE and Elk both hardcode, and  $J = 1.28$  eV.

```
&system
  hubbard_slater = 'yukawa'
/
HUBBARD {ortho-atomic}
U Ni-3d 5.0      # lambda is solved for; F2, F4 and J follow
```

**On a spinor** (`noncolin`, with or without `lspinorb`) the occupation matrix is one  $2 \times 2$  matrix in spin space rather than two real matrices, measured on projector columns that are themselves spinors, and its off-diagonal blocks are what let the shell's moment point anywhere. Both functionals work there. Against `pw.x`: **7.7e-9 Ry** on relativistic BN and **1.4e-9 Ry** on fully-relativistic iron carrying  $U = 2.20$ ,  $J = 1.75$  eV with its moment turned into the  $xy$  plane, with  $\text{Tr}[n^{\sigma\sigma}]$  matching every digit `write_ns` prints.

**Tensor moments** put that matrix in a basis where each element is a multipole rather than a number: an orthonormal, complete set  $\Gamma_t^{kpr}$  with  $k$  the rank in the orbital index,  $p$  the rank in spin and  $r$  their coupling, so that

$$n = \sum_{kprt} w_t^{kpr} \Gamma_t^{kpr},$$

with real  $w$ . The charge is  $w^{000}$ , the spin moment is  $w^{011}$ , and  $\mathbf{L} \cdot \mathbf{S}$  is  $w_0^{110}$  alone — to  $10^{-11}$  over all one hundred moments of a  $d$  shell.

**Fixing one** (a `TENSOR_MOMENTS` card) converges the field to a state carrying it, which selects an orbital or multipolar ordering the field would not find on its own. The penalty is written down and its potential is `jax.grad` of it, and, like a constrained moment's, its energy is not part of the reported total — so the number that comes back is the unconstrained functional at the constrained state.

```
&system
  tensor_moment_penalty = 20.0
/
TENSOR_MOMENTS
N-2p 1 1 0 0 -0.40      # hold L.S at -0.40
```

Being quadratic, the penalty settles where its gradient balances the material's own restoring force, so the moment *approaches* its target as the penalty stiffens rather than reaching it: on nitrogen's  $2p$ , whose free  $\mathbf{L} \cdot \mathbf{S}$  is  $-0.0006$ , holding it at  $-0.40$  gives  $-0.109$ ,  $-0.263$ ,  $-0.355$  and  $-0.388$  at  $\lambda = 1, 5, 20, 80$ , with the energy rising monotonically by 2.6, 15.3, 27.4 and 32.6 mRy.

**Refuses.** A *collinear* DFT+U run needs `nosym` when its magnetic group carries time-reversed operations: there the occupation matrix is two separate channels and such an operation exchanges them, which is not written. **A spinor run does not** — it keeps both spin indices, so the operation acts on the matrix itself, and both halves of that action are in: the  $SU(2)$  representation of each operation beside the rotation of the  $m$  indices, and, for a time-reversed one, the transpose that goes with an antiunitary operation. A fixed tensor moment needs a spinor: for a collinear run the spin-rank moments keep only their  $z$  component and the constraint degenerates into the fixed spin moment `constrained_magnetization` already is. `hubbard_slater = 'yukawa'` needs a **PAW** dataset, and this is a measurement rather than a preference: only PAW carries the all-electron partial wave. A norm-conserving  $\chi$  is normalised and gives a pseudo-orbital  $F^0$  far too small (silicon's  $3p$ : 8.8 eV), and an *ultrasoft*  $\chi$  is not even normalised — iron's  $3d$  has  $\int (r\chi)^2 = 0.41$ , the rest of the norm living in the augmentation charge, and its  $F^0$  comes out at 2.1 eV where the answer is above twenty. The norm that comes back with each manifold is what says whether it is bound inside the cutoff, and the cutoff defaults to the dataset's own augmentation radius because a partial wave *diverges* outside it. Elk's `inpdftu = 2` and `3` (Slater and Racah *input*) have no card syntax: they are a change of coordinates on the three numbers ( $U, J, B$ ) already spans, and QE's card has spent the names  $E_2$  and  $E_3$  already. The FLL/AMF interpolation (Elk's `dftu = 3`) is not here because it is not in Elk either any more — documented in the 11.0.2 manual, removed from the 11.0.2 source.

**Refuses.** The intersite  $V$ , background channels, the orbital-resolved variant, the `wf/pseudo` projector sets, and `noncollinear ns_nc`; and PAW + GGA + a noncollinear magnetization together.

A **magnetic field in a noncollinear run with no starting moment** is refused rather than run. `domag` comes from `starting_magnetization` and from nothing else (`setup.f90`), so such a run has `nspin_mag = 1` and a density with no magnetization channels for the field to act on. Set `starting_magnetization` for one species — any nonzero value will do, the field decides where the moment goes. Worth knowing: `pw.x` *accepts* this input and silently drops the field, because `add_bfield` writes it into the potential's magnetization channels and `vloc_psi_nc` applies those only IF (`domag`).

The **residual solver** cannot take a fixed occupation whose filled/empty boundary cuts a **degenerate multiplet** — which is exactly what an atom's Hund's-rule filling does. Which member of the multiplet the eigensolver returns is arbitrary, so the density map is not a function of the density and a Newton step has nothing to converge on; it is diagnosed by name rather than left as a NaN. The mixer damps its way past it, and a *gapped* fixed-occupation LSDA cell agrees between the two solvers to  $5e-12$  Ry.

## 6 Forces, stress and relaxation

$$\mathbf{F}_I = -\frac{\partial E}{\partial \tau_I} \Big|_{\psi \text{ fixed}}, \quad \sigma_{ij} = -\frac{1}{\Omega} \frac{\partial E}{\partial \varepsilon_{ij}} \Big|_{\psi \text{ fixed}}$$

Both are one gradient. With ultrasoft the orthonormality constraint  $\langle \psi_n | \hat{S} | \psi_m \rangle = \delta_{nm}$  is

carried explicitly, so its multiplier term — QE’s Pulay contribution — falls out of the same derivative rather than being added afterwards. A variable-cell relaxation minimises the **enthalpy**, which is why a relaxed crystal carries the applied pressure rather than having zero stress:

$$H = E + P \Omega, \quad \frac{\partial H}{\partial h} = \Omega (P \mathbf{1} - \sigma) h^{-T}$$

```
from defumat.forces import compute_forces
from defumat.stress import compute_stress
from defumat.workflows.relax import run_relax
from defumat.workflows.vc_relax import run_vc_relax

forces = compute_forces(calculation, result)      # method=... for QE's own terms
print(forces.forces)                             # (nat, 3) in Ry/bohr

stress = compute_stress(calculation, result, terms=True)
print(stress.pressure)                           # kbar

relaxed = run_relax(system, pseudos, forc_conv_thr=1.0e-4)
cell = run_vc_relax(system, pseudos, press=500.0) # kbar
print(cell.pulay_error)  # the frozen-basis error, reported not hidden
```

QE’s `force_lc`, `force_cc`, `force_ew`, `force_us`, `addusforce` and `force_corr` are transcribed too, behind the same registry — **as a cross-check, not as the implementation**. The two share no machinery, which is what makes disagreement informative; it is how the augmentation force’s sign and a missing gradient correction were found.

For a noncollinear or spin-orbit run the same expression is evaluated on two-component spinors: the nonlocal term takes the pseudopotential’s bare `dvan_so` (a complex  $2 \times 2$  matrix in spin space) and the orthonormality constraint takes `qq_so`, and the state is one coefficient vector of length  $2n_{\text{pw}}$  rather than two. Nothing else in the functional changes.

```
from defumat import Calculator

calculator = Calculator.from_file("pt2-soc-force.in") # noncolin, lspinorb
forces = calculator.get_forces()                     # (nat, 3) Ry/bohr
stress = calculator.get_stress()                     # (3, 3) Ry/bohr^3
print(forces.max_force, stress.pressure_kbar)
```

Validated against `pw.x` with `tpnfor/tstress` on a four-atom noncollinear hydrogen chain (norm-conserving) to  $8.9 \times 10^{-7}$  Ry/bohr, doubled fcc platinum with spin-orbit coupling to  $7.5 \times 10^{-6}$  (ultrasoft) and  $7.3 \times 10^{-7}$  (PAW), and the stress on those three plus QE’s own three `pw_spinorbit` cases to  $\leq 1.2 \times 10^{-6}$  Ry/bohr<sup>3</sup>; and, without any Fortran at all, against a central finite difference of the frozen energy to  $6.2 \times 10^{-9}$  Ry/bohr.

**With a gradient-corrected functional** the derivative is checked separately, because the gradient correction of a *vector* magnetization is a branch of its own: bcc iron with its moment in the plane, ultrasoft, PBE (`fe-noncolin-pbe-stress.in`) agrees with `pw.x` to  $6.7 \times 10^{-9}$  Ry in energy and  $1.6 \times 10^{-7}$  Ry/bohr<sup>3</sup> in **stress** — 0.019 kbar out of 152.71, and the same level the collinear ultrasoft cases reach. With one atom in a bcc cell the force is zero by symmetry, so the stress is the only derivative that geometry has, which is why it is the quantity compared. Quantum ESPRESSO resolves the density onto a *fixed* quantization axis whenever the starting moments are all parallel to one direction (`compute_ux`), using  $(n \pm \text{sign}(\mathbf{m} \cdot \hat{u}) |\mathbf{m}|)/2$  so that “up” stays up across a node where  $\mathbf{m}$  passes through zero;

that is the branch iron takes, and both codes pick the same axis. A genuinely **canted** cell takes plain  $|\mathbf{m}|$  instead, with a real kink at every node, and that branch has no external number yet — not for want of trying here: `pw.x` itself limit-cycles at  $5 \times 10^{-6}$  Ry on a canted PBE hydrogen chain that converges to  $10^{-11}$  under LDA in 62 iterations.

A noncollinear or spin-orbit run has forces, a stress and a relaxation, but only through the *autodiff* method: `method='analytic'` is refused by name, because `force_us/stres_knl` are transcriptions of QE's hand-derived expressions and have no spinor form. The force on an atom of a **spin spiral** is refused separately — its two spinor components live on different plane-wave spheres, so the nonlocal term needs the projectors of both; `dE/dq` is what a spiral has instead. **Elastic constants and electrostriction** stay refused for `noncolin`: they differentiate the same functional a third time, and while the Sternheimer solve their first-order wavefunctions come from now *does* have a spinor form, no third derivative has been run with one. The refusal is about the assembly rather than the solve, which is the distinction that cost this project a whole quantity once before.

## 7 Response: dielectric, phonons, Raman

The **velocity operator** comes first and everything else uses it: it is one jvp of  $\hat{H}(\mathbf{k})$  at a frozen sphere, because  $\partial \hat{H} / \partial k_\alpha = i[\hat{H}, r_\alpha]$  in the periodic gauge. `band_velocities` exposes it — `Calculator.get_band_velocities` is the short way — and because the overlap carries a velocity too, a band velocity is  $\langle \psi | \partial_k \hat{H} - \epsilon \partial_k \hat{S} | \psi \rangle$ .

```
calc = Calculator.from_file("scf.in", pseudo_dir="pseudo")

velocities = calc.get_band_velocities()          # .velocities is (nk, nbnd, 3)
elsewhere = calc.get_band_velocities(kpoints=path)
```

Every first-order quantity then comes from the Sternheimer equation — the response of an occupied orbital to a perturbation, projected outside the occupied manifold so that no empty states and no energy denominators appear:

$$(\hat{H} - \epsilon_n \hat{S} + \alpha \hat{Q}) |\Delta \psi_n\rangle = -\hat{P}_c^+ \Delta V |\psi_n\rangle$$

It is solved self-consistently, because  $\Delta V$  contains the response of the potential to the response of the density,

$$\Delta V_{\text{scf}} = \Delta V_{\text{ext}} + \int K(\mathbf{r}, \mathbf{r}') \Delta n(\mathbf{r}') d\mathbf{r}'$$

and the kernel  $K$  is one jvp of `v_of_rho` rather than a second implementation. From its solutions:

$$\epsilon_{\alpha\beta}^\infty = \delta_{\alpha\beta} + \frac{8\pi}{\Omega} \sum_{\mathbf{nk}} w_{\mathbf{k}} \langle \Delta_\alpha \psi_n | \partial_\beta \psi_n \rangle, \quad Z_{I,\alpha\beta}^* = \frac{\partial F_{I\alpha}}{\partial \mathcal{E}_\beta}$$

$$C_{I\alpha,J\beta} = \frac{\partial^2 E}{\partial \tau_{I\alpha} \partial \tau_{J\beta}}, \quad \omega^2 \mathbf{e} = \frac{1}{\sqrt{M_I M_J}} C \mathbf{e}, \quad \frac{\partial \epsilon_{ij}}{\partial \tau_{I\gamma}} \text{ (Raman)}$$

The Raman tensor is the second-order energy differentiated once more along a displacement, at *frozen first-order* wavefunctions — the  $2n+1$  theorem, which is why a third derivative costs one more jvp and not a new solve.

```

from defumat.response import (dielectric_tensor, dynamical_matrix,
                              raman_tensors, vibrational_spectrum)

eps = dielectric_tensor(calculation, scf.wavefunctions, scf.eigenvalues,
                       scf.density)
print(eps.epsilon)                # 3x3, cartesian

ph = dynamical_matrix(calculation, scf.wavefunctions, scf.eigenvalues,
                      scf.density, scf.becsum)
print(ph.frequencies)            # cm-1

raman = raman_tensors(calculation, scf, born_charges=True, keep_internals=True)
spectrum = vibrational_spectrum(calculation, scf) # per-mode Raman and IR
print(spectrum.raman_activity, spectrum.depolarisation)

```

`born_effective_charges` gives  $Z^*$  on its own, and `mode_activities` is the bare assembly — a transcription of `dynmat.x`'s RamanIR, kept a pure function of arrays so it is testable without a self-consistent run.

Ultrasoft and PAW work for the dielectric constant ( $<1.2\text{e-}4$  against `ph.x` on four cases) *and for the dynamical matrix* — silicon's optical mode comes out at **513.295** and **513.378**  $\text{cm}^{-1}$  against `ph.x`'s 513.275 and 513.404, which is tighter than the norm-conserving case's 0.05. Four things make the difference and all four switch themselves off when  $S$  is the identity: the source term is  $(dH/du - \epsilon dS/du)|\psi\rangle$ ; the first-order state has an occupied block the Sternheimer solve does not produce; the mixed state changes at *frozen* states, because the augmentation charge and the projectors travel with their atom; and the orthonormality multipliers are variables of the functional that move — as a **matrix**, since a diagonal one is not invariant under the occupied-manifold rotation the state tangent is free in. Metals work for the response and for a norm-conserving dynamical matrix; and a symmetry-reduced wedge works as of the rank- $n$  symmetriser.

**The Raman tensor is ultrasoft and PAW as well**, at **1.2e-4** on both against a central difference of  $\epsilon$  over re-converged displaced cells, where the norm-conserving control on the same script is  $6.8\text{e-}4$ . It took **two tangents that are only right together**, and either alone is worse than neither: the state tangent is  $P_c d\psi + \psi^{\text{ort}}$ , the occupied block again; and  $|b\rangle = P_c \mathbf{r}|\psi\rangle$  is *not* the solution of its own linear equation once `adddvepsi_us` has applied  $S$  to it and added the augmentation dipole, so  $db$  is the tangent of a *composition* rather than of a solve. What located them is that  $\partial\epsilon/\partial\tau$  is a sum of five partial derivatives and each one can be finite-differenced on its own against a re-converged run: three agreed to  $7\text{e-}4$  and two did not.

**A spinor run has a dielectric constant and Born charges**, on a norm-conserving dataset. A nonmagnetic one is the same physics on a doubled space, four bands of two electrons against eight of one, and it comes out the same to  $5 \times 10^{-14}$  in  $\epsilon$  and  $3 \times 10^{-15}$  in  $Z^*$ ; the symmetrised 8-point wedge of an unshifted grid gives the closed 64-point grid's tensor to  $7 \times 10^{-13}$ , and `ph.x`'s own number to  $4.3 \times 10^{-5}$ , which is the scalar route's figure to the digit.

```

silicon = Calculator.from_file("si-epsilon.in")
spinor = silicon.with_spin(4)                # 8 bands of one electron, not 4 of two
spinor.get_dielectric_tensor().isotropic    # 13.806641, against 13.806646

```

`with_spin` promotes the converged state rather than starting again, so the second ground state is not paid for twice.

**A magnetic spinor is refused, and the reason is a measurement rather than a miss-**

**ing term.** An iodine atom with spin-orbit coupling and its moment along  $z$  (i-atom-soc.in, nosym, a 0.164 eV gap) gives  $\epsilon = \text{diag}(1.3565721, 1.3565721, 1.5744824)$ : uniaxial along the moment, off-diagonals at  $10^{-15}$ , and *nothing imposes it* — the run carries no symmetry, so the two equal entries are a measurement. Turn the moment to  $x$  and the distinct axis moves with it, returning the same two numbers to nine digits, which is what makes the anisotropy a statement about spin-orbit coupling rather than about the box. And yet ph.x, on a ground state the two codes agree on to the printed digit, gives 1.494594 along the moment against this 1.574482 — **5.3%**, where the two components across it agree to  $5 \times 10^{-4}$ . Every threshold QE’s dmx\_c\_nc has that a derivative of the potential does not (a clamped  $\zeta$ , a zeroing above  $|\zeta| = 1$ , an  $|\mathbf{m}| \leq 10^{-10}$  rule) fires at **zero** of that cell’s 157464 grid points, so it is not a convention at an edge. It is unlocated, and a number that is 5% out while looking like a working calculation is what a refusal is for.

**A subset of the atoms can be displaced on its own.** The cost of a dynamical matrix is that  $3n_{\text{at}}$  bare perturbations and  $3n_{\text{at}}$  first-order wavefunctions are held at once, which is what stops it running on a large cell; atoms= solves only the perturbations asked for and returns that subset’s own block, diagonalised with its own masses. That is the frozen-substrate approximation — a molecule on a fixed slab — and the modes it leaves are the frustrated translations and rotations, which are the physics of such a calculation rather than an artefact. A on\_row callback receives each row as it is assembled, so a run that outlives its job keeps what it paid for.

```
phonons = calc.get_phonons(atoms=range(57))    # the molecule, slab held fixed
phonons.frequencies                          # (3 * 57,) in cm-1
```

It works on norm-conserving, ultrasoft and PAW datasets. The validation needs no reference of any kind, which is unusual here: a partial dynamical matrix is not an approximation — it solves fewer perturbations of the *same* equations — so the whole-cell run is exact for what the two share, and the subset’s rows reproduce it to **9.3e-15** (norm-conserving), **2.8e-14** (ultrasoft) and **4.9e-15** (PAW) Ry/bohr<sup>2</sup>. The physical check is that freezing one of silicon’s two atoms turns the optical mode’s reduced mass  $m/2$  into  $m$ , so the frequency falls by  $\sqrt{2}$ : 519.205  $\rightarrow$  367.146, a ratio of 1.414173.

**A soft mode converges in ecutrho, and the finite-difference route hides it.** The analytic Hessian and a finite difference of the forces are the same quantity in exact arithmetic and are not on a grid: the FFT box breaks continuous translation invariance, so the energy carries a small ripple as an atom moves across it (the “egg-box” effect). The analytic second derivative takes the *point* curvature of that ripple; a finite difference averages it over the displacement. The two therefore differ by the amount the dense grid is not translationally invariant — and because that error is roughly a fixed curvature, it matters in proportion to how *small* the mode’s own curvature is. A stiff mode barely notices; a soft one can be several per cent out.

It is ecutrho and not ecutwfc that fixes it, the ripple living on the dense grid where the density and the local potential do. Measured on N<sub>2</sub> in a 12 bohr cube (SG15 ONCV PBE, nosym, K\_POINTS gamma, one atom frozen, the other at 1.15 Å), with the finite difference taken as a central difference of the forces at  $\pm 0.03$  Å:

|                              | transverse (cm <sup>-1</sup> ) | gap  | stretch (cm <sup>-1</sup> ) |
|------------------------------|--------------------------------|------|-----------------------------|
| 40/160 Ry, analytic          | 314.12                         |      | 1483.49                     |
| 40/160 Ry, finite difference | 296.59                         | 17.5 | 1490.77                     |
| 40/320 Ry, analytic          | 306.35                         |      | 1488.1                      |
| 40/320 Ry, finite difference | 301.9                          | 4.4  | 1495.2                      |

Raising *only* `ecutrho` moves the frustrated-rotation pair by  $8 \text{ cm}^{-1}$  analytically and  $5 \text{ cm}^{-1}$  by finite difference, closing their gap from **17.5 to 4.4**  $\text{cm}^{-1}$ , while the stretch is 0.5% off either way at both cutoffs — which is the signature. The analytic asymmetry is  $3\text{e-}8$  and  $1.4\text{e-}8 \text{ Ry/bohr}^2$ , so neither number is a convergence failure of the solve.

**What makes this worth stating rather than discovering:** both routes converge, each agrees with itself, and they drift *together*, so nothing in either calculation announces the error. The practical rule is that a soft mode wants its own `ecutrho` convergence pair, and the shift between them is the error bar to quote — not the agreement between the analytic and finite-difference routes at one cutoff, which is a weaker statement than it looks. (Measurements contributed from a production study of a molecule on a surface, where the modes of interest are 10–15 meV.)

**A trap worth knowing:** a response on a reduced  $k$ -set is a *polar vector field* and must be symmetrised as one. Running the whole grid instead is sound only if the grid is closed under the point group — a **shifted** Monkhorst-Pack grid is not.

## 7.1 Phonons away from $\Gamma$

A displacement pattern  $\mathbf{u}_s(\mathbf{R}) = \mathbf{u}_s e^{i\mathbf{q}\cdot\mathbf{R}}$  is not a displacement of the unit cell — it is periodic only on a supercell commensurate with  $\mathbf{q}$ , which is exactly what a frozen-phonon calculation pays for. Linear response does not pay it, because the *first-order* quantities are still lattice periodic once their  $e^{i\mathbf{q}\cdot\mathbf{r}}$  is taken out. The perturbation is  $e^{i\mathbf{q}\cdot\mathbf{r}}$  times a periodic function, so  $\Delta V_{\mathbf{q}}|\psi_{\mathbf{k}}\rangle$  is a Bloch state at  $\mathbf{k} + \mathbf{q}$ : the calculation stays in the unit cell, with one extra plane-wave sphere.

That asymmetry is the whole of it. The operator, the projector and the preconditioner belong to  $\mathbf{k} + \mathbf{q}$ ; the eigenvalue subtracted is the one of the band being perturbed, at  $\mathbf{k}$ :

$$(\hat{H}_{\mathbf{k}+\mathbf{q}} - \epsilon_{n\mathbf{k}} \hat{S}_{\mathbf{k}+\mathbf{q}} + \alpha \hat{Q}) |\Delta\psi_{n\mathbf{k}}\rangle = -\hat{P}_c^{\mathbf{k}+\mathbf{q}} \Delta V_{\mathbf{q}} |\psi_{n\mathbf{k}}\rangle$$

and the response density is  $\Delta n_{\mathbf{q}}(\mathbf{r}) = (2/\Omega) \sum_{n\mathbf{k}} w_{\mathbf{k}} \psi_{n\mathbf{k}}^* \Delta\psi_{n\mathbf{k}}$ , which is complex — at  $\Gamma$  it collapses to twice a real part and nowhere else.

```
# nosym = .true., noinv = .true., on an unshifted grid -- see the box below
calc = Calculator.from_file("si-nosym.in", pseudo_dir="pseudo")

ph = calc.get_phonons_at_q(q=(0.5, 0.5, 0.5))          # crystal coordinates, L
print(ph.frequencies)                                  # cm^-1
print(ph.matrix.shape)                                # (nat, 3, nat, 3), complex

xpoint = calc.get_phonons_at_q(q=(0.0, -1.0, 0.0), q_cartesian=True)
```

$\mathbf{q}$  is in crystal coordinates of the reciprocal lattice unless `q_cartesian` is set, which reads it in the  $2\pi/a$  units `ph.x` prints. The matrix is **complex**:  $D(\mathbf{q})$  is hermitian rather than symmetric, and only at  $\mathbf{q} = 0$  — or where the crystal has inversion — does it collapse to a real one.

Silicon on a 64-point unshifted grid, against the vendored `ph.x`: at  $L$  the six modes are 101.789, 101.789, 382.235, 405.099, 488.028 and 488.028  $\text{cm}^{-1}$  against 101.843, 101.843, 382.241, 405.121, 488.019 and 488.019, the worst of them **0.054**  $\text{cm}^{-1}$ ; at  $X$  the worst is **0.038**. That is the same floor the zone centre has and for the same reason (QE interpolates its radial form factors from a  $dq = 0.01$  table where this integrates them directly). `ph.x` runs the eight-point wedge with its response symmetrised where this runs the whole

grid with nothing symmetrised, so the comparison is also the only check the two routes get against each other.

**Only the Ewald sum knows about  $\mathbf{q}$ ,** among the terms that do not involve the response. The second derivative of the external potential at frozen density is diagonal in the atom index — both derivatives are taken at the same site, so the two modulation phases cancel — and it is therefore the same matrix at every wavevector. The ion-ion sum is not diagonal, and it is the one piece rebuilt.

**Refuses.** A symmetry-reduced  $k$ -set: a response has to be averaged over the operations that leave the *perturbation* invariant, and for a phonon at  $\mathbf{q}$  those are the ones with  $S\mathbf{q} = \mathbf{q} + \mathbf{G}$  — the small group of  $\mathbf{q}$ , which is not the crystal's group and is not written. Run with `nosym` and `noinv` on an **unshifted** grid, which is the grid that is closed under the point group. There is no **dispersion** either: that needs the star of  $\mathbf{q}$  and `q2r/matdyn`'s Fourier interpolation on top of this. Ultrasoft and PAW are refused ( $S$  moves with the atoms, and the orthonormality multipliers carry a term whose bra is at  $\mathbf{k}$  and whose ket is at  $\mathbf{k} + \mathbf{q}$ , where  $q_{ij}$  pairs projectors on one sphere); so are metals, spin polarization, spinors, spin spirals, meta-GGA and DFT+U. A **nonlinear core correction** is refused for a reason that belongs to  $\mathbf{q}$  rather than to the dataset: two atoms' core charges overlap inside a nonlinear  $E_{xc}$ , so unlike every other frozen term that one is a function of  $\mathbf{q}$ .

**Refuses.** Born charges for PAW (`int3_paw` against `becsumort` is missing, and without it the expression is wrong in sign as well as size);  $\chi^{(2)}$  and the electro-optic tensor (the second-order source term has nothing to build the  $\langle u_i | r_k | u_j \rangle$  piece from — and it is **42%** of the answer, measured); a partial dynamical matrix (`atoms=`) of a *symmetrised* run, since `syndvscf` acts on the  $3n_{at}$  stack as one object — an operation rotates the cartesian direction *and* permutes the atom, so it mixes a mode inside the subset with one outside it, and such a run wants `nosym`; the dynamical matrix of an ultrasoft or PAW *metal* (the `wg/wk` weight split was derived for a response whose `becsum` dependence is entirely inside `dpsi`, and an insulator cannot say which weight the three further tangents belong with); any response under a potential-only meta-GGA; and a shifted grid with a reduced  $k$ -set.

A **spinor** run (`noncolin`, with or without `lspinorb`) has a dielectric constant and Born charges on a **norm-conserving** dataset, provided it carries *no net moment*. A **textured** one (`nspin_mag = 4`) is refused by name and the refusal carries its own measurement: it runs, its tensor is uniaxial along the moment and follows the moment when that is turned, and it is 5.3% from `ph.x` in the component along the moment with the two across it agreeing to  $5 \times 10^{-4}$ . An *ultrasoft or PAW* one is refused by name and the missing piece is exactly one object:  $dD_{ij}$  is a  $2 \times 2$  matrix in spin space (`set_int3_nc`) where a norm-conserving dataset has no  $dD$  at all. The **phonons, the Raman tensor, the strain response** and everything built on them stay refused for a spinor, and for their own reason rather than this one: `symmetrize_displacement` carries the axial landmine the field path had — three of the four `nspin_mag` channels are a magnetization and rotate with  $\det(R)$  and the time-reversal sign — with the atom permutation on top, and a symmetriser that has never been checked against the whole zone is worse than a refusal. That is not rhetoric: on the field path the wrong rotation was **66%**, worse than no symmetrisation at all.

Also refused: a ground state converged under a **magnetic field or a constrained moment**. The whole stack rebuilds its potential from the field the *input* asked for, and that is not the field the density belongs to — `reducebf` scales a field away as the run proceeds (7% of its input value after 25 iterations at 0.9) and the fixed-spin-moment

scheme replaces it outright, which is why SCFResult carries `magnetic_field` and `field_scale` separately. Threading that pair through is the missing piece for a field put in by hand; a constrained\_magnetization = 'fsm' moment needs the *induced* field as well, and its field comes from a feedback update rather than from a penalty, so it is not a derivative of anything. Re-run the ground state without the field.

## LO-TO splitting and the static dielectric constant

At  $\Gamma$  a polar crystal's dynamical matrix is not analytic. An optical mode that moves the ions against each other builds a macroscopic electric field, and what the field costs depends on the *direction* the zone centre is approached from, so the matrix acquires a rank-one term:

$$D_{I\alpha,J\beta} \rightarrow D_{I\alpha,J\beta} + \frac{4\pi e^2}{\Omega} \frac{(\mathbf{q} \cdot \mathbf{Z}_I^*)_\alpha (\mathbf{q} \cdot \mathbf{Z}_J^*)_\beta}{\mathbf{q} \cdot \epsilon^\infty \cdot \mathbf{q}}$$

It raises the longitudinal mode and leaves the transverse ones alone — the LO-TO splitting — and it vanishes identically where  $\mathbf{Z}^*$  does, so a non-polar crystal is untouched. Contracting the same two ingredients the other way gives the ionic screening of a *static* field, one oscillator per polar mode:

$$\epsilon_{ij}^0 = \epsilon_{ij}^\infty + \frac{4\pi e^2}{\Omega} \sum_\nu \frac{p_i^\nu p_j^\nu}{\omega_\nu^2}, \quad p_i^\nu = \sum_{I\gamma} Z_{I,i\gamma}^* z_{I\gamma}^\nu$$

where  $p$  is the same mode dipole the infrared activity is the square of. So  $\epsilon^0$  and  $\epsilon^\infty$  are two ends of one contraction, and together with the frequencies they satisfy the Lyddane-Sachs-Teller relation  $\epsilon^0/\epsilon^\infty = (\omega_{\text{LO}}/\omega_{\text{TO}})^2$ .

```
spectrum = calc.get_vibrational_spectrum(neutralize=True)
print(spectrum.static_permittivity)           # eps_0, 3x3
print(spectrum.ionic_permittivity)           # the polar modes' share of it

lo = calc.get_vibrational_spectrum(loto_direction=(1, 0, 0), neutralize=True)
print(lo.frequencies)                         # the LO modes along x
```

`nonanal` adds the term to a dynamical matrix on its own and `loto_modes` diagonalises the result; `polar_mode_permittivity` is the oscillator sum, and `neutral_born_charges` the sum-rule correction below. All four are pure functions of arrays, so they can be checked without a self-consistent run behind them.

Both halves agree with `dynmat.x` — which applies the same term and the same oscillator sum to the tensors this code writes to its `fildyn` — to every digit it prints. **The Lyddane-Sachs-Teller relation is the check neither half can pass alone**, since one side is a ratio of two diagonalisations and the other a sum of dipoles over  $\omega^2$ ; on AlAs it holds to **5.0e-11**.

**It needs a charge-neutral  $\mathbf{Z}^*$  and that is worth knowing before reading an LO frequency.** The sum rule  $\sum_I \mathbf{Z}_I^* = 0$  says a rigid translation builds no field; a computed  $\mathbf{Z}^*$  misses it by the basis-set error, which *charges the crystal* — AlAs at `ecutwfc = 10` misses it by  $-1.257$ , and the non-analytic term then lifts a longitudinal *acoustic* mode from  $1.8$  to  $33.8 \text{ cm}^{-1}$  and puts the LO frequency  $7.7 \text{ cm}^{-1}$  low. `neutralize=True` shares the violation out over the atoms, which is what `dynmat.x` does under any `asr` but 'no'. Comparing against another code does not see this: it is handed the same charges and reproduces the same wrong number. LST does.

**Refuses.** Everything the Raman tensors and the Born charges refuse, since it is built from both: PAW  $Z^*$ , metals, and the rest of the list above. The splitting is a  $\Gamma$  quantity by construction — it is what  $q \rightarrow 0$  leaves behind — so a dispersion through it needs phonons at  $q \neq 0$ , which are not implemented.

## Elastic constants, electrostriction and the elasto-optic tensor

The same machinery in the *strain* coordinate rather than the atomic one. **The strain response itself works on all three pseudopotential kinds** (`strain_response`): the augmentation charge  $Q_{ij}(\mathbf{r})$  is a function of the *cell*, so a homogeneous strain deforms the table it is tabulated on where a displacement only translates it, and on top of that come the four terms the dynamical matrix needed. Against a central difference of *re-converged* strained runs — which needs no `ph.x`, since `ph.x` has no strain perturbation — it agrees to **4.6e-4** (ultrasoft) and **4.7e-4** (PAW) against a norm-conserving 1.9e-4 on the same cell. The *derivatives* below are norm-conserving still — but the variational second-order energy they all stand on is not: it reproduces `dielec.f90`'s dielectric constant to **3.4e-10** (ultrasoft) and **6.9e-11** (PAW) against a norm-conserving 8.4e-10, which took four terms that each vanish when  $S$  is the identity. One of them is worth stating on its own, because it is a distinction rather than a term: a *state* is projected with  $1 - \sum |\psi\rangle\langle\psi|S$  and a *right-hand side* with  $1 - \sum S|\psi\rangle\langle\psi|$ , and collapsing the two — which is exact at  $S = 1$  — costs a factor of fourteen. The third derivative in this *strain* coordinate is what stays norm-conserving, and how far off it is has been measured rather than left unknown: both of the tangents that closed the Raman tensor transfer, and take it from **4.6e-2** to **1.3e-2** against a central difference over re-converged strained cells, where the norm-conserving control on the same script is **2.3e-4**. What is left is entirely the *db* term — the same  $-1.72$  of 112 on ultrasoft and on PAW, which is what says it is structural. One candidate for it is excluded by measurement: writing the commutator that sources *db* with the multiplier matrix rather than the frozen scalar eigenvalue closes this coordinate (1.7e-4) and breaks the *displacement* one, taking the Raman tensor from 1.2e-4 to 1.14e-3. `strain_response` solves the first-order problem for a strain — Abinit's metric-tensor formulation, obtained for nothing here because the cell was already written in reduced coordinates — and two things follow from it:

$$C_{ijkl} = \frac{\partial \sigma_{ij}}{\partial \varepsilon_{kl}}, \quad \frac{\partial \chi_{ij}}{\partial \varepsilon_{kl}} \text{ (elasto-optic)}$$

The second is a **third** derivative of the energy, and it comes from one `jvp` of the second-order energy at frozen first-order wavefunctions — the  $2n+1$  theorem again. Through the thermodynamic identity of Tanner, Bousquet and Janolin it gives the four electrostriction tensors  $m$ ,  $q$ ,  $M$  and  $Q$ .

```
from defumat.response import elastic_constants, electrostriction

es = electrostriction(calculation, scf)      # solves the strain response itself
print(es.m, es.q, es.M, es.Q)              # the four electrostriction tensors
print(es.photoelastic)                     # the elasto-optic tensor
print(es.elastic.voigt)                    # C_ij in Voigt notation, GPa

# or the elastic constants alone, from a strain response you already have
C = elastic_constants(calculation, scf.wavefunctions, scf.eigenvalues,
                      scf.density, es.strain)
print(C.tensor.shape, C.voigt)
```

Converged silicon gives  $C_{11} = 198.5$  GPa and  $C_{12} = 68.9$ , against a measured 165.7 and 63.9; the three independent components of  $\partial\epsilon/\partial\epsilon$  match a central difference over re-converged strained cells to **2e-4**, which is that difference's own floor, and the rank-4 tensor is cubic to 3e-14 with nothing imposing it.

**Refuses.** Both are norm-conserving, `nspin = 1`, insulators and **clamped-ion**. A diverged first-order solution is refused rather than consumed in silence (`require_converged_responses`) — at QE's `alpha_mix = 0.7` the strain response of a *slab* diverges, and consuming it gave a  $C_{ijkl}$  that was not even symmetric under  $C_{ijkl} = C_{klij}$ . The elastic constants additionally need the whole  $k$ -grid rather than a wedge, because their functional builds its own density and symmetrises it as a scalar inside the chain rule.

## 7.2 The piezoelectric tensor

The polarization a strain induces is the stress an electric field induces — one mixed second derivative read either way round:

$$e_{k,ij} = \frac{\partial P_k}{\partial \epsilon_{ij}} = \frac{\partial \sigma_{ij}}{\partial \mathcal{E}_k} = -\frac{1}{\Omega} \frac{\partial^2 E}{\partial \epsilon_{ij} \partial \mathcal{E}_k}$$

It is the Born charge's construction with one coordinate changed.  $Z^*$  is a jvp of the *force* along the field's response; the force is `jax.grad` of the frozen-state energy in the atomic positions and the stress is `jax.grad` of the same functional in a strain, so this is a jvp of the *stress* along the same response — three of them, one per field direction, on top of a dielectric constant that was going to be solved anyway. The orthonormality constraint contributes nothing here, because  $\langle \psi | \hat{S} | \psi \rangle$  is a sum over a sphere of integers and carries no cell.

```
calc = Calculator.from_file("alas.in", pseudo_dir="pseudo")

piezo = calc.get_piezoelectric_tensor()
print(piezo.voigt)           # (3, 6) in C/m^2, columns xx yy zz yz xz xy
print(piezo.e14)             # -0.7638: zincblende's one component
print(piezo.e.shape)         # (3, 3, 3) in e/bohr^2, e[k, i, j]
print(piezo.dielectric.epsilon) # the field response is shared, not re-solved
```

`pw.x` computes no piezoelectric tensor — the word occurs once in the whole distribution, in a citation in a comment — and Elk's task 380 reaches it by a finite difference of the Berry-phase polarization over one full ground state per strain tensor. So the validation is internal, and there are four independent statements. Silicon is centrosymmetric and its whole tensor vanishes (2.4e-5 C/m<sup>2</sup> against AlAs's 0.764 from the same code, with nothing imposing it on a `nosym` run); AlAs is *43m* and only  $e_{14} = e_{25} = e_{36}$  survive, to 1.7e-14; an eight-point wedge reproduces the sixty-four-point closed grid to **4.5e-9**; and the same mixed derivative contracted the other two ways — `zstar_eu.f90`'s expression with a strain label, and the strain response against the field's bare perturbation — agree to **6.2e-15** and **1.3e-7**. What anchors the sign and the normalisation is that the *same* assembly run in the position coordinate is the Born charge, and that is `ph.x`'s number.

**Refuses. Clamped-ion** — the atoms are carried by the strain and do not relax, which is also what Elk's task computes; the internal-strain term that makes the constant comparable with experiment is not here, and for zincblende it very nearly cancels this

one. And **polar crystals by name**: what the derivative above gives is the *improper* tensor, and the proper piezoelectric response differs from it by  $\delta_{ki}P_j - \delta_{ij}P_k$ , which needs the spontaneous polarization itself. Those terms vanish identically whenever the two Cartesian labels they pair are different — so  $e_{14}$  of a zincblende crystal never carries the ambiguity — and they vanish for every component of a class that admits no invariant vector, which is what is checked. **Ultrasoft and PAW**, and that one is a gap rather than a missing term: nothing in the assembly is norm-conserving and the displacement leg of it is the Born charge, which is validated on all three kinds, but every ultrasoft and PAW crystal here is centrosymmetric and agrees with zero whatever is wrong. Beyond that: insulators, `nspin = 1`, and everything the Sternheimer regime and the differentiable cell refuse (a metal, a spin spiral, a magnetic field, a shifted grid run with `nosym`).

### 7.3 Spin-polarized response

The Sternheimer equation is solved per spin channel,

$$(H_\sigma - \varepsilon_{n\sigma}S + \alpha_{pv}Q_\sigma) |\Delta\psi_{n\sigma}\rangle = -P_{c,\sigma}^+ \Delta V_\sigma |\psi_{n\sigma}\rangle,$$

with the occupied manifold  $P_{c,\sigma}$  built from *that channel's* filling:  $N_\uparrow^{\text{occ}} = \text{NINT}(n_\uparrow)$  and  $N_\downarrow^{\text{occ}} = \text{NINT}(n_\downarrow)$ , which for a magnetic insulator are different numbers. Quantum ESPRESSO never meets this because LSDA doubles `nks` there and each spin channel is a separate  $k$ -point.

```
# chi_0 for a spin-polarized run -- the part that is validated unconditionally.
from defumate.response.sternheimer import make_sternheimer

solver = make_sternheimer(calculation, result, spin_polarized=True)
drho = solver.chi0(dv)          # dv is (2, n1, n2, n3) on the dense grid
```

Validated against a central difference of the density under a spin-dependent probe potential —  $1.8 \times 10^{-6}$  on an antiferromagnetic hydrogen chain (a smeared metal) and  $1.1 \times 10^{-6}$  on triplet  $\text{O}_2$  (the sliced branch, ultrasoft) — and against the identity that a cell with no magnetization gives the same  $\varepsilon_\infty$  at `nspin = 1` and `nspin = 2`:  $6.2 \times 10^{-14}$  on 13.806646105. The probe must differ between the channels, because  $\chi_0$  is block diagonal in spin and one equal in both would not tell the blocks apart.

The *screened* response and the Born effective charges are validated for `nspin = 2` as well, on triplet  $\text{O}_2$  against the vendored `ph.x`:  $\varepsilon_\infty = \text{diag}(1.11091441, 1.11091441, 1.19799977)$  against  $(1.110915996, 1.110915996, 1.198004867)$ , and  $Z^*$  against the *raw* (no acoustic sum rule) block, 0.13372 and 0.20042 against 0.13367 and 0.20023. Two things had to be right for those and neither was a new derivation. The exchange–correlation kernel is *defined* to be zero where the magnetization reaches the charge — `dmxc_lsd` does not regularise  $|\zeta| = 1$ , it skips it — so what was missing was that convention, applied to the *argument* the derivative is taken at rather than to its value. And the orthonormality multipliers carry a weight on the column of  $\Lambda_{mn}$  and need a *mask* on its row: the solver keeps  $\max(N_\sigma^{\text{occ}})$  bands in every channel because the shape is static, so a magnetic insulator's minority channel carries empty bands whose rows are not removed by the zero weight. That leak is worth a factor of three on  $Z_{xx}^*$ , and nothing already validated can see it — the term it feeds is  $dS/du$ , which vanishes for a norm-conserving dataset, and the mask is all ones whenever the two channels are filled to the same depth.

**Refuses.** `tot_magnetization` together with a smearing: the two channels have separate Fermi levels and `smearing` carries one scalar `ef`. A filling whose boundary lands inside a *degenerate multiplet* — a Hund’s-rule atom is exactly this case — where the solve converges and returns an answer 100 % away from a finite difference, because which member falls below the cut is arbitrary; that is the physics, not a missing term. The dynamical matrix, the strain response, the elastic constants, the Raman tensor and the electrostriction tensors for `nspin = 2`: the solve is spin-polarized, their second-derivative assembly is not — *except* the Born effective charges, which are validated above. The *screened* response of a magnetic system with vacuum: for `nspin = 2` the kernel `dv_of_drho` is the second derivative of the LSDA energy in the two channel densities and diverges wherever a channel density reaches zero — measured on triplet  $O_2$ , 1504 of 91125 grid points and exactly 1504 NaN — *lifted for the LDA*, which is the paragraph above, and standing for a **GGA**, whose `dgcxc_spin` carries its own density and  $\zeta$  gates that nothing here has transcribed. And any potential-only meta-GGA (`tb09`, `bj06`), which has no total energy to differentiate — refused by name now rather than surfacing as `v_of_rho` asking for `tau`.

## 7.4 Magnons: the transverse spin susceptibility

A magnon is the collective precession of a magnet’s own magnetization: a spin wave. It lives at the pole of the transverse susceptibility, which obeys a Dyson equation with *no* Coulomb term — a transverse spin fluctuation moves no charge, and the Hartree interaction belongs to the charge channel:

$$\chi^{+-} = [1 - \chi_0^{+-} f_{xc}^{+-}]^{-1} \chi_0^{+-}, \quad f_{xc}^{+-}(\mathbf{r}) = \frac{B_{xc}(\mathbf{r})}{m(\mathbf{r})}.$$

The Kohn-Sham part  $\chi_0^{+-}$  is a sum over pairs with one member in each spin channel, the majority state at  $\mathbf{k}$  and the minority at  $\mathbf{k} + \mathbf{q}$ ; its own spectral weight is the **Stoner continuum** of independent spin flips, which costs at least the exchange splitting. The kernel pulls a mode out from underneath that continuum, and that mode is the magnon.

**The kernel is not a second derivative and this is why it works for a GGA.** Rotate the whole ground state rigidly by an angle  $\theta$ : then  $\delta \mathbf{m}_\perp = \theta \mathbf{m}$  and  $\delta \mathbf{B}_{xc,\perp} = \theta \mathbf{B}_{xc}$ , so the transverse response of the exchange-correlation field to the magnetization is  $B_{xc}/m$  — an exact consequence of rotational invariance of  $E_{xc}[n, |m|]$ , whatever the functional. It also means the Goldstone theorem holds *by construction*, so the residual below tests the assembly and the truncations rather than the physics of  $f_{xc}$ .

```
import numpy as np
from defumat import Calculator

calc = Calculator.from_file("ni-fcc-magnon.in")

# one wavevector: the spectral function, and the Stoner continuum under it
chi = calc.get_spin_susceptibility((0.0, 0.0, 0.0), np.linspace(0.0, 0.06, 9),
                                   nbnd=30, ecut_response=60.0)

chi.goldstone          # how far  $X_\theta B_{xc} = m$  is from holding: the error bar
chi.enhancement         #  $\lambda_{\max}(X_\theta F)$  at  $\omega = 0$ ; one at  $q = 0$ 
chi.magnon              # the pole, in Ry, or None where there is none
chi.magnon_mev
chi.spectral_function    #  $-Im \chi_{\theta 0} / \pi$ 
chi.kohn_sham_spectral_function
```

```

chi.method          # "crossing", "peak" or "unstable"

# a path: one fixed-density run serves every q
disp = calc.get_magnon_dispersion([(0, x, x) for x in (0.0, 0.25, 0.5)],
                                   np.linspace(0.0, 0.06, 9),
                                   nbnd=30, ecut_response=60.0,
                                   goldstone_correction=True)

disp.energies_mev   # (nq,), nan where the grid holds no pole
disp.enhancements   # > 1 is an instability, not a magnon
disp.kernel_scale

```

The functional entry points are `defumat.workflows.run_spin_susceptibility` and `run_magnon_dispersion`.

**q must be a difference of two k-points of the run's own grid.** That is where the states at  $\mathbf{k} + \mathbf{q}$  come from — there is no second diagonalisation — and it is Elk's requirement that `vecql` be commensurate with the mesh, reached by construction. What survives of the  $\mathbf{k} + \mathbf{q}$  problem is an umklapp shift of one gather index, and  $\chi_0(\mathbf{q} + \mathbf{G}) = \chi_0(\mathbf{q})$  under a relabelling of the matrix's own  $\mathbf{G}$  index holds to  $1.3 \times 10^{-17}$ , which is what says that shift is right.

**The pole is found as a root, not as a peak.** Below the Stoner onset  $\lambda_{\max}(\chi_0(\omega)f_{xc})$  is real and rises monotonically, so the magnon is the root of  $\lambda_{\max} = 1$  — no broadening enters it, a handful of frequencies suffice, and it reports *below zero* where the state is unstable instead of returning the frequency grid's own edge.

**The Goldstone residual is the number to read first.** A global spin rotation costs no energy, so  $\chi_0(\mathbf{q}=0, \omega=0) B_{xc} = m$  exactly — but only with a complete band set *and* a complete  $\mathbf{G}$ -set, so what `goldstone` reports is a double truncation. **Which axis binds depends on the element.** On hydrogen the band count does it:  $8.0 \rightarrow 0.50$  per cent over  $n_{\text{bnd}} = 12 \rightarrow 140$ . On fcc nickel the band count does nothing at all (10.02, 10.13, 10.22 per cent at 30, 60, 100 bands) and only `ecut_response` moves it:  $10.0 \rightarrow 6.1 \rightarrow 2.0$  per cent at 12, 30, 60 Ry, because  $B_{xc}$  of a 3d shell has structure a small sphere cannot represent.

**Against Elk's committed reference, nickel's magnon agrees to 0.8%.** Elk ships an Ni-magnetic-response example — fcc nickel at  $\mathbf{q} = (0.1, 0.1, 0)$  on a  $10^3$  shifted grid — whose pole sits at 117.92 meV. Its input applies a small magnetic field, which is worth 1.99 meV of Zeeman gap, so its field-free magnon is 115.93 meV against **115.04** here on the same cell, grid and wavevector — and 123.08 with the Goldstone correction applied, which moves it *away*, so neither is chosen for agreeing. That is across an all-electron LAPW basis versus a norm-conserving plane-wave one, and across a 25 Ry response sphere versus 60. The moment agrees on the same run (0.6178 against  $0.6152 \mu_B$ ), and so does the sign of the static  $\chi$ . **That 0.7% of kernel is worth 7% of magnon** is the number to take away: it is how steeply  $\lambda(\omega)$  crosses one, and so how far the Goldstone residual has to fall before a magnon energy is worth a second digit.

**An enhancement above one is an instability, and that is a result.** It says the collinear state is not a minimum at that wavevector — the transverse susceptibility has changed sign — so the ferromagnet is not the ground state there. Nickel's enhancement *falls* away from  $\mathbf{q} = 0$ , which is what a stable ferromagnet looks like. Most simple models do not: a half-filled band on any lattice prefers antiferromagnetic order, so an fcc hydrogen crystal comes out soft at finite  $\mathbf{q}$  at every spacing tried, and the calculation says so rather than producing a plausible dispersion anyway.

**Do not compare that against a spiral  $E(q)$  without checking what the spiral converged to.** It is the obvious cross-check and it does not work: on fcc hydrogen the 90-degree spiral at  $q_3 = \frac{1}{2}$  converges *off* the magnetic branch, so its energy gain is de-

magnetization and not a spin wave; a 15-degree cone collapses the same way, because nothing holds it. (The collapsed  $|m|$  is not a number to quote — two runs of the same input give 0.0001 and 0.0435, which is where a wandering SCF stopped rather than a property of the state.)

**A constraint does not rescue it on this cell, and the reason is the cell.** A spiral SCF accepts `constrained_magnetization` and a magnetic field — neither is refused — and the quantity a constraint acts on is the **rotated-frame** magnetization, which is the right object for a helix. But `'fsm'` does not converge here, and it does not converge at  $q = 0$  either, where there is no rotated frame at all: fcc hydrogen at this spacing is a *marginal* magnet, so  $m(B)$  is nearly a step — 0.057 Ry of field takes the moment from 0.027 to 0.719 — with very little in between for a constraint to hold. The moment lands saturated, the inner SCF then stops converging under the field that put it there, and since the field steps only on self-consistent pairs it freezes. Two numbers say that cell was the wrong choice before the run: its bare moment is 0.0435 and the target was 0.0273, so the constraint asked for a moment *below* the cell's own while the only other stable point was saturated in the opposite direction. Use a robust magnet, and give `electron_maxstep` room: it is **shared** between the inner SCF and the outer field loop.

**What to read when one does not converge.** `'fsm'` drives a *feedback field* rather than a penalty, so its `constraint_energy` is 0 by construction and tells you nothing. `SCFResult.constraint_residual` is  $\mathbf{m} - \mathbf{m}_{\text{target}}$ , signed and per component, with the same per iteration in history; a residual that has changed *sign* is an overshoot rather than a slow approach. And `converged = False` on such a run is often not a density failure — accuracy can sit below `conv_thr` while the constraint is unmet, which is a separate warning with its own advice. Elk's own magnon-spiral example runs `fsmtype = -1` with a large field along `momfix` instead: the *direction* fixed hard rather than a small magnitude pinned. That scheme is not implemented here.

`goldstone_correction=True` scales the kernel by  $1/\lambda_{\text{max}}(\mathbf{q}=0)$ , the same factor at every  $\mathbf{q}$ , so the uniform mode sits at exactly zero. It is what the magnon literature does, it is off by default, and it is honest only if the residual error is  $\mathbf{q}$ -independent — check that by running the dispersion at two (`n_bnd`, `ecut_response`) settings.

**Refuses.** `nspin != 2`: without a collinear magnetization the transverse block does not decouple and  $B_{xc}/m$  does not exist, and a **noncollinear** ground state needs the full  $4 \times 4$  spin-density response, which is a different object. **Ultrasoft and PAW**, because  $\langle u_i | e^{-i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} | u_j \rangle$  is not the plane-wave overlap once the charge is not all in  $|\psi|^2$ . A **spin spiral**, whose rotating frame is not the one the kernel was derived in. A **Hubbard  $U$** , which responds to the magnetization too. An **applied magnetic field or a constrained moment**, because the rotation argument assumes the energy is invariant under a global spin rotation and a field is exactly what breaks it — a magnon in a field is gapped by the Zeeman energy, which this construction does not carry. A **potential-only meta-GGA**, there being no  $E_{xc}$  for that argument to be about. And a **symmetry-reduced k-set**:  $\chi_0(\mathbf{G}, \mathbf{G}')$  is a matrix in two  $\mathbf{G}$  indices that nothing here symmetrises, and the  $\mathbf{k} + \mathbf{q}$  fold needs the grid closed under translation by  $\mathbf{q}$  in any case.

## 8 Optical spectra and excitons: TDDFT

The response above is a *static* one, solved orbital by orbital. An absorption spectrum needs the frequency axis and needs the susceptibility as a **matrix** in reciprocal lattice vectors, so this is the one place where a sum over states earns its keep. `run_absorption`

builds

$$\chi_{\mathbf{G}\mathbf{G}'}^0(\omega) = \frac{1}{\Omega} \sum_{\mathbf{k}} \sum_{ij} \frac{(f_i - f_j) \rho_{\mathbf{G}}^{ij} \rho_{\mathbf{G}'}^{ij*}}{\epsilon_i - \epsilon_j + \omega + i\eta}, \quad \rho_{\mathbf{G}}^{ij} = \langle u_i | e^{-i\mathbf{G}\cdot\mathbf{r}} | u_j \rangle$$

and solves the Dyson equation with an exchange-correlation kernel:

$$\epsilon^{-1} = 1 + X(1 - X - \tilde{f}_{\text{xc}}X)^{-1}, \quad X = v^{1/2}\chi^0 v^{1/2}, \quad \tilde{f}_{\text{xc}} = v^{-1/2}f_{\text{xc}}v^{-1/2}$$

The **bootstrap** kernel of Sharma, Dewhurst, Sanna and Gross (*PRL* **107**, 186401 (2011); Elk's `fxctype = 210`) is that equation's other half, so the two are solved together to self-consistency:

$$f_{\text{xc}}^{\text{BS}} = -\frac{\epsilon^{-1}(\mathbf{q}, 0) v(\mathbf{q})}{\epsilon_0(\mathbf{q}, 0) - 1}, \quad \epsilon_0 = 1 - v\chi^0$$

It is parameter-free and diverges as  $1/q^2$ , which is what binds an electron-hole pair; ALDA's kernel is finite at  $\mathbf{q} = 0$  where  $v$  diverges, so its head and wings vanish *identically* and no amount of it can bind one.

**Quantum ESPRESSO has no counterpart.** `TDDFPT/` is a Liouville-Lanczos solver with RPA and ALDA; it has no bootstrap kernel and no Dyson equation in  $\mathbf{G}$  space. The reference is Elk.

```
from defumat.workflows import run_absorption
import numpy as np

omega = np.arange(0.0, 0.6, 0.005)          # Ry
spectrum = run_absorption(system, pseudos, scf.density, omega,
                          kernel="bootstrap", # or rpa / alda / lrc / bootstrap-1
                          nbnd=60, ecut_response=8.0,
                          broadening=0.012, scissor=0.05)

print(spectrum.absorption)                  # Im eps_M, isotropic average
print(spectrum.frequencies_ev)             # the same grid in eV
print(spectrum.alpha, spectrum.iterations) # the LRC-equivalent head, and the
                                           # bootstrap's fixed-point count
print(spectrum.static_residual)            # how far the band truncation reaches
```

`chi_0` is the whole cost and the kernel is nearly free, so comparing kernels means building it once: `independent_response` returns it and `solve_dyson` screens it, which is what `run_absorption` does internally and what the notebook shows.

Three knobs here are this code's own. `ecut_response` (Elk's `gmaxrf`) selects the  $\mathbf{G}$ -set the matrix is built over and has its own convergence: too small and the local-field effect is missing, and the cost is quadratic in it. `scissor` is *part of the method* rather than a convenience — the paper's Eq. (3) replaces  $\chi^0$  by a gap-corrected model response, and the velocity matrix elements are renormalised by  $e_{ij}/(e_{ij} \mp \Delta)$  along with it. And `static_residual` is the diagnostic for the one error this phase cannot refuse: how many empty states the sum ran over. It is  $\epsilon_M(0)$  from this run, in RPA, minus the same quantity from the Sternheimer solve, which is band-*complete*; it is kernel-matched on purpose, since differencing a bootstrap number against a Sternheimer one that contains  $f_{\text{xc}}$  would measure the kernel instead.

**A trap that is invisible in the answer:** the macroscopic dielectric function is the inverse of the  $3 \times 3$  *head* of  $\epsilon^{-1}$ , not the head of the inverse of the whole matrix. The second is the microscopic  $\epsilon$ , whose head in RPA is exactly the no-local-field result; it is smooth, positive, has the right peaks and is 9% too large on silicon. Both are returned (`epsilon` and `epsilon_no_local_fields`), because the gap between them is the local-field effect.

Validated by an identity rather than against another code's spectrum: the same  $\epsilon_M(0)$  is reached by this sum over states plus a Dyson inversion and by the projected conjugate-gradient solve of `dielectric_tensor`, which shares no machinery with it and never sees an empty state. On silicon the two agree to **1.3e-2** on a constant of 22 — and that residue is the band truncation, which is why it is reported rather than tuned away. The screening = "hartree" switch on `dielectric_tensor` exists for this comparison: the identity holds only when both sides use the same kernel.

**Refuses.** Finite  $\mathbf{q}$  (the perturbed states would live at  $\mathbf{k} + \mathbf{q}$ ); ultrasoft and PAW (every matrix element gains the augmentation charge  $Q_{ij}(\mathbf{G})$ ); metals (the  $f_i - f_j$  weight kills the  $i = j$  term, so the intraband Drude response is absent entirely — Elk's `tdftlr` does not add it either); `nspin`  $\neq 1$ , noncollinear magnetism, spin-orbit coupling, spin spirals and a Hubbard  $U$ ; a symmetry-reduced  $k$ -set, because symmetrising  $\chi_{\mathbf{G}\mathbf{G}'}^0$ , rotates *two*  $\mathbf{G}$  indices at once and the rank- $n$  symmetriser is Cartesian; the `alda` kernel with a gradient-corrected functional, since ALDA's kernel is pointwise in the density and a GGA's is not; and **non-convergence of the bootstrap fixed point is an error**, as it is in `tdftlr.f90`, because a spectrum from a kernel still in motion is the spectrum of nothing.

## 8.1 The conductivity tensor, the Kerr angle and the anomalous Hall effect

The whole complex tensor, interband plus a Drude term:

$$\sigma_{ij}(\omega) = \frac{i}{\Omega} \sum_{\mathbf{k}} \sum_{n,m} \frac{W_n(1-f_m)}{\epsilon_{mn}} \left[ \frac{v_{nm}^i v_{mn}^j}{\omega - \epsilon_{mn} + i\eta} + \frac{\overline{v_{nm}^i v_{mn}^j}}{\omega + \epsilon_{mn} + i\eta} \right] + \frac{\omega_{p,ij}^2}{4\pi(\gamma - i\omega)}$$

with  $v = \partial H / \partial \mathbf{k}$  from one jvp of  $H(\mathbf{k})$  at a frozen sphere. *That* is the one thing here not transcribed from `dielectric.f90`, and it is not a refinement: `epsilon.x` accumulates  $\langle \psi_1 | \mathbf{G} | \psi_2 \rangle$ , and  $[H, \mathbf{r}] \neq \mathbf{p}$  when the pseudopotential is nonlocal.

What is new beside `run_absorption` is the **antisymmetric** part,  $\sigma_{xy}$  — the response that rotates light reflected off a magnet, and whose static limit is the intrinsic anomalous Hall conductivity. Time reversal forces it to zero, so a nonmagnetic crystal has none however strong its spin-orbit coupling; a *collinear or coplanar* magnet without spin-orbit coupling has none either, because a spin rotation composed with conjugation survives as an antiunitary symmetry. (A *noncoplanar* magnet does have one with no spin-orbit coupling at all — the topological Hall effect — so the usual pairing is a route and not a theorem.)

```
calc = Calculator.from_file("ni-soc.in", pseudo_dir="pseudo")

sigma = calc.get_optical_conductivity(nbnd=36, window=0.6, nw=200)
print(sigma.sigma_s_per_cm.shape)    # (nw, 3, 3) in S/cm
print(sigma.kerr)                    # complex Kerr angle, degrees, per frequency
print(sigma.hall_conductivity)       # the static antisymmetric part, S/cm
print(sigma.plasma_ev)               # hbar omega_p, eV, a (3, 3) tensor
print(sigma.dielectric)              # 1 + 4 pi i sigma / omega
print(sigma.fsum, sigma.band_cut_gap) # two of the three diagnostics
print(sigma.degenerate_pairs)        # the third: pairs the guard dropped, usually 0
```

There is no reference output for any of it, so the validation is internal and analytic. The *symmetric* part reproduces the independent-particle dielectric function of the TDDFT

stack to **3.6e-14**, which is a different assembly (a Dyson solve over a response sphere) reaching `ph.x` through `dielectric_tensor`. The **f-sum rule** is the absolute anchor:  $\int_0^\infty \text{Re } \sigma_{aa} d\omega$  must equal  $(\pi/2\Omega) \sum W_n \langle n | \partial^2 H / \partial k_a^2 | n \rangle$ , and the familiar  $\pi n_e/2$  is only the *local* Hamiltonian's special case — silicon's diamagnetic weight is measurably **0.943** and not one. Measured against a central difference of the velocity operator, the ratio is 1.258 on a  $4^3$  grid, 1.045 on  $6^3$  and **0.990** on  $8^3$ : it converges in the *k-grid*, because what separates the two is  $\sum_k w_k \partial^2 \epsilon_n / \partial k^2$ , which vanishes over the zone by periodicity and not on a coarse mesh. Aluminium's plasma frequency comes out at **12.98 eV** against a free-electron 16.27, which is the zone-boundary gaps removing Fermi surface, and its tensor is isotropic to **1.7e-4** eV with nothing imposing that.

**Refuses. Ultrasoft and PAW**, for the Kubo curvature's reason: with a moving  $\hat{S}$  the current operator carries  $\epsilon_n \partial S / \partial k$  off the diagonal, and that term is identically zero for a norm-conserving dataset, so nothing validated here can see whether its convention is right. A **spin spiral**, whose two spinor components live on different spheres. A **symmetry-reduced k-set**, because the antisymmetric part is an axial vector — the escape is the whole unshifted grid, which is closed under the point group. And **nspin = 2**, whose two channels are two band structures whose conductivities add; a spinor run is the regime a magneto-optical spectrum wants anyway, since  $\sigma_{xy}$  needs spin-orbit coupling and collinear spin has none.

**Three things are reported rather than refused, and all three must be read.** `band_cut_gap` is the gap between the last band summed and the first dropped: stopping *inside* a degenerate multiplet keeps some members and drops others, and silicon's antisymmetric  $\sigma$  — zero by time reversal — comes out at 4.0e-13 when the cut falls at a real gap and at 1.0e-5 when it does not. `degenerate_pairs` counts the occupied/empty pairs that came back closer together than `degeneracy_tol` (1e-5 Ry, what an SCF converges an *empty* band to) and were therefore dropped. Zero is the ordinary case. When it is not zero it usually still does not matter — on the smeared frequency route the two orderings of such a pair cancel analytically, because what survives the  $1/\epsilon_{mn}$  is  $w_k[f(\epsilon_n) - f(\epsilon_m)]$  and that is itself linear in the gap — and the code warns exactly where that cancellation fails: `method = "curvature"`, whose  $1/\epsilon_{mn}^2$  leaves a  $1/g$  divergence, and a fixed-occupation run, where  $f$  is not a function of energy at all. And the **anomalous Hall conductivity of a metal is mesh-hungry**: the static Berry-curvature integral is concentrated on near-degeneracies at the Fermi surface, so it is quoted with its *k*-convergence and not on its own.

## 8.2 The shift current: the bulk photovoltaic effect

Illuminate a crystal with no inversion centre and it carries a direct current, with no junction and no built-in field. The intrinsic part of that is the *shift current*: the photoexcited electron is born displaced, and what the current counts is the real-space shift between the valence and conduction Wannier centres. It is a second-order optical response,  $J^a = 2\sigma^{abc}(0; \omega, -\omega)E_b(\omega)E_c(-\omega)$ , with

$$\sigma^{abc}(\omega) = -\frac{i\pi e^3}{4\hbar^2} \int [d\mathbf{k}] \sum_{n,m} f_{nm} \left( I_{mn}^{abc} + I_{mn}^{acb} \right) [\delta(\omega_{mn} - \omega) + \delta(\omega_{nm} - \omega)], \quad I_{mn}^{abc} = r_{mn}^b r_{nm}^{c;a}.$$

The whole difficulty is  $r_{nm}^{c;a}$ , the *generalised derivative*  $\partial_a r_{nm}^c - i(A_{nn}^a - A_{mm}^a)r_{nm}^c$  — neither piece of which is separately computable, since the diagonal Berry connection is gauge dependent and is not a function of  $H$  at one  $\mathbf{k}$ . The escape is the Aversa-Sipe sum rule,

which writes the covariant object entirely in gauge-free quantities living at a single  $\mathbf{k}$ :

$$r_{nm}^{c;a} = \frac{i}{\omega_{nm}} \left[ \frac{v_{nm}^c \Delta_{nm}^a + v_{nm}^a \Delta_{nm}^c}{\omega_{nm}} - w_{nm}^{ca} + \sum_{p \neq n,m} \left( \frac{v_{np}^c v_{pm}^a}{\omega_{pm}} - \frac{v_{np}^a v_{pm}^c}{\omega_{np}} \right) \right],$$

with  $v_{nm}^a = \langle n | \partial_a H | m \rangle$  and  $w_{nm}^{ab} = \langle n | \partial_a \partial_b H | m \rangle$ . Both are derivatives of code that already exists:  $v$  is one jvp of  $H(\mathbf{k})$  at a frozen sphere and  $w$  is that jvp differentiated by a second one. Nothing is transcribed. The interband dipole  $r_{nm}^a = -i v_{nm}^a / \omega_{nm}$  is *exact* for an eigenstate of the full  $H(\mathbf{k})$ , which is why a plane-wave code needs no counterpart to Wannier90's position-operator matrices.

```
calc = Calculator.from_file("alas-raman.in", pseudo_dir="pseudo")

sc = calc.get_shift_current(nbnd=14, window=0.9, nw=180)
print(sc.sigma.shape)           # (nw, 3, 3, 3) in A/V^2, real
print(sc.component(0, 1, 2))    # sigma^xyz(omega), -43m's only free component
print(sc.frequencies_ev)       # the photon energy axis in eV
print(sc.truncation)            # the band-truncation diagnostic: read it
print(sc.band_cut_gap)          # where the band set was cut, in Ry
```

There is no reference binary — `pw.x` computes no photocurrent of any kind, and Elk's `nonlinopt.f90` is second-harmonic generation, a different response — so the validation is internal and it is three statements. The **sum rule against a parallel-transport finite difference** of the dipole, which shares nothing with it: no second derivative, no intermediate sum, only the phases of neighbouring  $\mathbf{k}$ -points fixed to one another. That comparison is decisive in exactly one form, because the sum rule is an *identity over a complete basis*: held on a frozen sphere of 158 plane waves, ALAs agrees to **1.8e-4** at 158 bands against 4.3e-2 at 120 and 6.0e-2 at 20. A residue that falls off a cliff at completeness is a correct assembly with a truncated sum; one that plateaus is a bug. Then the **crystal class**: ALAs comes out exactly  $\bar{4}3m$  on a `nosym` grid, only the components with all three labels different surviving, while centrosymmetric silicon gives zero from the same machinery. And the **absorption edge**:  $\sigma^{xyz}$  is 6.9e-25 A/V<sup>2</sup> at 2 eV against a peak of 1.1e-4 near 4.4 eV, which is the one check no index bookkeeping can pass by accident — no absorption, no photocurrent.

**All three are blind to an overall constant**, and this document says so rather than leaving it to be discovered: a  $\sigma$  wrong by a factor of two is still exactly  $\bar{4}3m$ , still zero on silicon and still zero below the gap. Two further checks close that gap. The **spin sum** — the same cell run `nspin = 1` and as a two-component spinor gives the same tensor to **4.6e-9**, peak ratio 1.000000. And the **overall scale against the published literature**, where the *ordering* is what makes the comparison sharp rather than merely consistent. ALAs's first peak converges to **35.0**  $\mu\text{A/V}^2$  at 4.17 eV (33.9 at `ecutwfc = 16`, then 33.7, 34.8 and 35.0 at 22 with 14, 22 and 30 bands), where calculations across the fourteen III-V and II-VI zincblende semiconductors span 14  $\mu\text{A/V}^2$  (CdSe, smallest) to 83 (AlSb, largest) and find the aluminium compounds the strongest of the family. A factor of two either way breaks that ordering rather than moving the number: 17 would put ALAs at the bottom of the family, below the cadmium chalcogenides, and 70 would put it beside AlSb, the heaviest and narrowest-gap member. The number to compare is the peak below about 6 eV, which is the range published spectra cover. That sweep also showed `truncation` to be *predictive* rather than merely indicative: it falls 3.5% to 1.0% over the band sweep while the value moves 3.9%.

**The convention is declared, because the literature has two.** What is implemented is  $j^a = 2\sigma^{abc}(0; \omega, -\omega) \text{Re}[\mathcal{E}_b(\omega)\mathcal{E}_c(-\omega)]$ , the definition of the paper the formulae above come

from. Some authors write  $J = \sigma \mathcal{E} \mathcal{E}$  and report numbers twice as large for the same physics, so a comparison against a paper that does not state its normalisation is worth nothing to better than a factor of two.

**Refuses. Ultrasoft and PAW**, for the Kubo curvature's reason one order further out: with a moving  $\hat{S}$  the dipole carries  $\partial S / \partial k$  and its derivative carries  $\partial^2 S / \partial k_a \partial k_b$ , both identically zero for a norm-conserving dataset, so nothing validated here can see whether their convention is right. A **spin spiral**. A **symmetry-reduced  $k$ -set**, because  $\sigma^{abc}$  is a polar rank-3 tensor and a wedge sum is not the cell's until it is averaged over the point group; the escape is the whole unshifted grid. A **metal**, whose partially filled bands contribute pairs with vanishing  $\omega_{nm}$  where every denominator above diverges — the Fermi-surface terms that replace them are a different assembly. **DFT+ $U$** , whose Hubbard term is built from  $\phi_U(\mathbf{k} + \mathbf{G})$  and so carries a velocity and a second velocity of its own. And **nspin = 2**. A **spinor** run is supported and tested. **The band count is reported rather than refused, and it is the largest approximation here.** The intermediate sum over  $p$  converges only as it approaches completeness, so a production band count carries a few per cent that no symmetry check sees — truncation is the in-run estimate and is taken over the top *quarter* of the bands, because dropping one band reports 5.3e-5 where the error is 4 per cent. The route that would remove it is written down in `NONLINEAR.md`: the two sums are a Sternheimer solve rather than a spectral sum, and what stops it being taken is that the operator to invert is indefinite. `band_cut_gap` carries the same warning it does for the conductivity.

### 8.3 Second-harmonic generation

Shine light of frequency  $\omega$  on a crystal with no inversion centre and some of it comes back out at  $2\omega$ . The tensor that says how much is the second-order optical susceptibility  $P^a(2\omega) = \chi^{abc}(-2\omega; \omega, \omega) E_b(\omega) E_c(\omega)$ , a polar rank-3 tensor, symmetric in its last two labels and identically zero in every centrosymmetric crystal. It is a sum of three physically distinct pieces — the pure interband term  $\chi_{II}$ , the intraband modulation of it  $\eta_{II}$ , and the modulation of the transition energy  $\frac{i}{2\omega} \sigma_{II}$  — each of which is a contraction of the interband dipoles  $r_{nm}^a = -i v_{nm}^a / \omega_{nm}$  over two resonant denominators, one at  $\omega$  and one at  $2\omega$ :

$$\chi^{abc}(-2\omega; \omega, \omega) = \frac{1}{\Omega} \sum_{\mathbf{k}, m, n} \left[ \frac{c_{mn}^{(1)}}{\omega_{mn} - \omega + i\eta} + \frac{c_{mn}^{(2)}}{\omega_{mn} - 2(\omega - i\eta)} \right],$$

where the coefficients  $c_{mn}^{(1,2)}$  carry a further sum over an intermediate state and are built once, before the frequency axis is touched. This is the same sum over states the shift current uses, contracted differently; it needs no second derivative of  $H$  at all, because that triple sum is the sum-rule expansion of the generalised derivative written out rather than collapsed.

```
calc = Calculator.from_file("alas-shg.in", pseudo_dir="pseudo")

shg = calc.get_shg(nbnd=22, window=0.6, nw=240, broadening=0.010)
print(shg.chi.shape)           # (240, 3, 3, 3), complex, in pm/V
print(shg.component(0, 1, 2)[0]) # chi^xyz at the lowest frequency
print(shg.d_coefficient(0, 1, 2)[0]) # d = chi/2, what the literature quotes
print(shg.frequencies_ev[:3])    # the FUNDAMENTAL photon energy, in eV
print(shg.truncation)            # the band-truncation diagnostic
print(shg.chi_ii, shg.eta_ii, shg.sigma_ii) # the three parts, kept apart
```

**There is a real reference here**, which is unusual in this corner of the package: Elk's `nonlinopt.f90` (task 125) computes the same tensor, and `tests/data/elk/` carries its output for the same AlAs crystal, the three parts separately as well as their sum. `pw.x` has none — its `el_opt.f90` is the *static* electro-optic response, and the branch that reaches even that is the one this document's Raman section records as broken in the vendored 7.5.

What the comparison is *not* is digit-for-digit, and saying so is part of quoting it: an all-electron LAPW basis is not a norm-conserving pseudopotential one, and a second-order susceptibility carries two energy denominators, so a gap difference is not a scale factor. On the same crystal and the same  $6^3$  mesh: the **resonance position** agrees to **0.5%** (2.152 eV against 2.163), the **peak height** to **7%** (27.52 a.u. against 25.70), and the **static value** to **11%** (−3.10 a.u. against −3.50). The last is the one that moves with the basis, and it is shown converged: `ecutwfc` 10, 30 and 45 give −2.71, −3.10 and −3.13, so the residue is the pseudopotential rather than the cutoff.

Four internal checks close what the reference cannot. `43m`: AlAs comes out exactly zincblende on a `nosym` grid with nothing imposing it. **Silicon**: an inversion centre forbids every component, and the centrosymmetric control runs through the same machinery. **The two-photon edge**:  $\text{Im } \chi$  turns on at *half* the smallest direct gap, not at the gap, because the second channel's denominator is  $\omega_{mn} - 2\omega$  — the only check here blind to both the scale and the symmetry, and a lost factor of two there moves the absorption edge by an octave. And **the spinor**, which is the scale's own anchor: the same cell run `nspin` = 1 and as a two-component spinor must give the same tensor, which is what catches a factor of two in the spin sum.

**The convention is declared, because the literature has two.** What is computed is  $\chi^{(2)}$ , defined by  $P^a = \chi^{abc} E_b E_c$ ; nonlinear-optics papers more often quote  $d^{abc} = \chi^{abc}/2$ , so a number compared against a published  $d$  without halving it is wrong by exactly two and nothing about its shape, symmetry or zeros would say so. `d_coefficient` returns the halved form.

**The scissors shift is the one knob worth explaining.** An LDA or GGA gap error does not scale a second-order susceptibility, it moves its resonances, and `scissor` applies the usual rigid shift to the empty states. The dipoles are built from the *unshifted* gaps — a scissors correction is a statement about energies, not about wavefunctions — which is what Elk reaches by scaling its momentum matrix elements instead. Validated against Elk's own scissored run: at  $\Delta = 0.05$  Ha the  $2\omega$  peak moves **0.0502** Ry against a half-scissor of 0.0500, and its height falls to **0.60** of the unscissored value against Elk's 0.58.

**Refuses.** The same five the shift current refuses, since this is the same velocity matrix elements contracted a different way and it inherits every reason: **ultrasoft and PAW** (a moving  $\hat{S}$  puts  $\partial S/\partial k$  in the dipole), a **spin spiral**, a **symmetry-reduced  $k$ -set** (a polar rank-3 tensor is not the cell's until averaged over the point group; the escape is the whole unshifted grid), a **metal**, **DFT+ $U$** , and `nspin` = 2. A *spinor* run is supported and tested.

**This does not lift the  $\chi^{(2)}$  refusal in the Raman section**, and the distinction matters. That one is about the  $2n + 1$  route through the Sternheimer solver, where a field enters only through a source term and the  $\langle u_i | r_k | u_j \rangle$  piece has nothing to build it from; a sum over states never needed that object. The electro-optic tensor and a truncation-free static  $\chi^{(2)}$  are still refused for the same missing term.

**The band count is reported rather than refused**, exactly as for the shift current and for the same reason: the sum over the intermediate state is an identity only over a complete basis. `truncation` is the in-run estimate, taken over the top quarter of the bands.

## 9 Topological invariants

Everything is built from one primitive — the overlap of occupied manifolds at neighbouring  $k$ -points,

$$M_{mn}^{(\mathbf{b})} = \langle u_{m\mathbf{k}} | \hat{S} | u_{n,\mathbf{k}+\mathbf{b}} \rangle$$

because a determinant of overlaps is blind to the unitary mixing a degenerate eigensolver leaves. The Berry curvature is the phase of a plaquette of them and the Chern number their lattice sum, which is an **exact integer on any mesh**:

$$F_{12}(\mathbf{k}) = \ln [U_1(\mathbf{k})U_2(\mathbf{k}+\hat{1})U_1(\mathbf{k}+\hat{2})^{-1}U_2(\mathbf{k})^{-1}], \quad C = \frac{1}{2\pi i} \sum_{\mathbf{k}} F_{12}(\mathbf{k})$$

$Z_2$  has two independent routes — the flow of Wannier charge centres, and, when there is an inversion centre, the parity product over the eight TRIM points:

$$(-1)^\nu = \prod_{i=1}^8 \prod_{n \text{ occ}}^{\text{Kramers}} \xi_{2n}(\Gamma_i)$$

```
from defumat.workflows.topology import (run_z2, run_z2_3d,
                                         run_berry_curvature)

z2 = run_z2(system, pseudos, scf.density, axis=2) # Wilson loop by default
print(z2.z2, z2.gap_step) # read gap_step before believing the integer

chern = run_berry_curvature(system, pseudos, scf.density, shape=(12, 12))
print(chern) # an exact integer on any mesh

nu = run_z2_3d(system, pseudos, scf.density) # the four 3D indices
print(nu) # (nu0; nu1 nu2 nu3)
```

**Refuses.** A collinear spin-polarized run (two Hamiltonians, so two independent sets of invariants and no statement of which is meant); a  $Z_2$  without spin-orbit coupling, where every invariant is zero for a reason that has nothing to do with the band structure; the parity route without an inversion centre; and a manifold that is not separated from the band above it (`gap_tol`). A PAW run needs the converged `becsum` passed beside the density — `run_z2(..., becsum = scf.becsum)`, which Calculator does for you — because `ddd_paw` is a property of the wavefunctions and cannot be rebuilt from a density; leaving it out is worth 1.03 eV in the eigenvalues, and an invariant computed from those is still an integer.

**Running both  $Z_2$  routes is the check.** Where they disagree the parity one is the answer, because it has no mesh and the Wilson route does. **Two things bite in a plane-wave code and both are silent:** neighbouring  $k$ -points do not share a  $G$ -sphere, so coefficients are aligned by Miller index; and the wrap at the zone edge is a *shift* of that index, without which the Chern number comes out smooth and non-integer.

### 9.1 The curvature as a map: the Kubo route

Two constructions of the same quantity, chosen by name:

$$\Omega_n^{12}(\mathbf{k}) = -2 \operatorname{Im} \sum_{m \neq n} \frac{A_{nm}^1 A_{mn}^2}{(\varepsilon_n - \varepsilon_m)^2}, \quad A_{nm}^a = \langle \psi_n | \partial_{k_a} H - \varepsilon_n \partial_{k_a} S | \psi_m \rangle$$

with `method="kubo"`, against the lattice field strength of Fukui, Hatsugai and Suzuki with `method="fhs"` (the default). The velocity operator is one jvp of  $H(\mathbf{k})$  at a frozen plane-wave sphere — no dense  $H(\mathbf{k})$  is ever formed — and the tangent for a crystal direction  $d$  is the reciprocal lattice vector  $\mathbf{b}_d$ .

```
from defumat.workflows.topology import run_berry_curvature

curvature = run_berry_curvature(
    system, pseudos, density, shape=(4, 4), nocc=4, nbnd=12, method="kubo"
)
curvature.curvature          # (4, 4),  $\Omega(k)$ 
curvature.curvature_by_band  # (4, 4, nocc) -- see the refusal box
curvature.truncation         # what the highest empty band was worth
```

Validated against the FHS flux, which shares no machinery with it: a centred plaquette shrunk around one  $\mathbf{k}$ -point of zincblende AlAs converges onto the pointwise Kubo value to  $3.2 \times 10^{-3}$ , and the whole  $24 \times 24$  mesh agrees plaquette by plaquette to  $1.45 \times 10^{-4}$ , improving 65-fold over a sixfold refinement. The velocity matrix elements themselves reproduce a central finite difference of  $H(\mathbf{k})$  to  $1.8 \times 10^{-9}$ , and on centrosymmetric silicon the curvature vanishes *pointwise* to  $3.5 \times 10^{-5}$ .

**Refuses.** `method="kubo"` is for the map, never for the integer: its denominator  $(\varepsilon_n - \varepsilon_m)^{-2}$  is singular at a degeneracy and its Brillouin-zone sum is an ordinary Riemann sum, which converges to an integer without ever being one — `method="fhs"` is the default and is exact on any mesh. **Ultrasoft and PAW are refused by name:** the  $\varepsilon_n \partial_k S$  term is identically zero for a norm-conserving dataset, so no validation here can see whether its off-diagonal convention is right, and  $q_{ij}(\mathbf{b})$  is the second missing piece; `method="fhs"` carries both correctly on all three dataset kinds. The sum over empty states is truncated at `nbnd` and the truncation is *reported* rather than hidden — read `BerryCurvature.truncation`, or `truncation_abs` wherever symmetry forces the curvature to vanish, since the ratio is then noise over noise. `curvature_by_band` is gauge invariant only for a non-degenerate band; inside a degenerate multiplet only the sum over the members is defined, and `curvature` (the manifold total) always is.

A mesh point where an occupied and an empty band actually *touch* — a Dirac cone, a Weyl node, the closing a topological transition runs through — is a genuine singularity and is **dropped, with a warning naming how many**. `BerryCurvature.singular_points` carries the count, and a nonzero one means the mesh sum is not a Chern number of that state set: the curvature diverges there and a zero was put in its place, which is smooth and plausible and wrong. Move the mesh off the point, or use `method="fhs"`, which needs no gap in a denominator. A band pair counts as touching below  $10^{-8}$  of the band width *over the whole mesh* — a fraction rather than an energy, since a model Hamiltonian carries no unit, and over the whole mesh because for a two-band model the spectrum at one  $\mathbf{k}$  is the gap.

## 9.2 The Berry-phase polarization

The polarization of a crystal is not the dipole of its unit cell — that integral depends on where the cell is cut. What is a property of the crystal is a *phase*: the Berry phase the occupied manifold accumulates as  $\mathbf{k}$  crosses the zone once along a reciprocal lattice

vector  $\mathbf{b}_i$  (King-Smith and Vanderbilt, PRB **47**, 1651 (1993)),

$$\Omega \mathbf{P} = \sum_i \frac{\varphi_i}{2\pi} \mathbf{a}_i, \quad \varphi_i = \underbrace{\text{Im} \ln \prod_j \det M^{(\mathbf{b}_i/N)}(\mathbf{k}_j)}_{\text{electronic}} + \underbrace{2\pi \sum_a Z_a^v s_{ai}}_{\text{ionic}}$$

with  $M$  the same overlap the invariants above are built from and  $s_{ai}$  the atom's crystal coordinate. It is the *determinant* of the overlap that is kept here where a Wilson loop keeps the eigenvalues, which is the whole difference between the two calculations.

**The value is defined modulo a quantum** and the result object carries it. A single number is not a physical statement: what is meaningful is a *difference* between two geometries taken on the same branch, which is what a Born effective charge and a piezoelectric constant are made of.

```
from defumat.workflows.polarization import run_polarization

p = run_polarization(system, pseudos, scf.density,
                    gdir=2,          # 0-based; pw.x's gdir = 3
                    npostr=6,        # as pw.x counts it
                    transverse=(2, 2))
print(p.total_phase, p.quantum)      # -0.2412383 (mod 1)
print(p.polarization_si)             # -0.34909 C/m^2, mod 1.44708
print(p.strings.phases)              # one phase per string, four here
```

`Calculator.get_polarization()` reads `lberry`, `gdir` and `npostr` straight off a `pw.x` input, so a polarization run transfers unchanged. `npostr` counts the repeated endpoint the way `kp_strings` does; the mesh built here carries `npostr - 1` distinct points and closes the string with the reciprocal lattice vector, which is the same links and one diagonalisation fewer per string.

**Refuses.** A **metal** — a Berry phase is a property of a gapped manifold, and a smeared occupation does not say which bands the string carries. **nspin = 2**: two channels are two independent string sets and one phase each, which is a layout this does not have rather than a missing term. And a **spin spiral**, whose two spinor components live on spheres centred at  $\mathbf{k} \pm \mathbf{q}/2$ , so a string overlap is not a single gather. A *spinor* run (`noncolin`) is supported and has its own `pw.x` reference; its quantum is half the scalar one, because a spinor band holds one electron. Ultrasoft and PAW are *not* refused, and both are validated: zincblende SiC matches `pw.x` string by string with an electronic phase of  $-0.009696$  against  $-0.00970$ . It had to be SiC rather than silicon, because a centrosymmetric crystal's electronic phase is zero whatever  $q_{ij}(\mathbf{b})$  does and so agrees with zero however wrong the augmentation is.

**Validated three ways, two of which need no other code.** Against `pw.x`'s own `lberry` run on AlAs it agrees *string by string* and on every digit printed (electronic phase 0.00876, total  $-0.24124$ ). The Zak phase of the SSH model is pinned to 0 or  $\pi$  by symmetry and comes out exactly that on a mesh of *any* size ( $10^{-10}$ ), and silicon is pinned by inversion to 0 or half a quantum, landing on the half ( $10^{-8}$ ). Silicon is also the second `pw.x` reference, and the one that exercises the quantum of 2: both its species carry an even valence, where AlAs's arsenic does not. The strongest is  $dP/du$  against the Born charges, which come from a Sternheimer solve and share no machinery with a string of overlaps: AlAs on an  $8 \times 8 \times 8$  ground state gives  $Z^*(\text{Al}) = +2.148$  against `ph.x`'s  $+2.142$ , charge-neutral to  $5 \times 10^{-5}$ .

### 9.3 The magnetoelectric tensor

How much electric polarization a magnetic field induces,

$$\alpha_{ij} = \frac{\partial P_i}{\partial B_j},$$

computed Elk’s way (task 390): a full ground state at  $B \mp \delta/2$  for each Cartesian direction, the Berry-phase polarization of each, and a central difference. The cheap route — one derivative of the magnetization along an electric field’s response — needs a noncollinear Sternheimer solve, and an electric field as a ground-state perturbation, which this package does not have.

**Two symmetries must both be broken or the answer is zero for a reason that has nothing to do with the calculation:** time reversal and inversion each map  $\alpha$  to  $-\alpha$ . Elk’s own example is built around the first, and so is the case here — a non-magnetic semiconductor has no linear tensor at all, so a *large* external field is applied to break time reversal by hand and the derivative taken against a small further change in it. The magnetism is induced rather than spontaneous, which is what makes a plain gapped semiconductor usable.

```
from defumat import Calculator

alas = Calculator.from_file("alas-magnetoelectric.in", pseudo_dir)
me = alas.get_magnetoelectric_tensor(delta=0.02, directions=(2,))
print(me.magnitude)          # 4.4376e-06
print(me.largest_step)       # how close a field step came to a branch
```

**Refuses.** An unconverged ground state, which is what makes the next limitation visible rather than silent. **Only the column parallel to the seeded magnetization is reachable** on the committed cell: a field along the direction `starting_magnetization` points converges in 12 iterations, and one with a transverse component does not converge in 150, because rotating the moment is the slow mode. It is not the tilt — a pure transverse field fails on its own. Seed the magnetization along the column you want. A run carrying no magnetization vector ( $n_{\text{spin,mag}} \neq 4$ ): the Zeeman term then multiplies nothing and every polarization comes out identical, which looks like a symmetry result. And a field step that moves the Berry phase by more than a quarter of its quantum, since a difference taken across a branch is the quantum divided by  $\delta$  — large, smooth and meaningless. Everything the polarization refuses is refused here too.

**The validation is the spin-orbit null.** Without the coupling a global spin rotation maps  $B$  to  $-B$  and leaves every charge observable fixed, so  $\alpha$  vanishes identically. On the same cell, same field, same ultrasoft family at the same valence, with only `lspinorb` and the scalar/fully relativistic datasets changed:  $3.43 \times 10^{-9}$  against the coupled  $4.44 \times 10^{-6}$ , a ratio of **1293**. Two things had to be right together to get there, and neither helps alone: `chain` (seeding one field run from the other, which breaks the very  $B \rightarrow -B$  symmetry the null rests on — off by default here, unlike Elk) and the *polarization’s* threshold rather than the self-consistent field’s. **The absolute value is not calibrated against another code:** Elk’s `bfieldc` and this package’s `B_field` differ in normalisation by about two orders of magnitude, and Elk’s own tensor on this cell is not converged.

## 10 Effective masses, site angular momenta, nesting, structure factors and STM images

Quantities Elk computes and `pw.x` mostly does not. The first two cost one NSCF at a single k-point; the nesting function costs one NSCF on a dense grid, and nothing after it; the structure factors cost one transform of a density that already exists, and an STM image one band sum with different weights.

### 10.1 The effective mass tensor

$$\left(\frac{1}{m^*}\right)_{ab} = \frac{1}{2} \frac{\partial^2 \varepsilon_n(\mathbf{k})}{\partial k_a \partial k_b},$$

in units of  $1/m_e$ . The factor of a half is Rydberg atomic units and not a normalisation:  $\hbar^2/2m_e$  is exactly 1 Ry bohr<sup>2</sup>, so a free electron has  $\varepsilon = |\mathbf{k}|^2$  and the tensor is the identity.

**The first derivative is analytic and only the second is a difference.**  $\partial \varepsilon_n / \partial k$  is the generalised Hellmann–Feynman expression built from one jvp of  $H(\mathbf{k})$ ; its second derivative is not reachable the same way, because an individual band’s  $|\partial_k \psi_n\rangle$  is not what the Sternheimer projector produces (it removes the whole occupied manifold, and the band whose mass is wanted is usually empty). So the construction is one central difference of an *exact* first derivative — six stencil points against Elk’s twenty-seven, and no polynomial fit. Elk’s route, differencing eigenvalues, is kept beside it as `method="eigenvalue"` because it shares nothing with the velocity operator.

```
from defumat import Calculator

calc = Calculator.from_file("si.scf.in")
mass = calc.get_effective_mass((0.0, 0.0, 0.0), nbnd=10) # crystal coordinates

mass.inverse_mass      # (nbnd, 3, 3) in 1/m_e, cartesian
mass.principal(band=7) # the three principal masses and their axes
mass.density_of_states_mass(band=7) # (m_x m_y m_z)^(1/3)
mass.truncation        # the O(h^2) the Richardson step removed
mass.multiplets        # per multiplet, including the degenerate ones
```

On two-atom silicon at  $\Gamma$  against the vendored **Elk** binary (all-electron LAPW, PBE,  $a = 10.26$  bohr) with a PAW dataset here: the non-degenerate  $\Gamma_{1v}$  agrees to **0.02%** (0.8600902 against 0.8603044, i.e.  $m^* = 1.16267 m_e$  against 1.16238) and  $\Gamma_{2'c}$  to 0.36% ( $m^* = 0.170439 m_e$ ). The two routes here agree with each other to  $1.2 \times 10^{-5}$ , and the tensors come out isotropic to  $2.9 \times 10^{-8}$  with nothing imposing cubic symmetry.

**Refuses.** A band inside a **degenerate multiplet** has no tensor of its own — the eigen-solver’s arbitrary rotation within the manifold rotates it — so those bands come back as `nan` and what is reported instead is the multiplet’s *summed* inverse mass, which is the trace and is invariant. Silicon’s threefold  $\Gamma_{25'}$  is the case, and Elk’s own output is the evidence: it prints  $-5.8938, -3.7690, -3.7690$  for three states whose energies agree to  $5 \times 10^{-8}$  Ha. A **spin spiral** is refused, since a k-stencil moves both spheres. `delta` is the stencil’s outermost displacement in both routes; the  $O(h^2)$  truncation is removed by a Richardson step and then *reported* in `truncation` rather than tuned away.

## 10.2 $\langle L \rangle$ , $\langle S \rangle$ and $\langle J \rangle$ on each atom

Where the orbital moment of a spin-orbit magnet actually sits. From one site density matrix per atom and shell, built out of the same projection a projected density of states uses,

$$\rho_{(ms),(m's')}^a = \sum_{n\mathbf{k}} w_{n\mathbf{k}} c_{ms,n\mathbf{k}}^a \overline{c_{m's',n\mathbf{k}}^a}, \quad c = \langle \phi | S | \psi \rangle,$$

and then  $\langle L_i \rangle = \sum_s \text{Tr}_m(L_i \rho_{ss})$  and  $\langle S_i \rangle = \frac{1}{2} \sum_m \text{Tr}_s(\sigma_i \rho_{mm})$ , with  $J = L + S$ . The  $L$  matrices are the complex-basis ones conjugated into the *real* spherical harmonics the code works in, using the same `rot_ylm` unitary the spin-orbit projectors are built from.

```
from defumat import Calculator

calc = Calculator.from_file("ni.scf.in") # noncolin, lspinorb, nosym
momenta = calc.get_angular_momenta()

print(momenta.table()) # what Elk writes to LSJ.OUT
momenta.atoms[0].l      # <L> on atom 1, cartesian, units of hbar
momenta.atoms[0].j      # <J> = <L> + <S>
momenta.total_orbital   # summed over the cell
```

**Validated by identities, not by another code’s number** — Elk integrates over a muffin tin of a stated radius and this projects onto an orbital set, so the two differ by definition the way Löwdin and muffin-tin charges do.  $\langle L \rangle$  is *quenched to zero* without spin-orbit coupling, measured at  $1.7 \times 10^{-16}$  on silicon; with the coupling on, fully-relativistic nickel gives  $\langle L_z \rangle = 0.0364767 \hbar$  against a measured orbital moment of about  $0.05 \mu_B$ , with  $|L|/|S| = 0.11665$  against an experimental  $m_L/m_S \approx 0.1$  and  $\langle L \rangle \parallel \langle S \rangle$  (Hund’s third rule). Driving the moment along  $z$ ,  $x$  and  $y$  gives  $|\langle L \rangle| = 0.0364767$  in all three, with a spread of  $7.3\text{e-}11$  over the three cubic axes, with nothing imposing that a magnitude is a scalar.

**A symmetry-reduced k-set works**, and what makes it work is that  $\langle L \rangle$  and  $\langle S \rangle$  are averaged over the group as **axial** vectors — each operation carrying  $\det(R)$  and the time-reversal sign beside the rotation, the site charge carrying only the atom permutation. The distinction is not a refinement: the *polar* average of an axial vector is exactly zero on any centrosymmetric crystal, so writing the obvious one would return a clean and plausible nothing rather than a wrong number.

**Refuses.** A **fully-relativistic ultrasoft or PAW** dataset, because the spinor overlap’s off-diagonal spin blocks are `qq_so` and the projection here applies the scalar  $S$  to each component; a fully-relativistic *norm-conserving* dataset has  $S = 1$  and is exact. And a **spin spiral**, whose two components do not share a set of atomic orbitals. `pw.x`’s `lorbm` computes the *cell*’s orbital magnetization and has no per-atom counterpart to check this against — that quantity is the next subsection, and the two are worth reading together.

## 10.3 The orbital magnetization of the cell

The other half of a magnet’s moment, and the half no integral over the unit cell can give. A Bloch state’s current circulates through the whole crystal, so  $\int \mathbf{r} \times \mathbf{j}$  depends on where the cell is cut — the same difficulty the electric polarization has. What is well defined is a k-space quantity (Thonhauser *et al.*, PRL **95**, 137205 (2005); Ceresoli *et al.*, PRB **74**,

024408 (2006)):

$$\mathbf{M} = \frac{e}{2\hbar c} \text{Im} \sum_n \int [d\mathbf{k}] \langle \partial_{\mathbf{k}} u_n | \times (H_{\mathbf{k}} + E_{n\mathbf{k}} - 2\mu) | \partial_{\mathbf{k}} u_n \rangle,$$

a *local* circulation inside each cell plus an *itinerant* one along the boundary. Neither is separately measurable and their sum is. The derivative is the covariant finite difference over a uniform mesh — the dual states  $|w_m\rangle = \sum_n (M^{-1})_{nm} |u_{\mathbf{k}+\mathbf{b},n}\rangle$  of the neighbouring manifold — so nothing is divided by  $E_n - E_m$  and the answer is invariant under any unitary mixing of the occupied bands.

```
from defumat import Calculator

calc = Calculator.from_file("i-atom-soc.in") # noncolin, lspinorb, gapped
orbm = calc.get_orbital_magnetization(divisions=(3, 3, 3))

orbm.lc, orbm.ic # pw.x's two Kubo terms, Bohr magnetons per cell
orbm.total      # their sum, at mu = 0
orbm.chern      # the Chern vector; zero for an ordinary insulator
```

**Against pw.x's lorbm** on the same  $3 \times 3 \times 3$  grid, an iodine atom in a twelve-bohr box:  $M_{LC} = -0.5986388$  against  $-0.5986370$  and  $M_{IC} = -0.5757995$  against  $-0.5757987$ , so the total is  $-1.1744384$  against  $-1.1744357$ . What is left at  $2 \times 10^{-6}$  is the difference between two separately converged densities: re-running at conv\_thr of 1e-6, 1e-8 and 1e-12 moves  $M_{LC}$  by 5e-6, 2e-7 and 2e-6 against the same reference. **The physics check is independent of both:** iodine is  $5s^2 5p^5$ , one hole in the p shell, and Hund's rules give  $L = 1$  locked parallel to  $S = 1/2$ . The site projection of the previous subsection measures  $\langle L_z \rangle = 0.99977 \hbar$  on the same run, and the modern theory gives  $1.1744 \mu_B$  — 17% more, which is what a projection onto atomic orbitals does not see: the circulation outside those orbitals, and the nonlocal pseudopotential's own contribution to the velocity, since  $\langle L \rangle$  is  $\mathbf{r} \times \mathbf{p}$  where this contracts  $H$ .

**Two conventions are QE's and are worth knowing.** What both codes print is  $M(\mu = 0)$ : orbm\_kubo imports the Fermi level and never uses it, and the term it drops is  $\mu$  times the Chern vector, which vanishes for any topologically trivial manifold. And the two printed terms are *not* the papers' LC/IC split — the Fortran prints  $\text{Im}\langle \partial u | H | \partial u \rangle$  and  $\text{Im}\langle \partial u | E | \partial u \rangle$ , which differ from  $(H - E)$  and  $2(E - \mu)$  term by term and agree in the sum. This follows the Fortran, so the two halves can be compared and not only the total.

**Refuses. Ultrasoft and PAW**, which pw.x refuses in the same place (setup.f90: 'Orbital Magnetization not implemented with USPP/PAW'): the overlap between neighbouring manifolds would need  $q_{ij}(b)$  and the dual states would have to be dual in the  $S$  metric. A run **without spin-orbit coupling**, where the orbital moment is quenched identically — measured here as  $10^{-9}$  with soc\_scale = 0 in the same relativistic dataset. **nspin = 2**, which is the same statement: each channel is separately time-reversal symmetric. A **metal**, or any manifold not separated from the band above it, which the gap check refuses by name. A **spin spiral**. And a mesh with **two divisions** along a direction that carries a derivative, where a point's two neighbours are the same k-point and the central difference is an alias; one division is allowed and sets that direction's derivative to zero, which is what a slab means.

## 10.4 The Fermi-surface nesting function

How much of the Fermi surface maps onto itself when translated by  $\mathbf{q}$ :

$$N(\mathbf{q}) = \frac{1}{N_k} \sum_{\mathbf{k}} g(\mathbf{k}) g(\mathbf{k} + \mathbf{q}), \quad g(\mathbf{k}) = g_s \sum_{s,n} \delta(\varepsilon_{sn\mathbf{k}} - E_F).$$

$g(\mathbf{k})$  is the density of states *at one  $k$ -point* — a picture of the Fermi surface on the grid — so  $N(\mathbf{q})$  is the  $\omega \rightarrow 0$  imaginary part of the Lindhard function stripped of its matrix elements: the *geometric* half of an instability. Where it is large, a perturbation of wavevector  $\mathbf{q}$  connects many occupied states to many empty ones at no energy cost, which is what softens a phonon, opens a charge-density-wave gap, or picks a spin spiral's pitch.

**It is a convolution.** Elk writes an  $O(N_q N_k)$  double loop with  $\mathbf{k} + \mathbf{q}$  folded back onto the grid; that fold makes the sum a cyclic cross-correlation, so  $N = \text{ifftn}(|\text{fft}(g)|^2)/N_k$  gives every  $\mathbf{q}$  in one transform. Measured on a  $12^3$  aluminium grid that is **0.001 s** here against Elk's 0.37 s. A symmetry-reduced wedge is *unfolded* onto the complete grid rather than refused, since  $\varepsilon_n(R\mathbf{k}) = \varepsilon_n(\mathbf{k})$ : 72 diagonalisations instead of 1728.

```
from defumat import Calculator

calc = Calculator.from_file("al.scf.in")
nest = calc.get_nesting(grid=(12, 12, 12)) # the grid is the convergence knob

nest.nesting      # (nq,) N(q), in (states/Ry/cell)^2
nest.qpoints      # (nq, 3) crystal coordinates, on the unshifted grid
nest.ratio        # N(q) / D(E_F)^2, dimensionless and 1 on average
nest.peak()       # the largest N(q) away from q = 0, and where it is
nest.along(2)     # (q_3, N) along one crystal axis, for a line plot
nest.as_grid()    # (n1, n2, n3), for a slice
nest.fermi_dos    # D(E_F) in states/Ry/cell
nest.elk_units    # the same array in NEST3D.OUT's normalisation
```

The functional entry point is `defumat.workflows.run_nesting`, which takes the converged density directly.

**Two identities come with the normalisation and both are worth reading.** The mean of  $N$  over the whole  $\mathbf{q}$ -grid is exactly  $D(E_F)^2$  (`sum_rule` reports the residue; it is  $3 \times 10^{-16}$  relative). And  $N(0) \geq N(\mathbf{q})$  for every  $\mathbf{q}$  by Cauchy-Schwarz, so **the nesting peak is always a peak away from the origin** — `peak()` excludes  $\mathbf{q} = 0$  for that reason, not as a plotting convenience.

Validated against a closed form and against a quantity computed by an unrelated route. For free electrons  $N(q) = \Omega/4\pi^2 q$  below  $2k_F$  and zero above: the code reproduces the  $1/q$  law to  $10^{-4}$  and the hard cut to  $10^{-16}$ . A half-filled hydrogen chain has  $2k_F = \pi/c$ , so it must nest at  $\mathbf{q} = (0, 0, \frac{1}{2})$  — it does, at **99.8%** of the Cauchy-Schwarz bound, and `relax_spiral_q` relaxes the corresponding spin spiral to 0.500014 by going downhill in a magnetic total energy, which shares nothing with this. Against the vendored Elk binary on aluminium,  $N(0)/D(E_F)^2$  agrees to 1.5%; the raw  $N(0)$  differs by 4.7%, which is the all-electron Fermi surface underneath and is quoted rather than tuned away.

**Refuses.** A constrained `tot_magnetization`, which has one Fermi level per channel, so  $g(\mathbf{k})$  is two different surfaces; a **fixed-occupation** run, which never searches for a level, so  $\delta(\varepsilon - E_F)$  has no argument; and a **spin spiral**, because the quantity is a statement about the state a spiral grows out of — compute it on the paramagnet and

compare its peak with `relax_spiral_q`'s answer. The delta defaults to a **Gaussian** even when the run used Methfessel-Paxton or cold smearing: those go negative on the wings, so  $g(\mathbf{k})$  can be negative and  $N(\mathbf{q})$  would have no sign at all. Pass `smearing=` to have the run's own anyway.

## 10.5 Magnetocrystalline anisotropy

The energy it costs to point a magnet's moment one way rather than another — what makes a hard magnet hard and what pins a spiral into a plane. It is tenths of a meV per atom against total energies of hundreds of Rydberg, so taking it as a difference of two self-consistent total energies asks two SCF runs to agree in their ninth digit. The *force theorem* does not ask that. Converge the magnet **without** spin-orbit coupling, rotate the converged density so its magnetization points along  $\mathbf{n}$ , and diagonalise **once** with the coupling switched on. At frozen density every term of the total energy except the band energy is a functional of  $\rho$  alone, so

$$E(\mathbf{n}_1) - E(\mathbf{n}_2) = E_{\text{band}}(\mathbf{n}_1) - E_{\text{band}}(\mathbf{n}_2), \quad E_{\text{band}} = \sum_{n\mathbf{k}} w_{n\mathbf{k}} \varepsilon_{n\mathbf{k}},$$

exactly, and the whole calculation is one diagonalisation per direction.

**The cheaper thing gives zero.** Freezing the wavefunctions as well and taking  $\langle \psi | H_{\text{SOC}} | \psi \rangle$  once returns no anisotropy at all: the coupling enters at first order as  $\xi \langle \mathbf{L} \rangle \cdot \mathbf{n}$ , and the orbital moment of a scalar-relativistic collinear state is quenched. The anisotropy is second order in the coupling, and what supplies it is the repulsion between levels that a diagonalisation performs and an expectation value does not. `frozen_expectation` computes that vanishing term anyway.

**Two pseudopotential files, one density.** The first leg runs with a scalar-relativistic dataset and the one-shot leg with the fully-relativistic dataset of the *same generation*; only the density crosses, and a density does not know which file made it. That is what makes an ultrasoft anisotropy reachable, the alternative — averaging one relativistic file's  $j$  channels back — being refused for ultrasoft and PAW by `pw.x` itself.

```
from defumat import Calculator

scalar = Calculator.from_file("scf.in")          # nspin = 2, Co.pbe-nd-rrkjus
spinor = Calculator.from_file("nscf.in")        # noncolin, lspinorb, rel- file
mae = scalar.get_anisotropy(spinor, directions="xyz")

print(mae.anisotropy_mev, mae.easy_axis)
print(mae.difference(0, 2))                    # in-plane minus out-of-plane
```

`directions` takes a list of vectors, an integer asking for that many points spread evenly over the sphere, or one of "cardinal", "xz" and "xyz". `projected=True` adds the decomposition of the band energy over atomic orbitals, resolved by atom,  $\ell$ ,  $m$  and spin, which is what says *which* orbitals supply the anisotropy; comparing its total against  $E_{\text{band}}$  measures the spilling.

`soc_scale` scales the spin-orbit part of the nonlocal potential and the overlap while keeping the same dataset and the same k-points. At 0 the anisotropy is exactly zero, which is the control for the whole calculation: without the coupling the Hamiltonian is invariant under a global spin rotation, so a band energy *cannot* depend on where the moment points.

**The torque: the same number from one angle.** Taking the anisotropy as  $E(\mathbf{n}_1) - E(\mathbf{n}_2)$  is  $10^{-5}$  Ry out of  $10^2$  — seven digits of cancellation. The torque takes it as a *derivative* evaluated once, where nothing cancels: for  $E(\theta) = K_1 \sin^2 \theta$ ,

$$-\frac{dF}{d\theta} = -K_1 \sin 2\theta, \quad \text{so at } \theta = 45^\circ \text{ the torque is } -K_1.$$

```
torque = scalar.get_torque(spinor)           # 45 degrees by default
print(torque.anisotropy_constant_mev)      # K1 in meV
```

**It differentiates the free energy, and for a smeared metal that is not the band energy.** A Hellmann-Feynman derivative at frozen occupations gives  $dF/d\theta$  with  $F = \sum w\varepsilon - TS$ ;  $\sum w\varepsilon$  carries an extra  $\sum (dw/d\theta)\varepsilon$  that the entropy cancels. The difference is not small. On tetragonal cobalt, as the smearing narrows:

| degauss (Ry) | band difference | free difference | torque |
|--------------|-----------------|-----------------|--------|
| 0.020        | 1.2353          | 0.5523          | 0.5523 |
| 0.005        | 0.5434          | 0.5291          | 0.5291 |
| 0.002        | 0.5308          | 0.5308          | 0.5308 |

all in meV. The torque tracks the free-energy difference at every width and both converge on  $K_1 = 0.531$ ; at a production degauss of 0.02 the *band*-energy difference reads 2.3 times that. Use `MagneticAnisotropy.free_energies` to compare like with like.

**Refuses. PAW**, because the handoff from the collinear run carries the density and nothing else, and a PAW Hamiltonian needs `ddd_paw`, which is built from a `becsum` belonging to a run with a different pseudopotential file (`pw.x` refuses it in the same place). A **Hubbard  $U$** , whose `ns` is not in the handoff either; a potential-only **meta-GGA**, whose  `$\tau$`  is not; a converged **magnetic field** or constrained moment, whose energy sits outside the reported total; and a **spin spiral**, which refuses spin-orbit coupling permanently and so has no coupling to switch on. A direction other than the system's own `angle1/angle2` needs `nosym`: a magnetic noncollinear run reduces its k-grid with the *magnetic* symmetry group, which depends on where the moment points, so two directions would otherwise be sampled on two different wedges. `soc_scale` takes 0 or 1 and nothing in between: the coupling-free end of the overlap is spin-independent, so the anisotropy vanishes identically there whatever else it is, but a blend of it with the true overlap partway is not the overlap of any set of projectors and  $S$  stops being a usable metric.

## 10.6 The relaxed anisotropy: total energies instead of band sums

The section above buys its precision by freezing the density: every term of the total energy except the band sum is a functional of  $\rho$  alone, so it cancels between two directions exactly. The price is that the density is not allowed to respond to the spin-orbit field. `run_relaxed_anisotropy` pays the other way round — one whole self-consistent noncollinear run per direction, and a difference of two **total** energies:

$$E_{\text{MAE}} = E_{\text{tot}}[\hat{\mathbf{n}}_1] - E_{\text{tot}}[\hat{\mathbf{n}}_2], \quad \text{each } E_{\text{tot}} \text{ converged at its own } \hat{\mathbf{n}}.$$

The extra energy the relaxation finds is variational, so it lowers *both* directions, and what survives in the difference is second order on a quantity that is already second order in the coupling. Nothing bounds it in general, which is why it is measured rather than argued about.

```

from defumat import Calculator

soc = Calculator.from_file("ni-tetragonal-relaxed-mae-paw.in",
                           pseudo_dir="tests/data/pseudo")
mae = soc.get_relaxed_anisotropy(directions="xz")

print(mae.energies_mev)      # relative to the first direction
print(mae.easy_axis, mae.anisotropy_mev)
print(mae.converged, mae.drifts)  # read these BEFORE the number

```

There is **one** calculator here and no second leg. That is the whole practical difference from the force theorem, and it is what lets this route reach **PAW** and a Hubbard  $U$ : the theorem hands a density from a scalar-relativistic run to a fully-relativistic one, and `ddd_paw` and `ns` are properties of *wavefunctions* that cannot cross that gap. A relaxed run hands nothing over — each direction converges its own `becsum` from its own states with one dataset throughout.

**Read .converged and .drifts first, and the second one is the trap.** Nothing holds the moment in a relaxed run, which is the point of it; so a direction that is not a stationary point of the anisotropy energy will turn towards one that is, and its total energy then belongs to a state other than the one asked for. A cardinal axis of a cubic or uniaxial crystal is stationary by symmetry and does not move; an oblique direction on a strongly anisotropic magnet need not be. A drift past one degree raises a warning naming the direction and where the moment went.

A **magnetic field** or a constrained moment, because the field's energy is deliberately outside the reported total, so two directions' totals would differ by a Zeeman term neither accounts for — and because a constraint defeats the drift check, the constraint being what holds the moment rather than the anisotropy. A potential-only **meta-GGA**, which has no energy functional at all, so its printed total is not the value of anything the run minimised. A **spin spiral**, which refuses spin-orbit coupling permanently. A **collinear** run, which has no direction to point. And `nosym` is required rather than merely advised: a magnetic noncollinear run reduces its k-grid with the *magnetic* symmetry group, which depends on where the moment points, and here that is worse than it is for the force theorem — a total energy carries the Ewald and Hartree terms too, so the contamination is not confined to the bands.

## 10.7 X-ray and magnetic structure factors

What a diffractometer measures is not a density but its Fourier coefficients, one per reflection:

$$F(\mathbf{H}) = \int_{\Omega} \rho(\mathbf{r}) e^{i\mathbf{H}\cdot\mathbf{r}} d^3r, \quad F_j(\mathbf{H}) = \int_{\Omega} m_j(\mathbf{r}) e^{i\mathbf{H}\cdot\mathbf{r}} d^3r$$

for every reciprocal lattice vector with  $|\mathbf{H}| \leq h_{\max}$  — X-rays scattering off the charge, neutrons off the magnetization. Elk's tasks 195 and 196; `pw.x` has neither. In a plane-wave code the integral is an array that already exists, up to the crystallographers' two conventions (the positive phase, and no  $1/\Omega$ ), so the whole step is one transform and a gather: **6 ms** against the seconds its SCF cost.  $F(0)$  is the electron count of the cell and  $F_j(0)$  its magnetic moment, which is what anchors both.

```

from defumat import Calculator

```

```

calc = Calculator.from_file("si.scf.in")
factors = calc.get_structure_factors(hmax=5.0)

factors.charge          # (nh,) complex F(H), in electrons
factors.magnetization    # (nh, ncomp) in Bohr magnetons, or None
factors.vectors.indices  # (nh, 3) the (h k l) after the output transform
factors.vectors.length   # (nh,) |H| in 1/bohr
factors.vectors.multiplicity # the size of each star that was collapsed
factors.of((2, -2, -2))  # one reflection, by the index the table prints
print(factors.table(limit=10))

# Elk's wsfac: rebuild the density from the states in an energy window (Ry),
# which is what makes this a probe of bonding rather than of the whole charge.
bonding = calc.get_structure_factors(hmax=3.0, window=(-1.0, 0.2))

```

The functional entry point is `defumat.workflows.sfac.run_structure_factors`. `transform="conventional"` labels the reflections on the conventional cubic axes (Elk's `vhmat`), `reduce=False` keeps every member of every star, and `core=True` adds a dataset's frozen core-correction charge.

**A pseudopotential density is valence-only, and the comparison with an all-electron code is structured by reflection class rather than by one tolerance.** On silicon, against the vendored Elk binary on the same cell and k-grid: the origin is 8 electrons here and 28 there, and every *allowed* reflection is core-dominated in the same proportion — (111) is 1.7495 against 15.14, and no tolerance makes that agree. What is comparable is a **forbidden** reflection, where the spherical part of every atom cancels and the core with it: silicon's (222) is **0.347406** here against **0.33416**, 4%, on a quantity whose entire value is the aspherical bonding charge. That last statement is checked without any other code, since this package's own SCF starting guess is a superposition of free atoms: it gives 0 at (222) to  $10^{-10}$ . The space-group forbidden reflections — diamond's (200), (420) — vanish to  $10^{-9}$  here and  $10^{-13}$  there.

**Norm-conserving, ultrasoft and PAW all work, and the difference between them is measurable.**  $F(0) = 8.0000000000$  on each, which is the check that the augmentation charge reached the transform at all — for a soft dataset most of the density near the nucleus is that charge. The like-for-like reference is not Elk's all-electron total but its **valence-only** run, which `wsfac` above the 2p core produces: against it the augmentation charge is worth a factor of two, (111) coming out  $-0.7\%$  for ultrasoft and PAW against  $-1.5\%$  norm-conserving, and (004)  $+31\%$  against  $+61\%$ . It is worth *nothing* on the forbidden (222), which is 0.3474 on all three to  $3 \times 10^{-4}$ : a silicon atom in diamond sits at  $\bar{4}3m$ , whose lowest non-spherical invariant is  $l = 3$ , and an *s, p* dataset's augmentation charge carries multipoles only to  $L = 2$ . That is also what the 4% against all-electron is, by elimination — not the k-grid, not either basis, not the functional (0.07%), and **not the core**, since Elk's own valence-only run gives the identical 0.334165.

**Magnetic reflections are a low-|H| quantity and the boundary is measured.** Ferromagnetic bcc iron against Elk's task 196, comparing the normalized  $F_{\text{mag}}(\mathbf{H})/F_{\text{mag}}(0)$  because the moments differ ( $2.2145 \mu_B$  here,  $2.0613$  there): 5.8% at the first reflection, 17% at the second, worse outwards. Raising `ecutwfc` from 30 to 45 moves those by less than 0.1%, so it is the pseudisation and not the basis — iron's moment is in the 3d shell, inside the radius the dataset smooths. The cheapest cell carries the sharpest magnetic statement instead: an antiferromagnetic hydrogen chain has charge factors 2.000000, 0, 0.8587, 0 along the chain and magnetic ones 0, 1.2254, 0, 0.3862, so it scatters neutrons exactly where it scatters no X-rays, and the zeros are exact.

**Refuses.** An  $h_{\text{max}}$  past  $\sqrt{\epsilon_{\text{cutrho}}}$ : the FFT box carries a coefficient at every frequency out to its corner, and past the density's own cutoff those are the aliased tail of  $|\psi|^2$  rather than Fourier components — small, smooth and entirely plausible, which is why this is a guard and not a check on the answer. An empty or inverted energy window, and `core=True` on a dataset with no nonlinear core correction. **Said rather than refused**, because the numbers are right and only a claim about them would be wrong: these are *valence* structure factors, and a star's multiplicity applies to the charge, the magnetic members of a star being related by  $\det(R) R$  rather than being equal.

## 10.8 Scanning-tunnelling microscopy images

Tersoff and Hamann's result is that the current an s-wave tip draws at  $\mathbf{r}$  is the sample's local density of states there, at the energy the bias selects. So an STM image is not a new calculation, it is the *density built from different occupations*:

$$\rho_{\text{STM}}(\mathbf{r}) = \sum_{n\mathbf{k}} w_{\mathbf{k}} \tilde{\delta}(E - \epsilon_{n\mathbf{k}}) |\psi_{n\mathbf{k}}(\mathbf{r})|^2,$$

with  $\tilde{\delta}$  the same smeared delta a density of states is built from. Elk's task 162 and `pw.x`'s `plot_num = 5` both say exactly that. Two energy selections: with no bias the weight is a delta at  $E$  (Elk's, a density of states per unit volume, in  $1/(\text{bohr}^3 \text{ Ry})$ ), and with one it is the window  $[E, E + V]$  (QE's, a density in electrons/ $\text{bohr}^3$ ) — a *negative*  $V$  imaging the filled states, which is a real experiment's sign convention and the mode a semiconductor surface needs.

```
from defumat import Calculator

calc = Calculator.from_file("graphene.scf.in")

# constant height: the tip plane at crystal coordinate 0.85 along c
image = calc.get_stm(height=0.85, shape=(64, 64))

image.values          # (64, 64) the tunnelling density on the plane
image.coordinates     # (64, 64, 2) in-plane cartesian coordinates, bohr
image.extent()        # (x0, x1, y0, y1) for imshow
image.integral        # int rho_STM d3r = D(E_F): the sum rule
image.density         # the same field on the FFT grid

# filled states, 0.15 Ry below the Fermi level
filled = calc.get_stm(height=0.85, bias=-0.15)

# constant current: the corrugation, in bohr along the surface normal
scan = calc.get_stm(height=0.70, mode="constant-current",
                    current=1.0e-5, heights=(0.0, 6.0), nheights=80)
scan.heights_bohr    # (n1, n2), nan where the set-point is not crossed

# a magnetic tip: the image depends on which way it points
up = calc.get_stm(height=0.85, spin="up")          # collinear
along_x = calc.get_stm(height=0.85, spin=(1, 0, 0)) # noncollinear

# Elk's three corners, in crystal coordinates, for any other plane
side = calc.get_stm(plane=((0.15, 0, 0), (0.15, 0.3, 0), (0.15, 0, 1)))
```

The functional entry point is `defumat.workflows.stm.run_stm`. A delta at the Fermi level wants a denser k-grid than the SCF's, which `grid=` re-solves the bands on and **recomputes the Fermi level for**; `smearing` picks the delta (a Gaussian by default whatever the run used, because Methfessel–Paxton and cold smearing go negative on their wings and a negative tunnelling current is meaningless); and `band_cutoff` is `stm.f90`'s fixed truncation at three widths, off by default because it is an approximation.

**The plane is read out of the grid exactly.** A density is a finite sum of plane waves, so its value at a point between grid points is that sum evaluated there rather than a spline through neighbours. It matters for the one thing an image is about: the corrugation in the vacuum is a small modulation on a quantity falling by orders of magnitude across one grid spacing, and an interpolant of it carries the grid's own periodicity as a false corrugation. The three corners are in *crystal* coordinates, Elk's `vc1p2d`, and the sampling excludes the far edge ( $i/n$ , not  $i/(n-1)$ ) so a whole-cell image tiles.

**Against pp.x:**  $6.7 \times 10^{-10}$  on every point of the  $15^3$  grid of QE's own fcc aluminium, reading its `filplot` text dump so that nothing is interpolated on either side. Three of QE's conventions are needed to make it the same sum: `stm.f90` divides by the volume and not by `degauss`, so its image is the zero-bias one times the width; it uses the run's own smearing; and it truncates the band sum at three widths. That truncation is worth 0.4% *and the complete sum is the smaller one* — Marzari-Vanderbilt's delta is negative for  $x > \sqrt{2}$ , so the states QE drops carry negative weight. The check that involves no other code is the sum rule  $\int \rho_{\text{STM}} d^3r = D(E)$  against `compute_dos`, which holds to  $10^{-10}$  on a scalar run and on a spinor one, and in its window form on an ultrasoft one — the augmentation charge follows the tunnelling weights, which is why an ultrasoft or PAW image works here and `stm.f90`'s does not.

**A magnetic tip is the part neither code has.** It counts the states whose spin is along its own moment, so with  $P$  the tip's polarization

$$\rho_{\text{STM}}(\mathbf{r}; \hat{\mathbf{n}}) = \frac{1}{2} [\rho(\mathbf{r}) + P \hat{\mathbf{n}} \cdot \mathbf{m}(\mathbf{r})],$$

and  $P = 1$  makes the image a genuine spin channel. On an antiferromagnetic hydrogen chain the charge image is flat to six figures while the two spin projections are each other's mirror ( $\pm 2.58\%$ , antisymmetric to  $10^{-3}$ ), and they add back to the charge to  $10^{-14}$ . On a chain whose four moments each turn 90 degrees from the last, a tip along  $x$  sees atoms 1 and 3 with opposite sign and atoms 2 and 4 **not at all** ( $10^{-16}$  against a contrast of 0.063), while a tip along  $z$  sees none of the four — with nothing imposing any of it.

**The physics case is graphite.** In AB-stacked bilayer graphene one sublattice of the surface layer sits above an atom of the layer below and the other above a hollow, so the two are inequivalent and only one is bright at the Fermi level:  $1.64\times$  here, on two atoms of the same element carrying the same charge. That is the textbook result that half the atoms of graphite are invisible to an STM, and it is a statement about the density of states *at the Fermi level* rather than about the charge. The tunnelling density falls off log-linearly into the vacuum ( $R^2 = 0.9992$  over five orders) at a rate of at least  $2\sqrt{V_{\text{vac}} - E_F}$ , which is Tersoff–Hamann's own statement and involves none of the assembly; the bound is one-sided because graphene's states at  $E_F$  sit at  $K$ , whose in-plane momentum steepens the decay (measured 1.86, against 0.98 for  $k_{\parallel} = 0$ ).

**Refuses.** A spin spiral (its two spinor components live on different plane-wave spheres, so  $|\psi(\mathbf{r})|^2$  is not the lattice-periodic object this sum builds), a constrained `tot_magnetization` (one Fermi level per channel, and a tip sees one energy), an applied magnetic field (its energy is outside the reported total, so the level the tip would be put at is not the field-free one the image would be read as), a spin direction on a run

with no magnetization, and a *transverse* direction on a collinear run — there  $m_x$  is absent rather than zero, and projecting onto the axis would be a claim the calculation cannot make. A constant-current scan longer than the lattice period along the plane's normal, where the tip meets the periodic image of the surface and the density rises again. **Said rather than refused:** a constant-current image returns nan at every point where the set-point is not crossed inside the scan, with a warning saying how many; and a zero-bias image of a gapped run is identically zero, the delta sitting in the gap, which is what bias is for.

## 10.9 Tunnelling spectra: $dI/dV$ over an energy axis

An image is the local density of states at one tip energy, which is one picture at one bias. The other section of the same function is the curve at one place over many biases, which is what a scanning tunnelling microscope measures when it is used as a spectrometer, and what resolves a gap, a band edge or a state sitting inside a gap — none of which an image shows, since an image at a single energy reports only how much weight there is and not where in energy it sits,

$$\frac{dI}{dV}(\mathbf{r}, V) \propto \rho(\mathbf{r}, E_F + eV) = \sum_{\mathbf{k}, n} w_{\mathbf{k}} \delta(E_F + eV - \varepsilon_{\mathbf{k}n}) |\psi_{\mathbf{k}n}(\mathbf{r})|^2,$$

the same sum `run_stm` builds, evaluated at every energy of an axis.

**This is the measurement a modulation is read with**, and it is why the quantity is here rather than only in the unit cell: a charge density wave is a *gap* that opens in antiphase with the charge maxima, a spin density wave moves the two spin channels in opposite directions from cell to cell, and a domain wall carries a state that lives nowhere else. Each of the three is an energy axis at every position, and an ultracell, the subject of *A modulation over many unit cells*, is the calculation that has them.

**The energy axis is nearly free, and that is the whole of the implementation.**  $\psi_{\mathbf{k}n}(\mathbf{r})$  does not depend on the energy, so the states are sampled at the tip points *once* and each energy after that is one matrix product against the per-state weight. On an eight-cell spin density wave with a  $96 \times 12$  map, one, twenty-one and forty-one energies cost 2.50, 2.56 and 2.61 s, against 2.47 s for a single image: forty-one energies are 4.4 per cent more than one, and 39 times less than the forty-one images that are the only other way to get them. Neither `pw.x` nor Elk has an axis — `stm.f90` takes one scalar `sample_bias` and `wfplot.f90` sets one `delta` — so both compute a spectrum by being run again at every bias, paying the whole sum each time.

```
import numpy as np
from defumat import Calculator

calc = Calculator.from_file("si-ultracell-mag.in")
field = lambda x: 0.02 * np.cos(2 * np.pi * x[...] / 8)
wave = calc.get_ultracell(supercell=(8, 1, 1), kgrid=(1, 2, 2), nbnd=32,
                        magnetic_field=field)

# a spectrum above the middle of each unit cell: one tip point per cell,
# which is the line cut across the modulation
tips = np.stack([np.arange(8) + 0.5, np.full(8, 0.5), np.full(8, 0.35)], axis=-1)
bias = wave.fermi_energy + np.linspace(-0.06, 0.06, 41)
sts = calc.get_ultracell_sts(energies=bias, tip=tips, width=0.02, spin="up")
```

```

sts.values          # (41, 8) dI/dV, one column per cell
sts.bias_axis       # (41,) the bias in Ry, measured from the Fermi level
sts.current         # (41, 8) I(V), the same axis integrated outwards from V = 0
sts.integral        # (41,) D(E) per unit cell -- the *charge's* sum rule, which
                   # is what the weights give exactly whatever the tip is

# a whole map at every energy instead, and the ordinary unit-cell spectrum
maps = calc.get_ultracell_sts(energies=bias, height=0.35, shape=(48, 8))
plain = calc.get_sts(energies=bias, tip=tips[:1], width=0.02, grid=(4, 4, 4))

```

The entry points are `run_sts` and `run_ultracell_sts`, or `Calculator.get_sts` and `Calculator.get_ultracell_sts`.

**Reading it.** `energies` is the axis in Rydberg and has no default: the axis *is* the measurement and a range has no natural width. What comes back is `values`, of shape `(nE,)` + whatever the sampling was — a row of tip points, or a plane. `current` is  $I(V)$  within this approximation, a tip whose own density of states is flat and a barrier that does not depend on the bias; the decay of each state into the vacuum *is* in it, the prefactor is not, and the experiment's sign convention is kept, so the filled states below the Fermi level come back negative. A magnetic tip works exactly as it does for an image, through spin and polarization, and on a spin density wave it is what is wanted: an unpolarized tip cannot see the wave at first order and sees its square instead.

**Refuses.** Everything an image refuses, and two more of its own, both because  $\psi(\mathbf{r})$  is sampled and squared here where an image sums into a density. A **symmetry-reduced k-set**: `run_stm` goes through the density, which is symmetrised, so a wedge gives it the whole zone's answer, while nothing symmetrises this one and a wedge would return the sum over that wedge alone — a smooth, plausible, wrong density of states everywhere off a symmetry axis. On silicon's  $4 \times 4 \times 4$  grid reduced to eight k-points the wedge sum differs from the whole grid's by 98 per cent of the peak, while its *integral* is right to 1 per cent and the image on that same reduced set is right to 0.3 per cent — so a sum rule would not catch it and the image is genuinely unaffected. Pass `grid` (which builds the *complete* grid here, where an image's reduces) or `run_nosym`. A tip inside an **augmentation sphere** of an ultrasoft or PAW dataset, where the pseudo-wavefunction is not the true one and the norm is short by 3 to 9 per cent. On a dense bulk crystal that leaves nowhere to put a tip at all — SiC's radius is 2.4 bohr and its layers are closer together than that — which is the right answer rather than an obstacle: a tip sits above a *surface*, in vacuum, where a pseudo-wavefunction is the true one. An ultracell is exempt from both: it requires `nosym` and refuses ultrasoft and PAW already. Also refused: `bias` together with an energy axis, since the window is the integral the axis already carries and `current` is it; and a constant-current spectrum, where the tip moves as the bias is swept, which is a different experiment and  $n_{\text{heights}}$  times the cost. **Said rather than refused:** `current` raises on an axis whose step is coarser than the smeared delta, because a trapezoid that steps over a level returns a number of the wrong order rather than a slightly wrong one — one level of width  $w$  on an axis of step  $h$  integrates to 1.00004 at  $h = w$ , 1.14 at  $2w$  and exactly zero by  $250w$ , which is what any sane axis does against the  $10^{-5}$  Ry width a fixed-occupation run gets by default. A run with no Fermi level takes the middle of the gap as the zero of bias, with a warning, rather than 0 Ry.

## 10.10 Vertical tunnelling transport through a two-dimensional material

An STM image says what the tip *sees*. This says what gets *through*: an electron enters at a point  $\mathbf{r}$  above the material and leaves into an infinite plane below it — the substrate the sheet sits on — so what decides the current is the *nonlocal* Green's function between the two rather than the local density of states at the tip. A **point** tip makes  $\Gamma_t$  rank one, which collapses the Landauer-Büttiker trace exactly:

$$T(\mathbf{r}; E) = \text{Tr}[\Gamma_t G^r \Gamma_s G^a] \propto \int_{\text{plane}} d^2 r' |G(\mathbf{r}, \mathbf{r}'; E)|^2.$$

The substrate is taken to be *infinite and featureless*, so it is invariant under every lateral lattice translation and conserves the lateral momentum. The exit integral is then diagonal in  $\mathbf{k}$ , and everything reduces to one quadratic form per  $\mathbf{k}$ -point,

$$T(\mathbf{r}; E) = \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{nm} a_{n\mathbf{k}}(\mathbf{r}) a_{m\mathbf{k}}^*(\mathbf{r}) S_{\mathbf{k}}[n, m], \quad S_{\mathbf{k}}[n, m] = \int_{\text{plane}} \psi_{n\mathbf{k}}^* \psi_{m\mathbf{k}} d^2 r',$$

with  $a_{n\mathbf{k}} = \psi_{n\mathbf{k}}(\mathbf{r}) \sqrt{\tilde{\delta}(E - \varepsilon_{n\mathbf{k}})/\eta}$ . So **the sum over  $\mathbf{k}$  is incoherent and the interference is between bands at the same  $\mathbf{k}$** , which is what "the substrate is featureless" means physically.  $S_{\mathbf{k}}$  is a Gram matrix, so the transmission is non-negative by construction and is blind to a rotation inside a degenerate multiplet.

**Either electrode can be magnetic.** A spin-polarized substrate is a  $2 \times 2$  acceptance  $P_s = (1 + P_s \hat{\mathbf{n}}_s \cdot \boldsymbol{\sigma})/2$  sitting *inside*  $S_{\mathbf{k}}$ , between two bands. A spin-polarized *tip* is the same matrix on the other side,  $\Gamma_t = \gamma_t P_t$ , and it is not a weight but a contraction: with the spinor Green's function  $G_{ss'}(\mathbf{r}, \mathbf{r}') = \sum_n a_{n,s}(\mathbf{r}) \psi_{n,s'}^*(\mathbf{r}')$  the Landauer trace is no longer taken over the tip's spin, and the two spinor components stop adding and start interfering,

$$T(\mathbf{r}) = \int_{\text{plane}} d^2 r' \text{Tr}_{\text{spin}}[\Gamma_t G \Gamma_s G^\dagger] = \sum_{\mathbf{k}} w_{\mathbf{k}} \text{Tr}_{\text{spin}}[P_t M_{\mathbf{k}}(\mathbf{r})], \quad M_{\mathbf{k}}[s, s'] = \mathbf{a}_s^\top S_{\mathbf{k}} \mathbf{a}_{s'}^*.$$

$M$  is itself a Gram matrix,  $A^\top S A^*$  for the  $(n_{\text{bnd}}, 2)$  amplitude block, so it is positive semi-definite and  $\text{Tr}[P_t M] \geq 0$ : the non-negativity survives. With *both* leads polarized the map depends on the angle between the two moments, which is a **tunnelling magnetoresistance** image. **Note the asymmetry of the names:** here *spin* is the *substrate* and *tip\_spin* the *tip*, which is the opposite way round from `get_stm`, where *spin* is the tip. The substrate was the polarizer first and the name was kept.

```
from defumat import Calculator

calc = Calculator.from_file("graphene.scf.in")

# tip plane above the sheet, substrate plane below it, atoms in between
t = calc.get_vertical_transport(exit_height=0.38, height=0.62,
                              shape=(48, 48), broadening=0.02,
                              grid=(6, 6, 1))

t.image          # (48, 48) the transmission map, arbitrary units
t.interference    # coherent minus incoherent: what no local picture gives
t.notes["channels"] # open transmission channels the substrate offers
t.extent()       # (x0, x1, y0, y1) for imshow

# a transport dI/dV at one place: the tip sampling is paid once
import numpy as np
```

```

spectrum = calc.get_vertical_transport(
    exit_height=0.38, tip=np.array([[0.0, 0.0, 0.62]]),
    energies=calc.get_scf().fermi_energy + np.linspace(-0.2, 0.2, 41),
    broadening=0.02, grid=(6, 6, 1))

# a spin-polarized substrate: only on a magnetic run, and the two
# directions partition the unpolarized answer exactly
magnet = Calculator.from_file("h-sheet-lsda.scf.in")
up = magnet.get_vertical_transport(exit_height=0.15, height=0.85,
    spin="up", broadening=0.05)
down = magnet.get_vertical_transport(exit_height=0.15, height=0.85,
    spin="down", broadening=0.05)

# a magnetic tip on a noncollinear run, and both electrodes at once:
# the map then measures the angle between the two moments
spinor = Calculator.from_file("h-sheet-noncolin.scf.in")
valve = dict(exit_height=0.15, height=0.85, broadening=0.05,
    shape=(24, 24), spin=(1.0, 0.0, 0.0), polarization=0.9)
parallel = spinor.get_vertical_transport(tip_spin=(1.0, 0.0, 0.0),
    tip_polarization=0.9, **valve)
antiparallel = spinor.get_vertical_transport(tip_spin=(-1.0, 0.0, 0.0),
    tip_polarization=0.9, **valve)
tmr = antiparallel.image.mean() / parallel.image.mean() # 5.29 on this cell

parallel.tip_spin # (1.0, 0.0, 0.0), beside .spin for the substrate
parallel.tip_polarization # 0.9

```

### The amplitude is the on-shell one and that is a measurement, not a preference.

The literal Landauer denominator  $1/(E - \varepsilon + i\eta)$  cannot be evaluated by a sum over states: the states far from  $E$  build the barrier's evanescent decay entirely by cancellation between themselves. On a cell small enough to diagonalise completely (367 plane waves, 367 exact bands) the running sum for one  $G(\mathbf{r}, \mathbf{r}')$  wanders over more than an order of magnitude and lands only at the complete basis, with a cancellation ratio of **349**. Replacing the denominator by its modulus — which is Bardeen's golden rule, the sample visited on shell — keeps the interference between bands degenerate at the tip energy, which is the interference a real experiment lets happen, and converges: to **3e-7** in the mean between 8 and 45 bands. `method="resolvent"` is reachable and warns.

**Validated by identities**, since neither `pw.x` nor `Elk` computes this. (QE's `pwcond.x` is a Landauer transmission of a different geometry: two semi-infinite crystalline leads with the current along one axis, one conductance per energy, and no point contact and so no map. `Elk` has none.) Widening the substrate from a plane to the whole cell makes  $S_{\mathbf{k}}$  the identity by orthonormality, so the quantity becomes the Tersoff-Hamann tunnelling density of states *exactly* — agreeing with `run_stm` to **4.6e-13** with no factor between them, which is the check a normalisation error could not hide in.  $S_{\mathbf{k}}$  swept along its own normal integrates to the identity to 1.4e-15, and agrees with a real-space quadrature to 5e-16. A substrate polarized along  $\hat{\mathbf{n}}$  plus one along  $-\hat{\mathbf{n}}$  is a substrate that takes both, to round-off; one across the moment takes exactly half. The same holds for the *tip*, to **3.1e-16**, with the substrate polarized along a third direction at the same time, and a tip at  $P = 0$  is exactly half the unpolarized map (1.5e-16), the convention a nonmagnetic tip has in the STM images. A *magnetic* tip in the whole-cell limit reproduces the spin-polarized tunnelling density of states to **5.8e-13**, on a sheet whose moment is put in a generic direction: the transposed index order of  $M$  would return the answer for a tip at  $(n_x, -n_y, n_z)$ , which is real, non-negative and — on a moment lying along an axis — identical, so the check is only a check with  $m_y \neq 0$ . On that cell the mirrored tip's image differs by a factor

of two. Both electrodes polarized at  $P = 0.9$  give a parallel-to-antiparallel contrast of a factor **5.3**, with a perpendicular tip at the mean of the two to 5e-6. The same cell run as  $n_{\text{spin}} = 1$ ,  $n_{\text{spin}} = 2$  and as a spinor agrees to 1e-10 and 1e-5. And the physics: graphene's two Dirac states at  $K$  are degenerate partners, so the symmetry of that point makes the substrate's overlap on the pair a multiple of the identity and leaves nothing off-diagonal for the current to interfere through; its map correlates with the STM image at **1.000000** with zero interference, while an AB bilayer — whose current must cross both layer-polarized sheets — falls to **0.16**, its coherent map a factor of 26 below the incoherent one.

**Refuses.** A k-set with more than one division along the stacking axis: the lateral momentum is conserved exactly and the perpendicular one is not, so two  $k_{\perp}$  at the same  $k_{\parallel}$  interfere with a phase that depends on where the exit plane sits. A *symmetry-reduced* k-set, because a wedge sums to the map symmetrised over the whole point group while only the subgroup that leaves the exit plane in place belongs to this geometry — a mirror through the slab exchanges the tip side with the substrate side; grid= builds the whole grid for that reason. A tilted exit plane, which has no closed-form in-plane orthogonality to stand on. A tip or exit plane *inside* an augmentation sphere on an ultrasoft or PAW dataset — in the vacuum a pseudo-wavefunction is the true one, which is why those datasets otherwise need nothing extra. A spin-selective substrate *or a spin-polarized tip* on a run with no magnetization, there being nothing for either moment to couple to and every direction taking half of everything, which is the charge map again; and a transverse direction on a collinear run for either lead, where  $m_x$  and  $m_y$  are absent rather than zero. And, inherited from the STM images unchanged: a spin spiral, a constrained tot\_magnetization, an applied magnetic field. **Said rather than refused:** a geometry whose atoms do not lie between the two planes warns — a cell is periodic, so with both planes on the same side the electron tunnels through the vacuum and around the periodic image, which is a real number and not this one. **Not claimed:** a finite contact patch (which breaks the k-diagonality), a tip with structure beyond an s-wave, and an absolute conductance — the tip and substrate couplings are unfixed prefactors, so what the result carries is the map and its contrast.

### 10.11 Which k-points the tunnelling current comes out of

The section above asks *where on the surface* the current goes. This asks the conjugate question: *which states* carry it. Replace the point tip by a plane parallel to the surface and the real-space map collapses; what is left is one number per k-point, and on a two-dimensional metal that number is the answer to "which pocket of the Fermi surface does an electron actually tunnel out of".

Integrating the point-tip map over one cell's worth of the tip plane turns  $\int a_{n\mathbf{k}}(\mathbf{r}) a_{m\mathbf{k}}^*(\mathbf{r}) d^2r$  into  $g_{n\mathbf{k}} g_{m\mathbf{k}} S_{\mathbf{k}}^{\text{tip}}[m, n]$ , where  $S^{\text{tip}}$  is the *same* Gram matrix as the exit plane's, evaluated at the tip's height. So

$$W(\mathbf{k}) = w_{\mathbf{k}} \text{Tr} \left[ D_{\mathbf{k}} S_{\mathbf{k}}^{\text{exit}} D_{\mathbf{k}} S_{\mathbf{k}}^{\text{tip}} \right], \quad D_{\mathbf{k}} = \text{diag}(g_{n\mathbf{k}}), \quad \sum_{\mathbf{k}} W(\mathbf{k}) = \int_{\text{tip}} T(\mathbf{r}) d^2r.$$

That last equality is a theorem, not an approximation: this is the same object as the map, and nothing is sampled in real space, so it costs a small fraction of one — two Gram matrices and an  $n_{\text{bnd}}^2$  trace, against an  $n_k \times n_{\text{bnd}} \times n_{\text{points}}$  sampling.

**Three columns come back together**, because they are the same trace with one or both matrices replaced by the identity, and *the physics is in their ratio*. weight is  $W(\mathbf{k})$ ;

tersoff\_hamann is its  $S^{\text{exit}} \rightarrow 1$  limit, which is the tip plane's local density of states and so what an STM would see; bare is the further limit in which the tip plane stops discriminating too, the plain Fermi surface  $\sum_n \delta(E - \varepsilon_n \mathbf{k})$ . A fourth, incoherent, is every channel tunnelling on its own, taken in the basis that diagonalises  $S^{\text{exit}}$  inside each degenerate multiplet — that column is a *diagonal* and so is not invariant under the rotation a degenerate eigensolver is free in, where  $W$  is a trace and needs no such care.

**Why the ratio is large.** A state at in-plane momentum  $\mathbf{k}_{\parallel}$  decays into the vacuum as  $\psi \sim e^{-\sqrt{\kappa_0^2 + k_{\parallel}^2} z}$ , so a zone-corner pocket is suppressed against a zone-centre one by a factor that grows exponentially with the gap. That is also the **check that shares no machinery with either code**: fit a decay constant to each pocket over a height sweep and they must satisfy  $\kappa_1^2 - \kappa_2^2 = |\mathbf{k}_1|^2 - |\mathbf{k}_2|^2$ .  $\kappa$  is the *amplitude* decay and  $W$  is quadratic in  $\psi$  on each plane, so a sweep of the tip alone has  $d \ln W / dz = -2\kappa$  and a sweep of both planes  $-4\kappa$ ; halving on one side only makes the identity come out four times too large, which is what it is for. A height sweep costs *one* band solve, not one per height — the bands do not know where the tip is.

```
import numpy as np
from defumat import Calculator
from defumat.transport.momentum import decay_constants, decay_identity, pocket_mask

calc = Calculator.from_file("nbse2.in")          # a 1H-NbSe2 monolayer

# the two planes straddle the slab; a sweep of tip heights costs one band solve
run = calc.get_momentum_transport(exit_height=0.212, height=[0.677, 0.732, 0.788],
                                   grid=(6, 6, 1), broadening=0.006,
                                   smearing="gaussian")

run.weight          # (nheights, nk) -- the transmission per k-point
run.tersoff_hamann  # the same states, substrate made structureless
run.bare            # the plain Fermi surface
run.interference    # coherent minus incoherent
run.as_grid("weight") # reshaped to the Monkhorst-Pack grid, for a BZ image
run.per_kpoint()    # with the k-weight divided out, for another code

# the zone partition: closer to a corner than to Gamma, which on a hexagon is
# exactly 2/3 of the zone by area
corners = pocket_mask(run.kcartesian, calc.system.cell)
run.shares(corners, index=2)          # each column's share, and the suppression
run.reweighting(corners, index=2)    # (W_G/W_K)/(D_G/D_K): how the junction re-ranks

# the vacuum decay, and the identity that checks it
sweep = run.sweep("tersoff_hamann")
heights = np.array([1.5, 2.5, 3.5]) / 0.529177210903      # bohr
kappa = decay_constants(heights, sweep[:, [0, 14]])        # Gamma and K
decay_identity(kappa[1], kappa[0], run.kcartesian[14], run.kcartesian[0])
```

**The index order is the whole difficulty, and it needs two distinct planes to see at all.** The elementwise-looking alternative  $\sum_{nm} g_n g_m S^{\text{exit}}[n, m] S^{\text{tip}}[n, m]$  is  $\text{Tr}[D S^{\text{exit}} D (S^{\text{tip}})^T]$ : real, non-negative, and *exactly* equal to the right answer whenever either matrix is diagonal — so it agrees in the Tersoff-Hamann limit, agrees on a single-band metal, and passes every check that does not involve two planes at once. On random orthonormal bands it is **3%** out, which is far above round-off and far below anything a plot would show. Only the literal plane integral separates them, and it does: the two routes agree to **1e-12**, one

evaluating a Green’s function on a real-space quadrature of the tip plane and the other never leaving reciprocal space.

**Validated against a second implementation, which is new here.** Neither `pw.x` nor stock Elk computes this: Elk’s `fermisurf.f90` (tasks 100/101) writes bands for a plotting program without weighting them by anything, and its only tunnelling task is 162, the Tersoff-Hamann image. The one other implementation known is `elkpy`’s Elk patch (task 9007), and it was written independently — same on-shell Bardeen amplitude, same full Gram matrices on both planes, same trace of a product, and the same index-order trap recorded on its own side. On 1H-NbSe2 the two agree on the physics; the numbers are in `PLAN.md` P77.

**Refuses.** A *magnetic plane tip*: a polarized tip contracts the two spinor components through its own  $2 \times 2$  projector, which the plane integral does not collapse the way it collapses the trace over position, so it is left to `get_vertical_transport`’s map rather than approximated here. The substrate’s own acceptance does go in, inside the Gram matrix, exactly as it does there. A tilted tip plane, where the point-tip map allows one: the closed-form Miller-index orthogonality both sides rest on needs a plane spanned by two lattice vectors, and it is what makes this cheap. **Inherited from the vertical transmission unchanged:** a k-set with more than one division along the stacking axis, a symmetry-reduced k-set (`grid=` builds the whole grid), a plane inside an augmentation sphere on an ultrasoft or PAW dataset, gamma-only storage, a spin-selective substrate on a run with no magnetization, and — from the STM images — a spin spiral, a constrained `tot_magnetization` and an applied magnetic field. Both planes on the same side of the slab **warns** rather than refuses. **Not claimed:** an absolute conductance. The lead couplings are unfixed prefactors, so what this carries is the weights and their ratios — which is also why `shares` reports shares of a total rather than absolute weights.

## 11 Things with no `pw.x` counterpart

**Spin spirals** by the generalized Bloch theorem: a spiral of wavevector  $\mathbf{q}$  is a spinor whose two components live on *different* plane-wave spheres,

$$\psi_{\mathbf{k}}(\mathbf{r}) = \begin{pmatrix} e^{i(\mathbf{k}-\mathbf{q}/2)\cdot\mathbf{r}} u^{\uparrow}(\mathbf{r}) \\ e^{i(\mathbf{k}+\mathbf{q}/2)\cdot\mathbf{r}} u^{\downarrow}(\mathbf{r}) \end{pmatrix}$$

so in the rotated frame the density is lattice periodic and the SCF, the functional and the mixer are untouched. Only two terms carry  $\mathbf{q}$  —  $|\mathbf{k} \mp \mathbf{q}/2 + \mathbf{G}|^2$  and  $v_{\text{nl}}$  — and  $dE/d\mathbf{q}$  at frozen coefficients is `jax.grad` of them.

```
from defumat.workflows.spiral import (run_spiral_scan, relax_spiral_q,
                                     heisenberg_exchange)

scan = run_spiral_scan(system, pseudos, wavevectors) # E(q)
best = relax_spiral_q(system, pseudos) # BFGS on the reciprocal metric
print(best.wavevector, best.scf.total_energy)

J = heisenberg_exchange(scan, system.cell, shells) # fit E(q) for the J(R)

scan = run_spiral_scan(system, pseudos, wavevectors, gradients=True)
print(scan.relative) # E(q) - E(q_0) in mRy, from the energies
print(scan.integrated) # the same curve, from dE/dq along the path
```

heisenberg\_exchange fits  $E(\mathbf{q}) - E(0) = m^2 \sum_{\mathbf{R}} J(\mathbf{R}) [1 - \cos(\mathbf{q} \cdot \mathbf{R})]$  over a set of lattice shells, which is how a spiral scan becomes a spin model. compute\_spiral\_gradient gives  $dE/d\mathbf{q}$  on its own.

**$E(\mathbf{q})$  two ways.** With gradients=True the scan also takes  $dE/d\mathbf{q}$  at every point, and scan.integrated accumulates it along the path,

$$E(\mathbf{q}_n) - E(\mathbf{q}_0) = \sum_m \int d\mathbf{q} \cdot \frac{\partial E}{\partial \mathbf{q}},$$

by the trapezoid rule on the segments between successive wavevectors (both factors in lattice coordinates, so the contraction needs no metric and the path may bend). The two curves are the same physics and the gap between them is the useful part: the energies are computed on a plane-wave sphere *rebuilt* at every point and step by the Pulay error of a finite basis wherever a plane wave crosses the cutoff, while the gradient is taken at a *frozen* sphere and does not see those steps. On the hydrogen chain of tests/data/qe/h-chain-spiral.in at ecutwfc = 25 the direct energies rise, fall and rise again over a 13-point scan where the integrated curve descends monotonically; the two agree to 0.005 mRy at ecutwfc = 60 once the quadrature error is removed, against 0.130 at 25.

Two error sources live in that gap and refining tells them apart: the trapezoid rule's own  $h^2$  shrinks when the  $\mathbf{q}$  path is refined (0.051  $\rightarrow$  0.016  $\rightarrow$  0.003 mRy for 7  $\rightarrow$  13  $\rightarrow$  25 points) and the basis noise does not (0.139, 0.138, 0.137). What the integrated route does *not* buy is worth stating, since the intuition that it might is a common one: it is not cheaper (every point still needs its own SCF), it does not converge on a coarser  $k$ -mesh (the gradient is the exact derivative of the *same* fixed-mesh energy, so it inherits the same Brillouin-zone error), and it needs a *tighter* conv\_thr rather than a looser one, an energy's error being second order in the density's where a derivative's is first.

Also here: **per-atom magnetic fields**, the **topological invariants** of the previous section, and **rank- $n$  tensor symmetrisation** — symme.f90's symmatrix3/symtensor3 written at any rank.

**Refuses.** A spiral with spin-orbit coupling, permanently — it breaks the theorem, and Elk refuses it too; a spiral with symmetry, until the spin space group is written, so a spiral needs nosym; a spiral with ultrasoft/PAW; and spiral relaxation under a magnetic field, whose energy is outside the reported total, so the state is stationary for a different functional than the one being differentiated.

## 12 A modulation over many unit cells: the ultracell

A spin density wave in chromium has a period of 21 unit cells. A charged impurity in silicon is screened over about 20. A skyrmion is hundreds across. Each of them is a slow *envelope* on a crystal that is still atomically periodic, and a supercell pays the same price per degree of freedom for the envelope as for the atoms — which is why that length scale is the one nobody computes from first principles.

The ultracell separates the two. Take  $N = n_1 n_2 n_3$  copies of the unit cell. The reciprocal lattice vectors of that ultracell lying inside the unit cell's Brillouin zone are  $N$  wavevectors  $\mathbf{Q}$ , and everything allowed to be long ranged is written

$$\rho(\mathbf{r}) = \sum_{\mathbf{Q}} \rho_{\mathbf{Q}}(\mathbf{r}) e^{i\mathbf{Q} \cdot \mathbf{r}}, \quad \rho_{\mathbf{Q}} \text{ lattice periodic.}$$

The ultracell's Kohn-Sham states at an ultracell Brillouin zone point  $\mathbf{k}$  are then expanded

in the *unit cell's own* Kohn-Sham states at the  $N$  points that fold onto it,

$$|\Psi\rangle = \sum_{\mathbf{Q},n} c_{\mathbf{Q}n} |\psi_{\mathbf{k}+\mathbf{Q},n}\rangle, \quad n \leq \text{nbnd},$$

and what is diagonalised is the  $N_{\text{nbnd}} \times N_{\text{nbnd}}$  matrix

$$H_{(\mathbf{Q}_m),(\mathbf{Q}'_n)} = \delta_{\mathbf{Q}\mathbf{Q}'} \delta_{mn} \varepsilon_{\mathbf{k}+\mathbf{Q},n} + \langle \psi_{\mathbf{k}+\mathbf{Q},m} | \delta V | \psi_{\mathbf{k}+\mathbf{Q}',n} \rangle,$$

with  $\delta V$  the ultracell potential *minus* the unit cell's — the eigenvalue on the diagonal already carries the whole unmodulated potential, so putting it in twice would be the one easy mistake here.

**Why this is cheap.** The unit-cell states at the folded k-points are computed **once, before the self-consistent loop**, and frozen. Every iteration after that builds a small dense matrix and diagonalises it; no plane-wave Hamiltonian is touched again. The variational dimension per k-point is  $N_{\text{nbnd}}$  where a real supercell's is  $N_{\text{npw}}$ , and the ratio  $\text{npw}/\text{nbnd}$  is typically  $10^2$  to  $10^3$ . It is a prefactor, however, not a change of scaling: the dense solve is still  $O((N_{\text{nbnd}})^3)$ , so this reaches a few thousand cells along one direction rather than a micron in three.

**What the approximation is, exactly.** As  $\text{nbnd}$  grows to the full plane-wave count, the basis above spans precisely the ultracell's own plane-wave basis. So the method is a *variational truncation of the exact  $N$ -cell supercell problem to  $\text{nbnd}$  bands per folded k-point*, and nothing else. An eigenvalue's convergence carries no sign — the Hamiltonian moves with its own truncated density, so an ultracell eigenvalue is not an upper bound on the supercell's, and neither is a density — but it does converge, which is the property the implementation is checked against: a two-cell silicon ultracell under an applied modulation reproduces a real four-atom supercell's induced density, and the disagreement falls monotonically with  $\text{nbnd}$  (tests/regression/test\_ultracell.py).

**The total energy is the exception, and it does carry a sign.** What the run reports as `total_energy` is the Kohn-Sham free energy of the state the loop converged to, and the bases are *nested* in  $\text{nbnd}$ , so it is a monotone non-increasing upper bound on the  $N$ -cell supercell's own energy:  $+8.15 \times 10^{-5}$ ,  $+3.52 \times 10^{-6}$ ,  $+3.62 \times 10^{-7}$  and  $+1.43 \times 10^{-7}$  Ry at  $\text{nbnd} = 12, 24, 48, 64$  on that same silicon cell. Comparing against a supercell needs the *same* FFT box on both sides, since two discretisations of one functional differ by about  $10^{-6}$  Ry per cell here, which is larger than the top of that ladder; within the ladder the box is the same one and the monotonicity is exact.

```
from defumat import Calculator
import numpy as np

cal = Calculator.from_file("si.in")          # the two-atom unit cell

# a potential of one ultracell period along a_1, in Ry; x runs over [0, n_i)
modulation = lambda x: 0.05 * np.cos(2 * np.pi * x[... , 0] / 8)

ulr = cal.get_ultracell(supercell=(8, 1, 1), kgrid=(1, 2, 2),
                        nbnd=24, external=modulation)

print(ulr.iterations, ulr.accuracy)         # dr2 on the ultracell box, in Ry
print(ulr.density.shape)                   # (nspin, 8*n1, n2, n3): the whole ultracell
print(np.abs(ulr.modulation).max())        # rho - tile(rho_0): the induced envelope
print(ulr.fermi_energy, ulr.band_energy)
```

supercell is three integers and the ultracell lattice vectors are derived from them; kgrid samples the *ultracell*'s Brillouin zone, which is  $N$  times smaller than the unit cell's, so the folded set the frozen states are computed on is `supercell * kgrid`. `external` is an applied potential in Ry, given either as an array on the ultracell grid or as a function of unit-cell crystal coordinates, which run over  $[0, n_i)$  across the ultracell. `ulr.density` is normalised *per unit cell*, so it integrates to  $N$  times the electron count over the ultracell and equals the tiled unit-cell density exactly when nothing is modulated.

**nbnd is the convergence parameter and it is the only one.** It is the size of the variational basis per folded k-point, so the default — which for an insulator is the occupied bands and nothing above them — leaves the envelope nothing to be built from. Pass it. `ulr.multiplet_gap` reports the smallest gap the truncation opens anywhere on the folded k-set: where that is small the cut has fallen inside a degenerate multiplet, which member of it survives is whatever the eigensolver returned, and the basis is arbitrary there. **Whether that matters depends on where the cut is**, and the two cases were measured rather than assumed. Low — no more empty bands than occupied ones — the arbitrary rotation is among the very states the response is built from, and `run_ultracell` warns: `nbnd = 8` on silicon cuts at  $2.6 \times 10^{-14}$  Ry and does not converge at all. High in the empty manifold a cut this degenerate is both common and harmless — `nbnd = 32, 48` and `80` each open one ( $6 \times 10^{-15}$ ,  $5 \times 10^{-12}$ ,  $4.7 \times 10^{-11}$  Ry) and each converged in ten iterations to a monotonically improving answer — so the gap is reported there without a warning. Raise `nbnd` until the cut lands in a gap, or until it lands high enough not to be one.

**Convergence.** The mixed quantity is the density on the ultracell box and the convergence test is the Hartree energy of its residual — QE's `dr2`, generalised to  $|\mathbf{G} + \mathbf{Q}|$  — so `conv_thr` means here what it means everywhere else in this package. Kerker preconditioning is on by default and matters more here than in a unit cell: the smallest non-zero  $|\mathbf{G} + \mathbf{Q}|$  an ultracell carries is  $N$  times smaller, so the Hartree kernel there is  $N^2$  larger and long-wavelength charge sloshing is the whole difficulty. Elk's own example of the method runs at a mixing parameter of 0.001 for up to two thousand iterations; a two-cell silicon ultracell under an applied modulation converges here in nine.

**The total energy, and what it is for.** `ulr.total_energy` is the Kohn-Sham free energy per unit cell, and it is what says whether a modulation is the ground state rather than merely what a modulation does: the energy *gain* of a spin density wave over the uniform state is a difference of two of these. Neither reference code has it — Elk's `energyulr.f90` computes the occupied eigenvalue sum alone and `pw.x` has no ultracell — and the eigenvalue sum is not the total: it double-counts the Hartree and exchange-correlation energies exactly as an ordinary `eband` does, because the frozen eigenvalue carries the whole self-consistent potential. The assembly is therefore QE's own,

$$E = (\text{eband} + \text{deband}) + E_{\text{H}} + E_{\text{xc}} + E_{\text{Ewald}} + \text{demet},$$

with `deband` the integral of the density against the potential over the ultracell, and `ulr.energy_terms` carries the pieces. The Ewald term is the *unit cell*'s, unchanged, because the atoms do not move.

```
# the same run as above -- the energy comes with it and costs nothing extra
print(ulr.total_energy)           # Ry, per unit cell
print(ulr.energy_terms)          # the pieces, summing to it
print(ulr.energy_history[-3:])    # one per iteration, approaching from above
print(ulr.field_energy)          # None without a magnetic field
```

**This is the one quantity in the method that converges with a sign.** An ultracell eigenvalue is not an upper bound on the supercell's and neither is a density, for the

reason above — the Hamiltonian moves with its own truncated density. The energy is a bound because the bases are *nested*: raising `nbnd` keeps every function already in the basis, and the union over **Q** of the folded k-point spheres is the supercell’s own plane-wave space. So `total_energy` falls monotonically towards the real supercell’s from above, and on two-cell silicon under an applied modulation it does:  $+8.15 \times 10^{-5}$ ,  $+3.52 \times 10^{-6}$ ,  $+3.62 \times 10^{-7}$  and  $+1.43 \times 10^{-7}$  Ry at `nbnd` = 12, 24, 48, 64.

**Comparing against a supercell needs the same FFT box on both sides.** Two discretisations of one functional differ by about  $10^{-6}$  Ry per cell here, which is larger than the top of that ladder, and a supercell picks its dense grid from its own cutoff: at the default `ecutwfc` = 12 it takes (32, 15, 15) where the tiled unit cell’s is (30, 15, 15), and the third rung then comes out  $1.49 \times 10^{-7}$  Ry *below* the supercell. Choose a cutoff at which the two grids agree — `ecutwfc` = 13 puts the unit cell on (18, 18, 18) and the two-cell supercell on (36, 18, 18) — and do it through `ecutwfc` rather than `ecutrho`, since an ultracell refuses a double grid. Within one ladder the box is the same on every rung and the monotonicity is exact.

**An applied field’s energy is not in the total**, which is QE’s and Elk’s shared convention and this package’s everywhere else: `ulr.field_energy` carries  $-\int B \cdot m$  beside it. An applied *scalar* potential is treated the other way and its energy stays in the total, once, because that is what `vltot` is — which is also what makes a comparison against a supercell carrying the same modulation a comparison of the same quantity.

**Under a field, the quantity that carries the sign is `total_energy` + `field_energy`**, not the total on its own. What the calculation minimises is the full energy with the Zeeman term in it, and the reported total has that term taken out, so the total alone is a bound on nothing — measured on a partly-polarized hydrogen cell under a uniform 0.02 Ry field, it sits  $7.4 \times 10^{-7}$  Ry above an ordinary SCF at `nbnd` = 12 and  $9.7 \times 10^{-7}$  and  $1.6 \times 10^{-6}$  Ry *below* it at 24 and 40, moving monotonically the wrong way, where the sum is  $+4.15 \times 10^{-6}$ ,  $+2.47 \times 10^{-6}$  and  $+5.15 \times 10^{-7}$  — above at every rung and falling. Add the field energy back whenever two magnetic states are being compared.

**A spin density wave.** `nspin` = 2 is in, and it is what the method was published for: a modulation of the *magnetization* over tens of cells is the case a supercell cannot afford. Without spin-orbit coupling the ultracell matrix is block diagonal in spin, so two channels cost two solves of the same size rather than one of twice it; what the channels share is a **single Fermi level** over all  $2N_{\text{nbnd}}$  states, and that sharing is the physics — it is what lets an electron cross from the minority channel in one cell to the majority channel in the next, which is what a spin density wave is.

Nothing in a collinear calculation breaks spin symmetry on its own, so a modulated moment has to be driven. `magnetic_field` is the knob: a collinear  $B(\mathbf{r})$  in Ry, entering as  $v_{\uparrow} = -B$  and  $v_{\downarrow} = +B$ , in the same two forms external takes.

```
from defumat import Calculator
import numpy as np

cal = Calculator.from_file("si-mag.in")      # the same cell at nspin = 2

# a field of one ultracell period along a_1, in Ry
field = lambda x: 0.05 * np.cos(2 * np.pi * x[...], 0) / 2)

ulr = cal.get_ultracell(supercell=(2, 1, 1), kgrid=(1, 2, 2),
                        nbnd=12, magnetic_field=field)
```

```

print(ulr.cell_moments())          # the moment of each cell: the wave itself
print(np.abs(ulr.magnetization).max())  # rho_up - rho_dw on the box
print(ulr.charge_accuracy, ulr.magnetic_accuracy)

```

`ulr.cell_moments()` integrates  $\rho_{\uparrow} - \rho_{\downarrow}$  over each unit cell of the ultracell, which needs no sphere and no radius: the cells tile the ultracell exactly, so the partition is the geometry. On the cell above the two come back equal and opposite, which is the two-cell wave the field asked for.

**Read `magnetic_accuracy`, not `accuracy`, on a magnetic run.** `dr2` is the Hartree energy of the density residual, so its charge half is weighted by  $1/|\mathbf{G} + \mathbf{Q}|^2$  and its magnetization half by a constant — and in an ultracell the gap between them is wider than in a unit cell, because the smallest  $|\mathbf{G} + \mathbf{Q}|$  is  $N$  times smaller and the charge half’s weight at the envelope’s own wavevector is  $N^2$  larger. A converged total can therefore leave the moment still moving; the two halves are reported apart for exactly that reason, and `ulr.history` carries both per iteration.

**What a magnetic run is checked against.** At  $N = 1$  a *uniform* applied field is the same physics as an ordinary SCF with `B_field(3)`, which shares nothing with this code path — it goes through `add_bfield.f90`’s expression inside a plane-wave SCF. On the silicon cell at  $B = 0.02$  Ry the reference moment is 0.71159883 and the ultracell gives it to  $9.6 \times 10^{-3}$  relative at `nbnd = 12`,  $3.2 \times 10^{-3}$  at 24 and  $7.7 \times 10^{-4}$  at 40: the same convergence in `nbnd` the charge shows, now for the moment and against a route with no ultracell in it. Against a real two-cell supercell of polarized hydrogen under a scalar modulation, the charge and the magnetization agree to  $1.7 \times 10^{-4}$  and  $1.4 \times 10^{-4}$  at `nbnd = 24`.

**A texture, not just an amplitude.** `nspin = 4` is in as well, and it buys the modulation a collinear run cannot write down: one in which the moment’s *direction* turns from cell to cell rather than its length growing and shrinking along a fixed axis. A helix, a cycloid, a domain wall and a skyrmion are all of that kind. **Spin-orbit coupling comes with it and costs nothing:** it is a term in the Hamiltonian the frozen unit-cell states were diagonalised with, and the ultracell only ever sees their eigenvalues and their coefficients, so there is no spin-orbit term anywhere in the method.

What changes is the shape of one matrix. A collinear potential is diagonal in spin, so the  $2N_{\text{nbnd}}$  problem is two blocks of half the size; a spinor basis function has two components and the potential acts on it as the  $2 \times 2$  matrix

$$V(\mathbf{r}) = v_0(\mathbf{r})\mathbb{K} + \mathbf{B}(\mathbf{r}) \cdot \boldsymbol{\sigma}, \quad (1)$$

whose off-diagonal entries couple the components at every point — and that coupling is exactly what lets the magnetization turn. There is one block, one Fermi level over all  $N_{\text{nbnd}}$  states, and each state holds *one* electron rather than two.

The applied field changes shape with the regime: `magnetic_field` is a scalar  $B(\mathbf{r})$  for `nspin = 2` and a **vector** one for `nspin = 4`, either as a (3, \*box) array or as a function of unit-cell crystal coordinates returning three components per point. A field that *turns* is what drives a helix, and it cannot be written down at all in the collinear form.

```

from defumat import Calculator
import numpy as np

cal = Calculator.from_file("h-noncolin-ultracell.in") # moments free to turn

# a field of 0.01 Ry turning once over the ultracell, in the xy plane
def turning(x):

```

```

phase = 2 * np.pi * x[..., 0] / 4
return 0.01 * np.stack([np.cos(phase), np.sin(phase),
                        np.zeros_like(phase)], axis=-1)

ulr = cal.get_ultracell(supercell=(4, 1, 1), kgrid=(1, 2, 2), nbnd=16,
                      magnetic_field=turning, mixing_beta=0.3)

print(ulr.cell_moments())      # (4, 3): one moment VECTOR per cell
print(ulr.magnetization.shape) # (3, *box): m_x, m_y, m_z on the grid

```

`cell_moments()` returns an  $(N, 3)$  array for a noncollinear run where it returns  $(N,)$  for a collinear one, and `magnetization` likewise becomes the three cartesian components on the box. Neither needs a sphere or a radius: the cells tile the ultracell exactly, so the partition is the geometry. On the cell above the four moments come back at  $27^\circ$ ,  $62^\circ$ ,  $-41^\circ$  and  $-48^\circ$  in the plane the field turns in, with nothing out of that plane — so they follow the field, but by less than the  $90^\circ$  per cell it asks for. That is not a convergence failure: a hydrogen lattice at 5.5 bohr is a close-packed ferromagnet whose inter-cell exchange is far larger than a 0.01 Ry field, and what the run is measuring is the stiffness of the magnet against a long-wavelength twist. A texture that turns freely wants a crystal whose modulated axis is weakly coupled.

**A noncollinear ultracell converges slowly when the field is weak, and lowering `mixing_beta` is the wrong response.** Without spin-orbit coupling or anisotropy, turning every moment together by the same angle costs no energy, so the  $\mathbf{Q} = 0$  transverse component of the magnetization has no restoring force and the self-consistent solution sits in a nearly flat manifold rather than at a point. The mixer has to *traverse* that manifold to reach the place the field pins, and `mixing_beta` is how fast it travels.

On the four-cell hydrogen ultracell above, every run given 300 iterations: at a 0.01 Ry turning field every beta converges to the same state and the best is 0.5 (36 iterations, against 48 at 0.7 and 115 at 0.2). At 0.002 Ry — a field five times weaker, pinning the direction five times less — `beta = 0.7` converges in **263** iterations and `beta = 0.3` does *not converge in 300*, its residual sitting between  $2 \times 10^{-5}$  and  $9 \times 10^{-5}$  for the whole run without ever leaving. The small beta is not stable where the large one is unstable; it is **stuck**.

So a slow noncollinear run wants more iterations, or a stronger field to pin the direction with — not a smaller `mixing_beta`. The run says so by name when it fails to converge.

**What a noncollinear run is checked against.** At  $N = 1$  a uniform vector field is the same physics as an ordinary SCF with `B_field(1:3)`, which goes through `add_bfield.f90`'s expression inside a plane-wave SCF and shares nothing with this code path — and for a vector field that check reaches the transverse components, which no collinear identity can. On a hydrogen cell whose moment is free, that reference run says something clean on its own: the converged moment's direction equals  $\hat{\mathbf{B}}$  to *six decimals*, which is what a magnet with no anisotropy does and what fixes the sign of the coupling. Against it, at  $\mathbf{B} = (0.005, 0.0005, 0)$  Ry, the ultracell reproduces the moment to  $1.9 \times 10^{-2}$  relative at `nbnd = 16`,  $6.6 \times 10^{-3}$  at 40,  $1.5 \times 10^{-3}$  at 64,  $4.5 \times 10^{-4}$  at 96 and  $1.6 \times 10^{-4}$  at 128 — monotone, and a factor of 116 across the range. What is converging is the *transverse* component:  $m_x$  is right to  $1.5 \times 10^{-3}$  at the first rung, and  $m_y$  walks 0.0601, 0.0634, 0.0691, 0.0694, 0.0732, 0.0740 towards the reference's 0.0743.

**Expect to need more bands than a collinear run, by about a factor of two.** A spinor state holds one electron and lives in a space of  $2 n_{\text{pwx}}$ , so a given `nbnd` is half as complete a basis as the same number would be for one collinear channel. The ladder above is at

37 per cent of the basis at `nbnd = 128`, which is where stage 1's charge ladder was at `nbnd = 80` on silicon. Convergence is also slower the larger the *rotation* asked for: a field that turns the moment by 57 degrees converges in `nbnd` several times more slowly than one that tilts it by 6, because a change of the local spin frame has to be built out of empty states of the opposite spin channel, which the exchange splitting puts far away.

**There is no total energy**, in this code or in Elk, whose `energyulr.f90` reports the sum of occupied eigenvalues and the Fermi level and nothing else. `ulr.band_energy` is that sum, per unit cell. The energy *gain* of a modulated state over the uniform one — the quantity that says whether a spin density wave is the ground state at all — is therefore a quantity neither code computes.

Ultrasoft and PAW datasets: the augmentation charge is a function of the density through  $D_{ij}$ , so the frozen unit-cell states stop being a fixed basis as soon as the modulation moves. Gradient-corrected functionals: the gradient of the ultracell density carries the envelope's own gradient, which a per-cell evaluation drops (Elk takes that approximation silently; it is refused here). Meta-GGA. An applied `magnetic_field` on a noncollinear cell that **carries no magnetization** — a spin-orbit run on a nonmagnetic crystal, where the density and the potential have one component apiece and a field has nothing to be added to; give the unit cell a small `starting_magnetization` so that it carries four, which it may then converge back to nearly zero, and what the run measures is the Pauli susceptibility. A ground state converged **under a magnetic field or a constrained moment**, because the frozen eigenvalues would carry the field the SCF ended with — which `reducebf` and the fixed-spin-moment scheme both change from the input — while  $\delta V$  is rebuilt here from the density alone; apply the field through `magnetic_field` instead. DFT+U, the occupation matrix being per atom where an ultracell has  $N$  copies of each. Spin spirals, as a second modulation of the same states. Gamma-only storage, the basis being built at  $N$  points none of which is  $\Gamma$ . A double grid, the wavefunctions and the density then living on different ultracell boxes. Symmetry is not used at all: a modulation breaks the crystal's point group, so the run needs `nosym` and the whole k-grid. And a non-integer ultracell is refused, where Elk's `avec` is an independent input with no consistency check against its Q-grid. An **unconverged** unit-cell ground state, because its density is the one thing an ultracell holds fixed and there is no later iteration in which a bad seed could work itself out. Two things are *not* refused and are limits on a claim rather than on a run. The total energy's monotone bound rests on the free energy being minimised, which Methfessel-Paxton smearing breaks — its occupation function is non-monotone, so the generalised entropy is not concave and the self-consistent point is not a minimum; gaussian, Fermi-Dirac and Marzari-Vanderbilt are safe, and the energy itself is computed either way. And a cell with more than one self-consistent solution has no ladder at all: the energy then *ranks* solutions at one `nbnd` and does not certify a sequence across several, since nothing says the same solution is found at each.

## 13 What a modulation looks like to a tip

A charge or spin density wave, a screened impurity, a domain wall: what all of these are actually seen with is a scanning-tunnelling microscope, so an ultracell that cannot be imaged is a calculation of something nobody measures directly. This section is the join between the ultracell and the two things a tip measures, the Tersoff-Hamann image and the vertical tunnelling transmission. The third, the *spectrum* at one place over many biases, is the same relabelling read along an energy axis and has its own subsection

under *Tunnelling spectra* — on a modulation it is usually the one to reach for, since a wave moves a band edge and an image at one bias cannot tell a level that moved from a level that gained weight.

Nothing new is computed, and the reason is one observation about the basis. An ultracell state is

$$\Psi_j(\mathbf{r}) = \sum_{\mathbf{Q},n} a_{\mathbf{Q},n}^j \psi_{\mathbf{k}_0+\mathbf{Q},n}(\mathbf{r}),$$

and in a plane-wave basis that sum is a relabelling rather than a transform: a plane wave with unit-cell Miller index  $\mathbf{G}$  belonging to  $\mathbf{Q}$  is the ultracell plane wave  $\mathbf{h} = n\mathbf{G} + \mathbf{q}$ , and no two of them collide, because reducing  $n(\mathbf{G} - \mathbf{G}') + (\mathbf{q} - \mathbf{q}') = 0$  modulo  $n$  forces  $\mathbf{q} = \mathbf{q}'$  and the sphere sits inside the dense grid. So the union over  $\mathbf{Q}$  of the  $N$  unit-cell spheres is the ultracell's own sphere at  $\mathbf{k}_0$ , the band index contracts away, and what is left is an ordinary coefficient vector that every routine written for a unit-cell wavefunction takes unchanged once it is handed the ultracell's Miller indices, its cell and  $\mathbf{k}_0$  in its reciprocal coordinates. On a modulated two-cell silicon ultracell the state sampled from that vector agrees with the box transform the density is built from to  $1.7 \times 10^{-16}$  against a peak of 0.14, and it integrates to 1 over the ultracell.

The image is then the ultracell density with a smeared delta at the tip energy in place of the occupations, which is what a Tersoff-Hamann image is in Elk's `wfplot.f90` and in QE's `stm.f90`, and the transmission is the same contraction `run_vertical_transport` uses with the ultracell's geometry in it. The entry points are `run_ultracell_stm` and `run_ultracell_transport`, or `Calculator.get_ultracell_stm` and `Calculator.get_ultracell_transport`.

```
from defumat import Calculator
import numpy as np

cal = Calculator.from_file("si.in")          # the two-atom unit cell

# a potential of one ultracell period along a_1, in Ry
cal.get_ultracell(supercell=(8, 1, 1), kgrid=(1, 2, 2), nbnd=24,
                  external=lambda x: 0.05 * np.cos(2 * np.pi * x[0] / 8))

img = cal.get_ultracell_stm(height=0.35, axis=2, shape=(96, 12),
                           bias=-1.5, width=0.01)
print(img.values.shape)                    # (96, 12): the plane spans all eight cells
print(img.integral)                        # 8.0039: the electrons in the window, per cell
```

**Two conventions, and the second is the one to know.** The plane is given in *unit-cell* crystal coordinates running over  $[0, n_i)$  across the ultracell, which is what `external` already takes, so `height = 0.35` means what it means in a unit-cell run and a map spans the whole modulation by default. And the image is a tunnelling density of states *per unit cell*, in the same units a unit-cell image reports, so the two are directly comparable: with nothing applied, the ultracell image is the unit cell's image tiled, to  $1.6 \times 10^{-7}$  of the peak at  $N = 2$ . The window above covers the whole valence band, so the integral comes back as silicon's eight valence electrons; a run with a Fermi level puts the tip at it instead, and one with fixed occupations warns and takes the middle of the gap, which is `stm.f90`'s own rule.

**A magnetic tip is what a spin density wave needs.** A collinear system is invariant under flipping every spin together with the sign of  $B$ , so the charge cannot respond at first order and a wave driven by `magnetic_field` is almost invisible in the charge *density*: on an eight-cell silicon ultracell carrying 0.12 Bohr magnetons the total density varies by  $1.9 \times 10^{-4}$  of its mean from cell to cell. What an unpolarized tip sees is neither that

small nor the wave itself, but the wave *squared*, two periods where the wave has one, at 37 per cent of its mean against a polarized tip's 82, since the leading charge response is at twice the wavevector and the tip energy is a smeared band edge rather than a count of electrons. spin and polarization project the channel, exactly as they do for a unit-cell image.

**The number.** The reference is the one the ultracell itself is checked against: a real  $N$ -cell supercell under the same applied potential, run through this package's own SCF and imaged the same way. The error on the induced corrugation falls with `nbnd`, which is the whole content of "a variational truncation to `nbnd` bands per folded k-point": 22.2, 5.2 and 2.0 per cent at `nbnd` = 12, 24 and 48 on a two-cell silicon ultracell, where the induced corrugation is itself 0.6 per cent of the image. Widening the exit region of the transmission to the whole ultracell must give the image back exactly, with no factor between them, and does so to  $10^{-11}$ ; that identity is what holds the two normalisations against each other, since the image carries its  $N$  in the weights and the transmission carries it in the volume the states are normalised over.

**Sampling the plane is exact and its set is the ultracell's own dense sphere,  $|\mathbf{G} + \mathbf{Q}|^2 \leq \text{ecutrho}$ ,** rather than Elk's choice of the unit cell's dense sphere at every  $\mathbf{Q}$ . Read back at the box's own grid points, the sphere reproduces a modulated silicon density to  $3 \times 10^{-16}$  where the other set is out by  $5 \times 10^{-9}$  at  $N = 2$  and  $2 \times 10^{-8}$  at  $N = 3$ . The box itself aliases its outermost shell from about  $N = 5$ , at  $10^{-11}$ , which is the ordinary `ecutrho = 4ecutwfc` corner aliasing one level out and is a property of the ultracell's grid rather than of the sampling.

Everything the ultracell itself refuses, since the run has to exist first. A result whose states were dropped (`keep_states = False`), which cannot be rebuilt, the diagonalisation on the folded k-set being the expensive step of the whole method. For the transmission, an ultracell that is more than one cell deep along the stacking axis: the exit plane would then sit between two of the stacked slabs, so the electron leaves through the material rather than out of it, and two  $\mathbf{Q}$  differing only along that axis share every in-plane index the exit integral has to tell apart. The same axis must carry one k-point, for the reason a unit-cell junction needs one. A denser k-set for the image, the way `grid` re-solves the bands for a unit-cell image: on an ultracell that is not post-processing but another run of the whole loop, so it is asked for with `kgrid` in `get_ultracell` instead. And a constrained `tot_magnetization`, whose two channels have their own Fermi levels where a tip sees one energy. An applied `magnetic_field` is **not** refused here, where a unit-cell image refuses it: there the field is something the ground state was converged under and the reported total does not carry, while here it is the modulation's own driver and the eigenvalues the tip energy is read from are the eigenvalues under it.

## 14 Continuing an all-electron ground state

Elk is an all-electron LAPW code and this is a pseudopotential plane-wave one, so their converged densities are different objects: Elk's carries the core electrons and the nuclear cusp, and this one carries a smooth pseudo valence density normalised to the valence electron count. What crosses between them is a *starting guess*, and the run's own SCF still happens on top of it.

Inside each muffin tin Elk stores the density as a real-spherical-harmonic expansion on a

logarithmic radial mesh, and everywhere else as a periodic function on a uniform grid,

$$\rho(\mathbf{r}) = \begin{cases} \sum_{lm} f_{lm}(r) R_{lm}(\hat{\mathbf{r}}), & |\mathbf{r} - \mathbf{R}_a| < r_a, \\ \sum_{\mathbf{G}} \tilde{\rho}(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{r}}, & \text{otherwise,} \end{cases}$$

so putting it on a plane-wave grid means deciding which region each grid point is in, then interpolating radially or summing the Fourier series. The muffin-tin part is written in pointwise and is never Fourier truncated, which is why the nuclear cusp transfers exactly and why nothing is lost to aliasing.

```
from defumat import Calculator

calc = Calculator.from_file("scf.in", pseudo_dir="pseudo")
seed = calc.get_elk_seed("elk_run/") # STATE.OUT beside GEOMETRY.OUT
result = calc.get_scf(starting_density=seed)
```

The argument is an Elk *run directory* rather than a file: STATE.OUT carries neither the lattice vectors nor the atomic positions, so GEOMETRY.OUT has to be beside it. And elk.in is not a substitute, because Elk's default tshift moves the origin onto the inversion centre and only GEOMETRY.OUT is written in that shifted frame. Pass report=[] to get back a SeedReport, which is what the reconstruction measured on the way: what it integrated to before renormalisation, how many of Elk's reciprocal lattice vectors this run's density cutoff kept, and what fraction of the norm sat outside it.

ElkState is the reader on its own, for looking at the state rather than seeding with it. read\_elk\_state is the same thing as a function and read\_elk\_geometry reads GEOMETRY.OUT alone, returning the ElkGeometry that state.geometry also holds; all four are exported from defumat.io.

```
from defumat.io import ElkState

state = ElkState.read("elk_run/")
print(state.rmt, state.efermi) # muffin-tin radii; e_F in Rydberg
print(state.muffin_tin_charges()) # the charge inside each sphere
print(state.interstitial_charge()) # and the charge outside all of them
rho = state.evaluate_at(points) # the density at arbitrary points
```

The pieces of the reconstruction are exported too, from defumat.io.elk\_density, because each is a separately checkable part of somebody else's convention rather than an internal: real\_spherical\_harmonics (the sign convention is Elk's and is not the textbook one), poly4 (the 4-point Lagrange radial window), spline\_weights (the quadrature the radial integrals use, where Simpson's rule is 9e-6 away), characteristic\_function (the Fourier-truncated step that separates the two regions) and interstitial\_coefficients (rhoir as a G-vector list). density\_on is what get\_elk\_seed calls.

**What it is worth, measured.** On simple-cubic hydrogen at  $a = 3.0$  bohr, the reconstruction agrees with Elk's own evaluation of its own density at every one of the 4096 points Elk printed, to  $4.5 \times 10^{-11}$ , which is the precision the file was printed at rather than any error in the transfer. The charge inside the sphere is 0.6125761996 and outside it 0.3874238004, both exact to every digit Elk printed. And a run started from that density and a run started from a superposition of atomic charges converge to the same state:  $-1.080181442650$  Ry either way, agreeing to  $6 \times 10^{-14}$  Ry with converged densities  $9 \times 10^{-8}$  apart.

**It buys no iterations, and that is the honest result.** Both runs take four. At a lower

cutoff the seeded one takes *one more*. For a hydrogen atom a superposition of atomic charges is already almost the answer, and the all-electron cusp inside the pseudisation radius is a place where Elk's density is further from the pseudo one than the atomic guess is. The reader is worth having because it bridges to an all-electron ground state, not because it is faster.

**Refuses.** A **spin-polarized** Elk state, and a **spin spiral** with it — a magnetization is a second field with a transfer rule of its own, and a seed that carries the charge while silently dropping the moment would start a magnetic run from a non-magnetic guess. A **DFT+U** or fixed-tensor-moment state, whose occupation matrices are trailing records nothing here reads. A state written by Elk older than **2.0.0**, which Elk's own reader refuses. A directory with no `GEOMETRY.OUT`, or one whose geometry is from a different run. A calculation on a **different cell** — a density is transferable between identical lattices and there is no interpolation here that would make anything else mean something. A calculation with more than one spin channel. And an element with a **core**: the muffin-tin density includes the core states and nothing in `STATE.OUT` separates them from the valence charge, so a run whose valence electron count is far from what the state integrates to is refused by name rather than scaled to fit. Hydrogen, with no core at all, is what this path is for. Passing an Elk density as a **fixed** density, with no SCF above it, is not offered: the two codes' converged densities are different functions wherever there is a core, and no splice of the two would be a solution of anything.

## 15 Performance: dials, memory and GPUs

**Two batching dials** control how much is in flight, and both defaults follow the platform: on a CPU, QE's loop — one  $k$ -point and one band at a time, which is what a *cache* wants — and on a GPU the whole of both axes, which is what a card wants. Either end is reachable everywhere:

```
DEFUMAT_K_BATCH=all DEFUMAT_BAND_BATCH=all python3 run.py
```

or per call, `run_scf(..., k_batch=...)`. The chunk size changes the order contributions are summed in and nothing else — round-off,  $\sim 1e-15$  Ry.

**The band dial is worth twice as much in a spinor run**, because a band in flight is then *two* real-space boxes rather than one: a noncollinear calculation with 403 bands on a  $240 \times 54 \times 200$  grid puts  $403 \times 2 \times 2,592,000$  complex numbers — 33 GB — into one transform if the whole block goes at once, and several such arrays are live inside a single application of  $H$ . `DEFUMAT_BAND_BATCH=16` makes that 1.3 GB. The answer is unchanged bit for bit, not merely to round-off, because chunking the band axis reorders nothing: it is a map, not a sum.

**Why the default is per platform.** A GPU has no cache to fit in and inverts the conclusion: on a ten-atom metal,  $k=1, b=1$  costs **4.5x** what  $k=all, b=all$  does on the same card, sixteen-atom silicon runs at **0.20x** — an outright loss against four CPU cores — and at thirty-two atoms  $b=1$  did not finish at all. So a run on a device gets both batched unless it says otherwise, and the plausible half-measure  $k=all, b=1$  is never the default: it is the *worst* setting available, because it buys the batched mode's memory with the looped mode's kernel launches. An explicit argument beats the environment variables, which beat the platform.

**A third dial says where the wavefunctions are kept** (`wfc_store`, `DEFUMAT_WFC_STORE`), which is a different question from how many are in flight. The store is  $n_{\text{spin}}n_kn_{\text{bnd}}n_{\text{pwx}}n_{\text{pol}}$  complex — 11.3 GB on a 45-atom slab at six  $k$ -points, 45 GB at twenty-four — and it stays resident all through the SCF because it is the next iteration’s starting guess. Quantum ESPRESSO does not keep it on the device: `c_bands.f90:get_buffer` and its `save_buffer` partner hold one  $k$ -point’s worth and `io_level` says where the rest lives. `wfc_store='host'` is that buffer.

*What it buys, and what it does not.* Parked, the store is off the accelerator while the energy terms, the potential on the dense grid, the mixer’s history, the PAW one-centre terms and the checkpoint write are running — which is where the other large allocations are. It is *on* the device for the diagonalisation, for `becsum` and for the density, because it is an input to all three, so this does **not** lower a peak that sits inside the eigensolver. The default follows the platform for a measured reason: on a CPU the host *is* the device, and a round trip of the store runs at 1.3 GB/s here, which at slab scale would be 18.6 s per iteration for nothing. On an accelerator the same transfer is PCIe, around a second. So device on a CPU, host on anything else, and the answer is bit-for-bit identical either way — a transfer is not arithmetic.

**A fourth dial says whether the projectors are kept** (`projectors`, `DEFUMAT_PROJECTORS`), and it is the one that has been measured to move a peak rather than a stage.  $\langle k + G|\beta \rangle$  is  $n_kn_{\text{pwx}}n_{\text{kb}}$  complex and resident for the whole run: on the 45-atom slab above that is **13.96 GB**, larger than the wavefunctions themselves. Quantum ESPRESSO does not hold it either — `c_bands.f90:111` calls `init_us_2` *inside* its  $k$ -loop — and `projectors='rebuild'` is that: only the phase-free core is kept, one column per *species* channel where the full array has one per *atom* channel, and each  $k$ -point’s is formed on demand. The expression is the same one either way, so the result is bit-for-bit identical.

*Why it moves a peak where the store dial does not.* The other three dials bound what is *in flight* or move what is held between two points; this one removes what is *held*, so it sits underneath every stage. Measured on that slab, two runs on the same card differing only in this flag: every stage bracket falls by 13.96 GB, and the peak with them — **79.14 GB to 65.18 GB**, with the converged energies identical to the last printed digit. **The saving is the whole of  $\langle k + G|\beta \rangle$** , not that array minus a core: the core is built unconditionally and both routes pay for it, so what the dial chooses between is the whole- $k$  array and one rebuilt chunk. The saving is therefore  $n_k/k_{\text{batch}}$  times the chunk — it grows with how many  $k$ -points are *not* in flight, and at a single  $k$ -point it is exactly neutral, buying nothing and costing nothing. It is the ratio  $n_{\text{kb}}/n_{\text{cs}}$ , atoms per species, that decides how much smaller the core is than the array, which is why the core stays cheap on a large cell. Measured cost:  $n_{\text{at}}n_{\text{pwx}}$  complex exponentials each time a  $k$ -point’s projectors are used, which on that slab is **+1.5% per SCF iteration** (46.1 s against 45.4 s) and 10 s more to compile. The default is `store`.

Nothing is refused, but two things are worth knowing before switching. The default is `store` and every validated number in this document was measured with it. And a size estimate follows the dial only from 433abd7 onward: an older `defumat` size reports the stored array for a rebuilt run and so overstates the floor by the largest single line in its own table.

**Measured GPU speedups** (one H200 against one CPU core, per SCF iteration): 2x on a cheap metal, 3–5x on norm-conserving silicon, 9–21x with an augmentation charge or a magnetization, and **83–115x on spin-orbit**. Energies agree to  $\sim 1\text{e-}9$  Ry — the same agreements the CPU already has.

**Memory.** Spin-orbit is where it bites: a 20-atom bismuth cell with a relativistic ultra-

soft dataset peaks at **34.7 GB**, which fits a 141 GB card and does not fit a 32 GB one. Everything else in that sweep sits under 2 GB.

**A response costs memory according to how it is differentiated**, which is worth knowing before starting one. A *forward* response — the Sternheimer solves behind  $\epsilon$  and  $Z^*$  — costs 0.7–1.0 GB over the SCF that produced the state, and a *forward-over-reverse* one — the dynamical matrix, the Raman tensors — 1–2 GB. The *stress* is the exception and it is a *reverse* derivative through the radial transforms: **10.5 GB** on eight-atom ultrasoft silicon, ten times any of the others, and an order more than the SCF it came from. Measure your own with `tools/gpu/phase5.py <case> --property <name>`.

**The augmentation charge is the largest object in an ultrasoft or PAW run on a large cell**, and there are two ways of keeping it. Below `DEFUMAT_AUG_MAX_BYTES` (default 2 GB)  $Q_{ij}(\mathbf{G})$  is built on the whole  $G$  sphere once and kept, which is what every number in the accuracy table below was measured with. Above it, Quantum ESPRESSO’s scheme takes over: the *radial* transforms

$$Q_{nm}^L(q) = \frac{4\pi}{\Omega} \int_0^{r_{\text{kkbeta}}} dr j_L(qr) [r^2 Q_{nm}^L(r)]$$

are tabulated on a grid in  $|q|$  at  $dq = 0.01$  and  $Q_{ij}(\mathbf{G})$  is rebuilt from them a block of  $\mathbf{G}$  at a time, every iteration, exactly as `addusdens` and `newd` do it. On a 45-atom slab with 21 Å of vacuum that is **8.4 MB** against **65 GB** for one relativistic nickel dataset.

```
DEFUMAT_AUG_MAX_BYTES=off python3 run.py # never tabulate
DEFUMAT_AUG_MAX_BYTES=2G  python3 run.py # the default
DEFUMAT_AUG_CHUNK=8192   python3 run.py # G vectors per block
```

`DEFUMAT_AUG_CHUNK` sizes the rebuild’s working array,  $(n_h, n_h, \text{chunk})$ , and defaults to putting it near 256 MB; like the batching dials it is a loop bound over an exact sum and is invisible in a result. The two schemes agree to **4e-10 Ry** on a total energy and exactly at  $\mathbf{G} = 0$ . On eight-atom ultrasoft silicon the two contractions cost about **4.5x** more from the table, and the whole run comes out level, because the table is cheaper to build than the stored array is.

**Past the end of the table is NaN, not an extrapolation.** The table reaches  $2\sqrt{\text{ecutrho}}$  — Quantum ESPRESSO’s own `cell_factor` — so that a strained cell, whose  $G$  vectors move while the  $G$  set does not, still lands on it. A strain large enough to take  $|\mathbf{G}|$  past that gives NaN rather than a clamped value, because a clamped  $Q^L(q)$  would be smooth, of the right order and wrong, and would show up only as a bad stress.

**A PAW run’s other large object is its one-centre quadrature, and what it costs is a derivative rather than a run.** Each atom’s sphere carries  $(n_{\text{spin}}, n_x, \text{mesh})$  radial fields, and inside a force or a stress every atom’s reverse pass is live at once: **272 MB per atom** for a relativistic nickel dataset with a magnetization, so 4.1 GB for a fifteen-atom sublattice and no ceiling. `DEFUMAT_PAW_ATOM_BATCH` is how many spheres are in flight together — all for one `vmap`, an integer for `paw_onecenter.f90`’s own loop over atoms:

```
DEFUMAT_PAW_ATOM_BATCH=1  python3 run.py # one sphere at a time
DEFUMAT_PAW_ATOM_BATCH=all python3 run.py # the whole sublattice at once
```

Left alone it chunks above three atoms per species and not below, because that is where the crossover was measured: chunked, the tape stops growing with the sublattice altogether (4.1 GB → 0.80 GB at fifteen nickel atoms) and the run gets *faster* — a ten-atom

PAW cell's force is 0.505 s against 0.418 s and its SCF iteration 1.40 s against 1.28 s. On two atoms it is the other way, 0.761 s against 1.286 s, which is what the threshold is for. Like the other batching dials it is a loop bound over an exact sum and is invisible in a result beyond round-off.

**Precision.** Every dtype comes from `config.Precision`; x64 is enabled before any array exists, and all validation is float64. float32 is a performance mode and **never one a correctness claim is made in**.

## 16 Accuracy summary

Against Quantum ESPRESSO on the same input:

| quantity                                  | agreement                                 |
|-------------------------------------------|-------------------------------------------|
| total energy, norm-conserving LDA         | $\sim 1\text{e-}9$ Ry                     |
| ultrasoft, PAW                            | $\leq 3\text{e-}9$ Ry                     |
| PBE / revPBE / PBEsol                     | $\leq 6\text{e-}9$ Ry                     |
| metals, every smearing and tetrahedra     | $2.5\text{e-}8$ Ry                        |
| collinear spin                            | $1.2\text{e-}9$ Ry                        |
| noncollinear magnetism                    | $2.8\text{e-}9$ Ry                        |
| spin-orbit coupling                       | $\leq 1.3\text{e-}8$ Ry                   |
| DFT+U                                     | $\leq 6.7\text{e-}9$ Ry                   |
| band structure                            | 0.0002 eV                                 |
| forces                                    | $\leq 2\text{e-}5$ Ry/bohr                |
| stress (13 cases)                         | $\leq 2.7\text{e-}7$ Ry/bohr <sup>3</sup> |
| relaxed geometry                          | $1\text{e-}6$ bohr                        |
| dielectric constant (NC, US, PAW)         | $\leq 1.2\text{e-}4$                      |
| Born effective charges (NC)               | every printed digit                       |
| phonons at $\Gamma$ , silicon             | $0.05\text{ cm}^{-1}$                     |
| phonons at $\Gamma$ , silicon (US, PAW)   | $0.02\text{--}0.03\text{ cm}^{-1}$        |
| phonons at $\Gamma$ , aluminium (metal)   | $0.002\text{ cm}^{-1}$                    |
| Raman/IR spectra vs <code>dynmat.x</code> | every printed digit                       |

**One case does not close, and is recorded rather than explained.** `bi10-soc` — ten bismuth atoms with a fully-relativistic dataset — sits **1.9e-4 Ry** from QE on a total of  $-1477.737$ , which is  $1.3\text{e-}7$  relative. Both codes choose the same symmetry operations, the same  $k$ -point, the same grid and the same  $G$ -vectors, and both converge. It is in the test suite at that tolerance, not hidden.

## Where to go next

- `README.md` — the short tour and a first calculation
- `notebooks/` — 26 worked examples on concrete systems, executed and committed, each with a `.md` export beside it
- `PERFORMANCE.md` — the running performance log, CPU and GPU
- `PLAN.md` — the architecture, the phase breakdown, the trap catalogue
- `GPU.md` — the GPU roadmap